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Mathematical Modelling for Engineering and Science



MASTER'S THESIS

Investigations on some gradient elastoplasticity models: existence results, qualitative analysis and numerics

Supervisor:

Prof. Filip Rindler
University of Warwick

Candidate:

Bernardo Ardini
2146900

Co-supervisor:

Prof. Giulio Giusteri
Università di Padova

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Premise

Introductory notions regarding mechanical modeling and plasticity theory

1. A general framework for developing mechanical models

In this section we describe a general framework that allows to systematically develop mechanical models. All the models presented in this thesis will be formulated in agreement with this framework.

The approach presented here is based on the principle of virtual powers, which is a powerful and elegant tool to describe the equilibrium of the “forces” acting on a mechanical system. The “forces” must be intended in a generalized sense here. In fact, the configurations of the mechanical system are described as elements of a space (possibly infinite dimensional) of configurations, and therefore the generalized forces must be understood as actions that “push” the configuration of the system within this space. Moreover the configuration of the mechanical system can include also the description of internal or microscopic conditions of the system, so that a generalized forces does not necessarily correspond to a mechanical macroscopic force in the usual sense.

Some historical information about the birth of the principle of virtual powers can be found in [26] and in [29]. Modern works where the principle of virtual powers is used to develop systematically mechanical model are [18], [24], [22], where second gradient theories are considered, and [20], [21], [17] that treat materials with microstructure.

1.1. Kinematics. Kinematics is concerned with the description of the possible *configurations* of a mechanical system. We assume that they can be described as the elements of a Banach space V , the *space of configurations*.

EXAMPLE 1.1. Consider N material points in $\mathbb{R}^d$. Their configuration can be described by the matrices $x = (x_1, \dots, x_N) \in \mathbb{R}^{d \times N}$, where $x_\alpha \in \mathbb{R}^d$ with $\alpha \in \{1, \dots, N\}$ is the position of the α th material point.

EXAMPLE 1.2. The configurations of a deformable body at large strain in $\mathbb{R}^d$, can be identified with the deformations $\chi \in W^{1,p}(\Omega)^d$ with $p \in (1, \infty)$, where $\Omega \subset \mathbb{R}^d$, open and bounded subset of $\mathbb{R}^d$ is some reference configuration of the body. Every $x \in \Omega$ is the reference position of a material point and $\chi(x)$ is the position occupied by that material point in the deformed configuration. The condition of incompressibility, $\det \text{Grad} \chi > 0$ a.e. in Ω , was purposely excluded from the definition of the configuration space in order to preserve the linear structure. Even though we do not consider it at the level of kinematical description, it can be imposed in a later moment, as we will show.

EXAMPLE 1.3. Consider a body that can be subject to damage. This can be modeled considering an internal damage variable $\alpha \in W^{1,p}(\Omega)$ with $p \in (0, 1)$. Where $\alpha = 1$ the material is sound, while where $\alpha = 0$ the material is completely damaged. The material is partially damaged where $\alpha \in (0, 1)$. One should add the constrain $\alpha \in [0, 1]$, but this is not required in the definition of the space of configurations, again in order to preserve the linear structure.

Examples 1.2 and 1.3 suggest that the assumption that space of configurations has a linear structure is not restrictive in most mechanical models, even when the truly attainable configurations actually should lie in a space M with the structure of a manifold. Indeed in most cases, M can be seen as a subset of a larger linear space of configurations V . Then we can formulate the model on V and impose that the evolution of the configuration of the system is restricted to M by including suitable reactions, as we will see. Thus the geometric complexity of M can be avoided and reduced to the analytical complexity of managing the reactions within V required to enforce the constraint

REMARK 1.4. As Example 1.3 show, the configuration of the mechanical system can include also the description of an internal state of the system that does not necessarily correspond with the macroscopic observable placement of the system in the space. This is particularly true also in the case of plasticity theory, that will be discussed in the next section.

1.2. Statics. With statics we mean the treatment of equilibrium of forces, in a generalized sense, acting on a mechanical system. Any *generalized force* acting on the mechanical system, can be described as a linear functional $\mathcal{F} \in V^*$. Given any $\tilde{v} \in V$, the duality pairing $\langle \mathcal{F}, \tilde{v} \rangle$ can be interpreted as the projection of $\mathcal{F}$ along the direction $\tilde{v}$. If we interpret $\tilde{v}$ as a virtual rate, that is $\tilde{v} = \dot{\tilde{u}}(0)$ for an arbitrary evolution $\tilde{u} \in C^1([0, 1], V)$ of the configuration of the system, then $\langle \mathcal{F}, \tilde{v} \rangle$ is called the virtual power of $\mathcal{F}$ along the virtual rate $\tilde{v}$.

EXAMPLE 1.5. (Example 1.1 continued) A generalized force can be seen as an element $F = (F_1, \dots, F_N) \in (\mathbb{R}^{d \times N})^* = \mathbb{R}^{d \times N}$. The virtual power acts on $\tilde{v} = (\tilde{v}_1, \dots, \tilde{v}_N)$ as

$$\langle F, \tilde{v} \rangle = \sum_{\alpha=1}^N F_\alpha \cdot \tilde{v}_\alpha.$$

Here F_α is the force acting on the α th material point.

EXAMPLE 1.6. (Example 1.2 continued) Consider the functional $\mathcal{F} \in W^{-1,p^*}(\Omega)^d$ of the form

$$\langle \mathcal{F}, \tilde{v} \rangle = \int_{\Omega} P : \text{Grad } \tilde{v} dX + \int_{\Omega} f \cdot \tilde{v} dX + \int_{\partial\Omega} h \cdot \tilde{v} dX,$$

where $P \in L^{p^*}(\Omega, \mathbb{R}^{d \times d})$, $f \in L^{p^*}(\Omega)^d$ and $h \in L^{p^*}(\partial\Omega)^d$. The first integral represents a generalized force that acts in the direction of the local bending of the body, the second integral represents a volume force distributed over Ω , the last integral a surface force distributed over $\partial\Omega$.

Consider now a time interval $[0, T]$ and an actual evolution of the configuration of the system, $u : [0, T] \rightarrow V$. For simplicity, in this discussion we assume that $u \in \text{AC}([0, T], V)$ even though less regular evolutions, for example exhibiting jumps, can be considered as we will see subsequently in this thesis. Along its evolution, the mechanical system is subject to various generalized forces and we call $\mathcal{F} : [0, T] \rightarrow V^*$ their sum. We require that equilibrium holds, that is

$$\mathcal{F}(t) = 0 \quad \text{for a.e. } t \in (0, T).$$

This can be written also as

$$(1.1) \quad \langle \mathcal{F}(t), \tilde{v} \rangle = 0 \quad \text{for all } \tilde{v} \in V \text{ and a.e. } t \in (0, T).$$

Equation (1.1) is the statement of the principle of virtual powers, that is, the virtual power of the generalized force acting on the system along all virtual rates must vanish.

1.3. Constitutive relations. To get an evolution equation, able to predict the evolution of the mechanical system, it is necessary to link $\mathcal{F}$ to the evolution u through constitutive relations. We focus on constitutive relations of the form

$$\mathcal{F}(t) \in \hat{\mathcal{F}}(u(t), \dot{u}(t), t) \quad \text{for a.e. } t \in (0, T),$$

where $\hat{\mathcal{F}} : V \times V \times [0, T] \rightsquigarrow V^*$. We postulate that

$$(1.2) \quad \hat{\mathcal{F}}(u, v, t) = -\partial_u \mathcal{E}(u, v, t) - \partial_v \mathcal{R}(u, v, t),$$

where $\mathcal{E} : V \times V \times [0, T] \rightarrow (-\infty, \infty]$ and $\mathcal{R} : V \times V \times [0, T] \rightarrow [0, \infty]$ are called respectively the *energy* and the *dissipation potential*. The ∂ symbol can have different definitions according to the regularity of $\mathcal{E}$ and $\mathcal{R}$. If Fréchet differentiability is assumed then ∂ denotes the Fréchet differential, while if only convexity is assumed it denotes the subdifferential. Less regularity requires to use extended notions of subdifferentials such as the Fréchet subdifferential. In view of (1.2), we can split $\mathcal{F} = \mathcal{F}_c + \mathcal{F}_d$, where $\mathcal{F}_c(t) \in -\partial_u \mathcal{E}(u(t), \dot{u}(t), t)$ is the conservative generalized force and $\mathcal{F}_d(t) \in -\partial_v \mathcal{R}(u(t), \dot{u}(t), t)$ is the dissipative generalized force. The requirement that $\mathcal{F}_d$ has a dissipative behavior can be formulated as

$$\langle \mathcal{F}_d(t), \dot{u}(t) \rangle \leq 0 \quad \text{for a.e. } t \in (0, T).$$

This means that the projection of $\mathcal{F}_d$ along the direction of the rate is negative, or equivalently that the power of $\mathcal{F}_d$ is nonpositive. This is automatically satisfied if

$$\langle \xi, v \rangle \geq 0 \quad \text{for all } (u, v, t) \in V \times V \times [0, T] \text{ and } \xi \in \partial_v \mathcal{R}(u, v, t).$$

This last condition is called *dissipation inequality*.

In most situations one can split $\mathcal{F} = \mathcal{X} - \mathcal{J}$, where $\mathcal{X}$ and $\mathcal{J}$ from $[0, T]$ to V^* are the external generalized force and the internal generalized force respectively. The internal generalized force is due to the internal response of the mechanical system, while the external generalized force represents the external actions imposed on it. In this case the constitutive relations are usually specialized to

- (i) $\mathcal{E}: V \times [0, T] \rightarrow (-\infty, \infty]$ of the form $\mathcal{E}(u, t) = \Psi(u) - \langle \mathcal{X}(t), u \rangle$, where $\Psi: V \rightarrow (-\infty, \infty]$ is the free energy, describing the conservative internal response, and $\mathcal{X}: [0, T] \rightarrow V^*$ is the external generalized force,
- (ii) $\mathcal{R}(u, v, t) = \Phi(u, v)$, where $\Phi: V \times V \rightarrow [0, \infty]$ is the dissipation potential describing the dissipative internal response.

Then the constitutive relations can be written as $\mathcal{J}(t) \in \hat{\mathcal{J}}(u(t), \dot{u}(t))$ for a.e. $t \in (0, T)$, where $\hat{\mathcal{J}}: V \times V \rightsquigarrow V^*$ is given by

$$(1.3) \quad \hat{\mathcal{J}}(u, v) = \partial \Psi(u) + \partial_v \Phi(u, v).$$

In this case the equilibrium condition reads

$$\partial \Psi(u(t)) + \partial_v \Phi(u(t), \dot{u}(t)) \ni \mathcal{J}(t) = \mathcal{X}(t) \quad \text{for a.e. } t \in (0, T)$$

or

$$(1.4) \quad \partial_u \mathcal{E}(u(t), t) + \partial_v \Phi(u(t), \dot{u}(t)) \ni 0 \quad \text{for a.e. } t \in (0, T).$$

This is known as the *doubly nonlinear equation*. Now the evolution problem can be formulated as follows.

PROBLEM 1.7. Let $u_0 \in \text{dom } \mathcal{E}(\cdot, 0)$ and $\mathcal{X}: [0, T] \rightarrow V^*$. Find $u \in AC([0, T], V)$ such that (1.4) holds.

REMARK 1.8. Before imposing particular constitutive relations, $\mathcal{J}$ and $\mathcal{X}$ are completely undefined. However we notice that, before making specific constitutive assumptions, it is useful to assume that $\mathcal{J}$ and $\mathcal{X}$ have a more specific structure as we will see in the following chapters. We notice that, in order to keep the notation simple, we will sometimes write $\tilde{\mathcal{S}}$ instead of $\langle \mathcal{S}, \tilde{v} \rangle$ for $\mathcal{S} \in \{\mathcal{J}, \mathcal{X}, \mathcal{F}\}$.

REMARK 1.9. In terms of Φ , the dissipative inequality is

$$(1.5) \quad \langle \xi, v \rangle \geq 0 \quad \text{for all } (u, v) \in V \times V \text{ and } \xi \in \partial_v \Phi(u, v).$$

A possible condition ensuring (1.5) is that $\Phi(u, \cdot)$ is convex and $\Phi(u, 0) = 0$. Indeed if $(u, v) \in V \times V$ and $\xi \in \partial_v \Phi(u, v)$, then by definition of subdifferential, $0 = \Phi(u, 0) \geq \Phi(u, v) + \langle \xi, 0 - v \rangle$, so that $\langle \xi, v \rangle \geq \Phi(u, v) \geq 0$.

1.4. Adding constraints with the method of Lagrange multipliers. To include a kinematical constraints on the process, say $u(t) \in K$ for all $t \in [0, T]$, one can set $\mathcal{E}(u, t) = \infty$ if $u \notin K$. This can be done by including ι_K in $\mathcal{E}$. In this way $\mathcal{F}_c(t)$ will include the reaction $\mathcal{P}(t) \in -N_K(u(t)) \subset V^*$ required to enforce the constrain. Here $N_K = \partial \iota_K$.

EXAMPLE 1.10. (Example 1.2 continued) Let $\Phi: W^{1,p^*}(\Omega)^d \rightarrow (-\infty, \infty]$. To describe an elastic response, we choose Ψ of the form

$$\Psi(\chi) = \int_{\Omega} W(X, \text{Grad } \chi(X)) dX,$$

where $W: \Omega \times \mathbb{R}^{d \times d} \rightarrow (-\infty, \infty]$ is a continuous free energy density. Incompressibility impose that $W(X, F) = \infty$ if $\det F \leq 0$.

EXAMPLE 1.11. (Example 1.3 continued) To ensure that $z(x) \in [0, 1]$ for a.e. $x \in \Omega$, one choose $\Phi: W^{1,p}(\Omega) \rightarrow (-\infty, \infty]$ such that $\Phi(z) = \infty$ if $z \neq 0$ on a set of positive measure.

However there are situations in which it would be desirable to treat the constraint reaction $\mathcal{P}$ as an independent variable instead. In this sense $\mathcal{P}$ can be considered as an additional descriptor of the configuration of the system. This is particularly useful if one wants to consider constitutive relations depending on the reaction $\mathcal{P}$ itself, as it happens for the model that is discussed in Part 2. Then we split the generalized force as $\mathcal{F} = -\mathcal{J} + \mathcal{X} + \mathcal{P}$. The constitutive relations are imposed on $\mathcal{J}$ only as

$$\mathcal{J}(t) \in -\partial \Psi(\mathcal{P}(t), u(t)) - \partial \Phi(\mathcal{P}(t), u(t), \dot{u}(t)) \quad \text{for a.e. } t \in (0, T),$$

where now $\Psi: V^* \times V \rightarrow (-\infty, \infty]$ and $\Phi: V^* \times V \times V \rightarrow [0, \infty]$. Instead $\mathcal{X}$ is assigned while $\mathcal{P}$ is included among the unknowns of the evolution together with u . Then the evolution equation can be written as

$$(1.6) \quad \begin{aligned} &u(t) \in K \\ &\mathcal{P}(t) \in -N_K(u(t)) \quad \text{and} \quad \partial_u \mathcal{E}(\mathcal{P}(t), u(t), t) + \partial_v \Phi(\mathcal{P}(t), u(t), \dot{u}(t)) \ni \mathcal{P}(t) \quad \text{for a.e. } t \in (0, T), \end{aligned}$$

where now $\mathcal{E}: V^* \times V \times [0, T] \rightarrow (-\infty, \infty]$ is defined by $\mathcal{E}(\mathcal{P}, u, t) = \Psi(\mathcal{P}, u) - \langle \mathcal{X}(t), u \rangle$. Then the evolution problem becomes as follows.

PROBLEM 1.12. Let $u_0 \in \text{dom } \mathcal{E}(\cdot, 0)$ and $\mathcal{X}: [0, T] \rightarrow V^*$. Find $u \in AC([0, T], V)$ and $\mathcal{P}: [0, T] \rightarrow V^*$ such that (1.6) holds.

2. A review of plasticity theory

Solution concepts and general existence results

1. An existence result for evolutionary variational inequalities

Let V be a separable and reflexive Banach space. Consider a mechanical system with space of configurations V and assume that

- (E) the free energy $\Phi: V \rightarrow \mathbb{R}$ is quadratic in the sense that $\Psi(u) = \frac{1}{2}\langle \mathcal{A}u, u \rangle$ for some $\mathcal{A} \in \mathcal{L}(V, V^*)$, and Ψ has superlinear growth, i.e.,

$$\lim_{\|u\| \rightarrow \infty} \frac{\Psi(u)}{\|u\|} = \infty,$$

- (D) the dissipation potential $\Phi: V \rightarrow [0, \infty]$ is convex, proper, l.s.c., positively 1-homogeneous and Lipschitz on its domain.

Assumption (E) immediately implies that Ψ is Fréchet differentiable with Fréchet differential $\partial\Psi(u) = \mathcal{A}$ for all $u \in V$. Since Φ is convex and Lipschitz continuous on its domain then it admits a non empty subdifferential $\partial\Phi(v)$ for all $v \in \text{dom } \Phi$. **Per il teorema di Rockafeller sappiamo che è subdifferenziabile nell'interno del suo dominio, poi però data la Lipschitzianità su tutto il bordo immagino che si possa dimostrare che è subdifferenziabile su tutto il dominio in realtà.** In this case Problem 1.7 can be formulated explicitly as follows.

PROBLEM 2.1. Given $\mathcal{X} \in H^1((0, T), V^*)$ with $\mathcal{X}(0) = 0$, find $u \in AC([0, T], V)$ such that $u(0) = 0$ and

$$(2.1) \quad \langle \mathcal{A}u(t), v - \dot{u}(t) \rangle + \Phi(v) - \Phi(\dot{u}(t)) \geq \langle \mathcal{X}(t), v - \dot{u}(t) \rangle \quad \text{for all } v \in V$$

for a.e. $t \in (0, T)$.

Inequalities of the form (2.1) are called *evolutionary variational inequalities*. The following theorem ensure the existence and uniqueness of solutions to Problem 2.1.

THEOREM 2.2. Let V a separable and reflexive Banach space and let assumption (E) and (D) hold. Then for every $\mathcal{X} \in H^1((0, T), V^*)$ with $\mathcal{X}(0) = 0$, there exists a unique $u \in AC([0, T], V)$ solution to Problem 2.1.

PROOF. See [25]. **Riportarla.** □

2. Energetic solutions

2.1. Motivation. Let Q be a separable and reflexive Banach space. Let $\mathcal{E}: Q \times [0, T] \rightarrow \mathbb{R}$ Fréchet differentiable such that $\mathcal{E}(\cdot, t)$ is convex for all $t \in [0, T]$ and let $\Phi: Q \times [0, \infty]$ be convex, lower semicontinuous and positively 1-homogeneous. Consider the following doubly nonlinear equation

$$(2.2) \quad \partial_q \mathcal{E}(q(t), t) + \partial\Phi(\dot{q}(t)) \ni 0.$$

We say that $q \in AC([0, T], V)$ is a differentiable solution if (2.2) holds for a.e. $t \in (0, T)$. From Proposition B.28 this is equivalent to

$$(2.3) \quad -\partial_q \mathcal{E}(q(t), t) \in \partial\Phi(0) \quad \text{and} \quad \Phi(\dot{q}(t)) = -\langle \partial_q \mathcal{E}(q(t), t), \dot{q}(t) \rangle = -\frac{d}{dt} \mathcal{E}(q(t), t) + \partial_t \mathcal{E}(q(t), t)$$

for a.e. $t \in (0, T)$. The first condition is the requirement that the conservative forces belong to the admissibility set $\partial\Phi(0)$. From the definition of subdifferential, this is equivalent to

$$\Phi(\hat{q} - q(t)) \geq \Phi(0) - \langle \partial_q \mathcal{E}(q(t), t), \hat{q} - q(t) \rangle \quad \text{for all } \hat{q} \in Q.$$

Exploiting the convexity of $\mathcal{E}(\cdot, t)$ it follows that

$$\mathcal{E}(\hat{q}, t) - \mathcal{E}(q(t), t) \geq \lim_{\epsilon \rightarrow 0} \frac{\mathcal{E}(q(t) + \epsilon(\hat{q} - q(t)), t) - \mathcal{E}(q(t), t)}{\epsilon} = \langle \partial_q \mathcal{E}(q(t), t), \hat{q} - q(t) \rangle.$$

Then the first condition in (2.3), implies

$$(2.4) \quad \mathcal{E}(q(t), t) \leq \mathcal{E}(\hat{q}, t) + \Phi(\hat{q} - q(t)) \quad \text{for all } \hat{q} \in Q,$$

Since Φ is 1-homogeneous, also the inverse implication holds, indeed

$$\langle \partial_q \mathcal{E}(q(t), t), \hat{q} - q(t) \rangle = \lim_{\epsilon \rightarrow 0} \frac{\mathcal{E}(q(t) + \epsilon(\hat{q} - q(t)), t) - \mathcal{E}(q(t), t)}{\epsilon} \geq \frac{\Phi(\epsilon(\hat{q} - q(t)))}{\epsilon} = \Phi(\hat{q} - q(t)).$$

The mechanical meaning of condition (2.4) is the following: the energy $\mathcal{E}(q(t), t) \leq \mathcal{E}(\hat{q}, t)$ released by the system while passing from $q(t)$ to $\hat{q}$ cannot exceed the dissipation $\Phi(\hat{q} - q(t))$. This is a stability condition. The second condition in (2.3) is the power balance. Integrating it on the interval $(0, t)$ for $t \in (0, T]$ we get the energy balance

$$\mathcal{E}(q(t), t) = \mathcal{E}(q(0), 0) + \int_0^t \partial_t \mathcal{E}(q(s), s) ds + \int_0^t \Phi(\dot{q}(s)) ds.$$

The first integral is the work of external forces, while the second integral is the dissipation. The following lemma gives an alternative way to express the dissipation.

LEMMA 2.3. For all $0 \leq s < t \leq T$ let

$$\text{Diss}_\Phi(q, [s, t]) = \inf \left\{ \sum_{k=1}^N \Phi(q(t_k) - q(t_{k-1})) \mid s = t_0 < \dots < t_N = t \right\}.$$

Then

$$\text{Diss}_\Phi(q, [s, t]) = \int_s^t \Phi(\dot{q}(r)) dr.$$

PROOF. Consider any subdivision $\{t_k\}_{k=0}^N$ of $[s, t]$. By Jensen inequality and in view of the 1-homogeneity of Φ , it holds that

$$\Phi(q(t_k) - q(t_{k-1})) = \Phi \left((t_k - t_{k-1}) \frac{1}{t_k - t_{k-1}} \int_{t_{k-1}}^{t_k} \dot{q}(r) dr \right) \leq \int_{t_{k-1}}^{t_k} \Phi(\dot{q}(r)) dr \quad \text{for all } k.$$

Then the inequality

$$\text{Diss}_\Phi(q, [s, t]) \leq \int_s^t \Phi(\dot{q}(r)) dr$$

holds. To prove the other inequality we split the interval $[s, t]$ into intervals

$$I_k^n = \left[s + \frac{k-1}{n}(t-s), s + \frac{k}{n}(t-s) \right]$$

for $k \in \{1, \dots, n\}$ with $n \in \mathbb{N}$. Then, by lower semicontinuity of Φ and by the Fatou's lemma,

$$\begin{aligned} \int_s^t \Phi(\dot{q}(r)) dr &\leq \int_s^t \liminf_{n \rightarrow \infty} \frac{n}{t-s} \Phi \left(q \left(r + \frac{t-s}{n} \right) - q(r) \right) dr \leq \\ &\leq \liminf_{n \rightarrow \infty} \frac{n}{t-s} \int_s^{t-\frac{t-s}{n}} \Phi \left(q \left(r + \frac{t-s}{n} \right) - q(r) \right) dr. \end{aligned}$$

Now

$$\begin{aligned} \int_s^{t-\frac{t-s}{n}} \Phi \left(q \left(r + \frac{t-s}{n} \right) - q(r) \right) dr &= \sum_{k=1}^{n-1} \int_0^{\frac{t-s}{n}} \Phi \left(q \left(\frac{k}{n}(t-s) + r \right) - q \left(\frac{k-1}{n}(t-s) + r \right) \right) dr \leq \\ &\leq \int_0^{\frac{t-s}{n}} \sum_{k=1}^{n-1} \Phi \left(q \left(\frac{k}{n}(t-s) + r \right) - q \left(\frac{k-1}{n}(t-s) + r \right) \right) dr \leq \\ &\leq \frac{t-s}{n} \text{Diss}_\Phi(q, [s, t]). \end{aligned}$$

We have proved that

$$\int_s^t \Phi(\dot{q}(r)) dr \leq \text{Diss}_\Phi(q, [s, t])$$

as desired. $\square$

Summarizing, we have seen that (2.2) is equivalent

$$\mathcal{E}(q(t), t) \leq \mathcal{E}(\hat{q}, t) + \Phi(\hat{q} - q(t)) \quad \text{for all } \hat{q} \in Q \text{ and all } t \in [0, T],$$

$$\mathcal{E}(q(t), t) = \mathcal{E}(q(0), 0) + \int_0^t \partial_t \mathcal{E}(q(s), s) ds + \text{Diss}_\Phi(q, [0, t]) \quad \text{for all } t \in [0, T].$$

This conditions makes sense also if $\mathcal{E}$ is not Fréchet differentiable and q is not absolutely continuous.

This results suggest to define energetic solutions directly as evolutions $q: [0, T] \rightarrow Q$ satisfying (2.5) also when less regularity is assumed. This is discussed below.

2.2. Introduction. Let Q be a separable and reflexive Banach space and $\mathcal{E}: Q \times [0, T] \rightarrow [0, \infty]$. Rather than using a dissipation potential, we describe the system's dissipation in terms of a dissipation distance. This is defined as a map $D: Q \times Q \rightarrow [0, \infty]$, where $D(q_0, q_1)$ denotes the minimum energy dissipated by the system when it jumps from q_0 to q_1 . The dissipation along an evolution $q: [0, T] \rightarrow Q$ in the time interval $[s, t]$ is defined as

$$\text{Diss}_{\mathcal{D}}(q, [s, t]) = \sup \left\{ \sum_{k=1}^N \mathcal{D}(q(t_{k-1}), q(t_k)) \mid s = t_0 < \dots < t_N = t \right\}.$$

Then we say that q is an energetic solution if

$$\mathcal{E}(q(t), t) \leq \mathcal{E}(\hat{q}, t) + \mathcal{D}(q(t), \hat{q}) \quad \text{for all } \hat{q} \in Q \text{ and all } t \in [0, T],$$

$$\mathcal{E}(q(t), t) = \mathcal{E}(q(0), 0) + \int_0^t \partial_t \mathcal{E}(q(s), s) ds + \text{Diss}_{\mathcal{D}}(q, [0, t]) \quad \text{for all } t \in [0, T].$$

In the following, the definition will be made more precise.

REMARK 2.4. Given a dissipation potential $\Phi: Q \times Q \rightarrow [0, \infty]$, $(q, \dot{q}) \mapsto \Phi(q, \dot{q})$, such that $\Phi(q, \cdot)$ is positively 1-homogeneous for all $q \in Q$, we can define the dissipation distance as

$$(2.7) \quad \mathcal{D}(q_0, q_1) = \inf \left\{ \int_0^1 \Phi(q(t), \dot{q}(t)) dt \mid q \in C^1([0, T], Q), q(0) = q_0 \text{ and } q(1) = q_1 \right\}.$$

Notice that the range of the integral can be choosen arbitrary due to the positive 1-homogeneity of Φ . The particular choice (2.7) of $\mathcal{D}$ is motivated by the following *formal* consideration. Let $q: [0, T] \rightarrow Q$ be an energetic solution. Fix $t \in [0, T]$, $w \in Q$, $\epsilon > 0$ and let $\tilde{q}(r) = q(t) + r\epsilon w$. Then

$$\mathcal{E}(q(t), t) \leq \mathcal{E}(q(t) + \epsilon w, t) + \mathcal{D}(q(t), q(t) + \epsilon w) \leq \mathcal{E}(q(t) + \epsilon w, t) + \int_0^1 \Phi(\tilde{q}(r), \dot{\tilde{q}}(r)) dr$$

so that

$$-\frac{\mathcal{E}(q(t) + \epsilon w, t) - \mathcal{E}(q(t), t)}{\epsilon} \leq \frac{1}{\epsilon} \int_0^1 \Phi(q(t) + r\epsilon w, \epsilon w) dr = \int_0^1 \Phi(q(t) + r\epsilon w, w) dr.$$

Assuming enough regularity, we let $\epsilon \rightarrow 0$ and we get

$$-\langle \partial_q \mathcal{E}(q(t), t), w \rangle \leq \Phi(q(t), w).$$

This means that $-\partial_q \mathcal{E}(q(t), t) \in \partial \Phi(q(t), \cdot)(0)$. Differentiating the energy balance

$$-\partial_q \mathcal{E}(q(t), t) = \Phi(q(t), \dot{q}(t)).$$

This means that $-\partial_q \mathcal{E}(q(t), t) \in \partial \Phi(q(t), \cdot)(q(t))$. Thus (2.7) formally implies that energetic solutions are also solutions to the Biot inclusion.

2.3. Assumptions and preliminary results. Assume that

(S1) Y and Z separable and reflexive Banach spaces,

(S2) $\mathcal{Y}$ and $\mathcal{Z}$ weakly closed subsets of Y and Z respectively.

Let $Q = Y \times Z$ and $\mathcal{Q} = \mathcal{Y} \times \mathcal{Z}$. The assumptions on the energy functional are that

(E1) $\mathcal{E}: [0, T] \times \mathcal{Q} \rightarrow (-\infty, \infty]$,

(E2) for all $t \in [0, T]$ and $E \in \mathbb{R}$ the set $\mathcal{D}_E := \{q \in \mathcal{Q} \mid \mathcal{E}(q, t) \leq E\}$ is bounded and weakly closed

(E3) there exists $\mathcal{D} \subset \mathcal{Q}$ such that $\text{dom } \mathcal{E}(\cdot, t) = \mathcal{D}$ for all $t \in [0, T]$,

(E4) for all $q \in \mathcal{D}$, the function $\mathcal{E}(q, \cdot)$ is $C^1([0, T])$ and

(i) there exist C_1^E and C_0^E such that $|\partial_t \mathcal{E}(q, t)| \leq C_1^E (C_0^E + \mathcal{E}(q, t))$ for all $q \in \mathcal{Q}$ and $t \in [0, T]$,

(ii) for each $E \in \mathbb{E}$ there exists a modulus of continuity ω^E such that for all $q \in \mathcal{D}_E$ and $t, s \in [0, T]$

$$|\partial_t \mathcal{E}(q, t) - \partial_t \mathcal{E}(q, s)| \leq \omega^E(|t - s|).$$

Notice that (E2) is equivalent to say that $\mathcal{E}$ is weakly lower semicontinuous and has bounded sublevels. Moreover (E4, i) implies

$$\mathcal{E}(q, t_1) + C_0^E \leq e^{C_1^E |t_1 - t_0|} (\mathcal{E}(q, t_0) + C_0^E) \text{ for all } q \in \mathcal{D} \text{ and } t_0, t_1 \in [0, T].$$

This follows from the following lemma.

LEMMA 2.5. Let $f \in C^1([0, T])$ and there exists $a \in \mathbb{R}$ and $b > 0$ such that $|\dot{f}(t)| \leq b(f(t) + a)$ for all $t \in [0, T]$. Then for all $t_0, t_1 \in [0, T]$ it holds $f(t_0) + a \leq e^{b|t_1 - t_0|} (f(t_1) + a)$.

PROOF. Let $z(t) = f(t) + a$. Then $|\dot{z}| \leq bz$. It follows that $\dot{z} \leq bz$ and

$$z(t) \leq z(t_0) + \int_{t_0}^t bz(s)ds \quad \text{for all } 0 \leq t_0 \leq t_1 \leq T.$$

By (B.1) it follows that $z(t_1) \leq e^{b(t_1 - t_0)} z(t_0)$ all for all $0 \leq t_0 \leq t_1 \leq T$. But it holds also $\dot{z} \leq -bz$. If we set $\zeta(t) = z(T - t)$ then $\dot{\zeta} \leq b\zeta$. Then reasoning as before we get $z(t_1) \leq \zeta(T - t_1) \leq e^{b(t_0 - t_1)} \zeta(T - t_0) \leq e^{b(t_0 - t_1)} z(t_0)$ for all $0 \leq t_1 \leq t_0 \leq T$. This concludes the proof. $\square$

The assumptions on the dissipation distance are

(D1) $\mathcal{D}: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty]$ is a quasidistance, that is $\mathcal{D}(z_0, z_1) = 0$ if and only if $z_0 = z_1$ and $\mathcal{D}(z_0, z_2) \leq \mathcal{D}(z_0, z_1) + \mathcal{D}(z_1, z_2)$ for all $z_1, z_2 \in \mathcal{X}$,

(D2) $\mathcal{D}$ is weakly lower semicontinuous, meaning that $z_n \rightarrow z$ and $\hat{z}_n \rightarrow \hat{z}$ implies $\mathcal{D}(z, \hat{z}) \leq \liminf_{n \rightarrow \infty} \mathcal{D}(z_n, \hat{z}_n)$.

For a given $q = (y, z) \in \mathcal{Q}$, sometimes we write $\mathcal{D}(q)$ to mean $\mathcal{D}(z)$.

DEFINITION 2.6. A sequence $(q_n, t_n) \in \mathcal{Q} \times [0, T]$ is said to be stable if

$$\sup_n \mathcal{E}(q_n, t_n) \leq \infty \quad \text{and} \quad q_n \in \mathcal{S}(t_n)$$

where the stable set is defined by

$$\mathcal{S}(t) = \{q \in \mathcal{Q} \mid \mathcal{E}(q, t) < \infty \text{ and } \mathcal{E}(q, t) \leq \mathcal{E}(\hat{q}, t) + \mathcal{D}(q, \hat{q}) \text{ for all } \hat{q} \in \mathcal{Q}\}$$

We add the following assumptions regarding the stable set

(C1) for all stable sequences $(q_n, t_n) \in \mathcal{Q} \times [0, T]$ such that $q_n \rightarrow q$ and $t_n \rightarrow t$, it holds that $\partial_t \mathcal{E}(q_n, t) \rightarrow \partial_t \mathcal{E}(q, t)$,

(C2) the stable sets are closed in the sense that, if $(q_n, t_n) \in \mathcal{Q} \times [0, T]$ is a stable sequence such that $q_n \rightarrow q$ and $t_n \rightarrow t$, then $q \in \mathcal{S}(t)$.

2.4. Definition of energetic solutions. We define energetic solutions as follows.

DEFINITION 2.7. A map $q: [0, T] \rightarrow \mathcal{Q}$ is called energetic solution if it is stable at every instant, that is

$$q(t) \in \mathcal{S}(t) \text{ for all } t \in [0, T],$$

if $t \mapsto \partial_t \mathcal{E}(q(t), t)$ belongs to $L^1((0, T))$ and the energy identity

$$\mathcal{E}(q(t), t) + \text{Diss}_{\mathcal{D}}(z, [s, t]) = \mathcal{E}(q(0), 0) + \int_0^t \partial_t \mathcal{E}(q(s), s) ds$$

holds for all $t \in [0, T]$. Here the dissipation is defined as

$$\text{Diss}_{\mathcal{D}}(z, [s, t]) = \sup \left\{ \sum_{k=1}^N \mathcal{D}(z(t_{k-1}), z(t_k)) \mid s = t_0 < \dots < t_N = t \right\}$$

for all $0 \leq s < t \leq T$.

2.5. Existence theorem.

THEOREM 2.8. (existence of energetic solution) If the hypotheses (S), (E), (D) and (C) holds, then for all $q_0 \in \mathcal{S}(0)$ there exists a bounded energetic solutions q such that $q(0) = q_0$.

2.6. Tools to verify the compatibility conditions.

PROPOSITION 2.9. Assume that the following property holds

(PC1) for all stable sequence (q_k, t_k) such that $(q_k, t_k) \rightharpoonup (q, t)$ and for all $\hat{q} \in \mathcal{Q}$, there exist a sequence $\hat{q}_k$ such that

$$\limsup_{k \rightarrow \infty} (\mathcal{E}(\hat{q}_k, t_k) + \mathcal{D}(q_k, \hat{q}_k)) \leq \mathcal{E}(\hat{q}, t) + \mathcal{D}(q, \hat{q}).$$

Then for all stable sequence (q_k, t_k) such that $(q_k, t_k) \rightharpoonup (q, t)$ it holds that $\lim_{k \rightarrow \infty} \mathcal{E}(q_k, t_k) = \mathcal{E}(q, t)$. Moreover (C2) holds.

Assume that in addition to (PC1) also the following property holds

(PC2) for all $\epsilon > 0$ and $E \in \mathbb{R}$, there exists $\delta > 0$ such that for all $t, s \in [0, T]$ and $q \in \mathcal{Q}$ the implication

$$\mathcal{E}(q, t) \leq E \text{ and } |t - s| < \delta \text{ implies } |\partial_t \mathcal{E}(q, t) - \partial_t \mathcal{E}(q, s)| < \epsilon$$

holds.

Then (C1) holds.

3. Balanced-viscosity solutions

3.1. Introduction. Let

V a separable and reflexive Banach space.

Consider an evolution $u: [0, T] \rightarrow V$ and the doubly nonlinear equation for the evolution

$$\partial_u \mathcal{E}(u(t), t) + \partial \Phi(\dot{u}(t)) \ni 0 \quad \text{for } t \in (0, T).$$

Here $\mathcal{E}$ is the energy and Φ is the dissipation potential. In application usually $\mathcal{E}(u, t) = \Psi(u) - \langle \mathcal{F}(t), u \rangle$ where Ψ is the free energy and $\mathcal{F}$ is the external action. For a given $\eta > 0$ consider the time-rescaled evolution $u: (0, \eta^{-1}T) \rightarrow V$ and the time-rescaled equation

$$(2.8) \quad \partial_u \mathcal{E}(u(t), \eta t) + \partial \Phi(\dot{u}(t)) \ni 0 \quad \text{for } t \in (0, \eta^{-1}T).$$

For $\eta \rightarrow 0$ this describe the situation in which the external actions are much slower with respect to the characteristic relaxation time of the system. Equation (2.8) can be written as

$$(2.9) \quad \partial_u \mathcal{E}(u_\eta(t), t) + \partial \Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for } t \in (0, T)$$

where $u_\eta(t) = u(\eta^{-1}t)$ and $\Phi_\eta(v) = \eta^{-1}\Phi(\eta v)$.

3.2. Assumptions. The energy $\mathcal{E}$ is supposed to satisfy the following assumptions.

$$\mathcal{E} \in C^1(V \times [0, T])$$

for all $E > 0$ the sublevel $D_E = \{u \in V \mid \sup_{t \in [0, T]} \mathcal{E}(u, t) \leq E\}$ is weakly compact

$$(E.3) \quad \text{there exists } C > 0 \text{ such that } |\partial_t \mathcal{E}(u, t)| \leq C(1 + \mathcal{E}(u, t)) \text{ for all } u \in V \text{ and } t \in [0, T]$$

The dissipation potential is defined by

$$\Phi(v) = \mathcal{R}(v) + \mathcal{V}(v)$$

and for $\eta > 0$ its rescaled version is

$$\Phi_\eta(v) = \eta^{-1}\Phi_1(\eta v) = \mathcal{R}(v) + \eta^{-1}\mathcal{V}(\eta v) = \mathcal{R}(v) + \mathcal{V}_\eta(v).$$

Here $\mathcal{R}$ is a rate-independent dissipation potential and $\mathcal{V}$ is a rate-dependent dissipation potential that models viscosity. It is assumed that $\mathcal{R}$ and $\mathcal{V}$ satisfy the following assumptions.

$$(\Phi.1) \quad \mathcal{R} \text{ and } \mathcal{V} \text{ from } V \text{ into } [0, \infty) \text{ are continuous, convex and strictly positive in } V \setminus \{0\}$$

$$(\Phi.2) \quad \mathcal{R} \text{ is positively 1-homogeneous}$$

$$\mathcal{V}(v) = \phi(\|v\|) \text{ with } \phi \in C^1([0, \infty)) \text{ convex such that } \phi(0) = 0, \phi'(0) = 0, \phi(r) > 0 \text{ for } r > 0 \text{ and } \lim_{r \rightarrow \infty} \phi'(r) = \infty$$

Notice that from Proposition B.28 it holds $\mathcal{R}(0) = 0$. Moreover notice that ϕ can be extended to an even $C^1(\mathbb{R})$ function.

3.3. Preliminary results regarding the dissipation potential. Let $K = \partial\mathcal{R}(0)$. By Proposition B.28 it follows that $\mathcal{R} = \sigma_K$ and $\mathcal{R}^* = \iota_K$. Notice that there exists a constant $M > 0$ such that $\mathcal{R}(v) \leq M\|v\|$ for all $v \in V$. Indeed by contradiction assume that there exist $v_n \in V \setminus \{0\}$ such that $\mathcal{R}(v_n) \geq n\|v_n\|$. Since $\mathcal{R}$ is positively homogeneous the inequality implies $\mathcal{R}(n^{-1}v_n/\|v_n\|) \geq 1$ so $\liminf_{n \rightarrow \infty} \mathcal{R}(n^{-1}v_n/\|v_n\|) \geq 1$. But $n^{-1}v_n/\|v_n\| \rightarrow 0$ so by continuity of $\mathcal{R}$ it follows $\lim_{n \rightarrow \infty} \mathcal{R}(n^{-1}v_n/\|v_n\|) = 0$. This is the desired contradiction.

Since Φ_1 is convex and $\Phi_1(0) = 0$ then for all $v \in V$ the functions $\eta \mapsto \Phi_\eta(v)$ are increasing. Then it is well defined

$$\Phi_0(v) = \lim_{\eta \rightarrow 0^+} \Phi_\eta(v) = \inf_{\eta > 0} \Phi_\eta(v) = \mathcal{R}(v).$$

Last inequality holds true since $\phi(0) = 0$ and $\phi'(0) = 0$.

It simple to check that $\Phi_\eta^*(\xi) = \eta^{-1}\Phi^*(\xi)$. It is possible to obtain a more explicit representation formula for Φ^* . By the inf-convolution duality formula

$$(2.10) \quad \Phi_\eta^*(\xi) = (\mathcal{R} + \mathcal{V}_\eta)^*(\xi) = \inf_{\zeta \in V} \left(\mathcal{R}^*(\zeta) + \mathcal{V}_\eta^*(\xi - \zeta) \right) = \eta^{-1} \inf_{\zeta \in K} \mathcal{V}^*(\xi - \zeta) = \eta^{-1} \min_{\zeta \in K} \mathcal{V}^*(\xi - \zeta).$$

Last equality holds true because, since K is closed and convex, it is also weakly closed, $\mathcal{V}$ is convex and coercive, then the infimum is actually a minimum. Now $\mathcal{V}^*(\xi) = \phi^*(\|\xi\|)$.

LEMMA 2.10. The function ϕ^* satisfies $\phi^*(0) = 0$, $(\phi^*)'(0) = 0$ and is monotone.

PROOF. Let $r = \rho(\gamma)$ the solution to $\gamma - \phi'(\rho(\gamma)) = 0$. Clearly $\rho \in C^1(\mathbb{R})$ and $\rho(0) = 0$. Then $\phi^*(\gamma) = \gamma\rho(\gamma) - \phi(\rho(\gamma))$. Now $\phi^*(0) = -\phi(0) = 0$ and $(\phi^*)'(0) = -\phi'(0) = 0$. $\square$

Using that lemma and (2.10)

$$(2.11) \quad \Phi_\eta^*(\xi) = \eta^{-1} \min_{\zeta \in K} \phi^*(\|\xi - \zeta\|).$$

3.4. Absolutely continuous and BV evolutions. Let $\mathcal{R}: V \rightarrow [0, \infty)$ satisfying (Φ.1) and (Φ.2). We say that a map $u: [0, T] \rightarrow V$ is $AC_{\mathcal{R}}([0, T], V)$ if there exists $m \in L^1((0, T), [0, \infty))$ such that

$$(2.12) \quad \mathcal{R}(u(t_1) - u(t_0)) \leq \int_{t_0}^{t_1} m(s)ds \quad \text{for every } 0 \leq t_0 < t_1 \leq T.$$

LEMMA 2.11. Let $u \in AC_{\mathcal{R}}([0, T], V)$. Then

$$\mathcal{R}[\dot{u}](t) \doteq \lim_{h \rightarrow 0} \mathcal{R}\left(\frac{u(t+h) - u(t)}{h}\right)$$

exists for a.e. $t \in (0, T)$ and $\mathcal{R}[\dot{u}] \in L^1((0, T))$. Moreover for all m as in (2.12)

$$\mathcal{R}[\dot{u}](t) \leq m(t) \quad \text{for a.e. } t \in (0, T).$$

PROOF. For any $s \in (0, T)$ and $t \in (0, T)$ let $\delta_s(t) = \mathcal{R}(u(t) - u(s))$. Then

$$\delta_s(t_1) - \delta_s(t_0) \leq \mathcal{R}(u(t_1) - u(t_0)) \leq \int_{t_0}^{t_1} m(s)ds \quad \text{for all } 0 \leq t_0 < t_1 \leq T.$$

The second inequality holds since $\mathcal{R}$ is sub additive by Proposition B.28 so

$$\delta_s(t_1) = \mathcal{R}(u(t_1) - u(s)) \leq \mathcal{R}(u(t_1) - u(t_0)) + \mathcal{R}(u(t_0) - u(s)) = \mathcal{R}(u(t_1) - u(t_0)) + \delta_s(t_0).$$

Then the map $t \mapsto \delta_s(t) - \int_0^t m(r)dr = \beta_s(t)$ is non increasing on $(0, T)$ and in particular it is differentiable a.e. on $(0, T)$. This in turns implies that δ_s is differentiable a.e. on $(0, T)$ and

$$\dot{\delta}_s(t) \leq \delta(t) \doteq \liminf_{h \rightarrow 0^+} \mathcal{R}\left(\frac{u(t+h) - u(t)}{h}\right) \quad \text{for a.e. } t \in (0, T).$$

Note that

$$0 \leq \delta(t) \leq \liminf_{h \rightarrow 0^+} \frac{1}{h} \int_t^{t+h} m(s)ds = m(t) \quad \text{for a.e. } t \in (0, T)$$

and this implies $\delta \in L^1((0, T))$. Since $\delta_s(t) = \beta_s(t) + \int_0^t m(r)dr$ and β_s is non increasing then the singular part of the distributional derivative of δ_s is non positive and

$$(2.13) \quad \mathcal{R}(u(t) - u(s)) = \delta_s(t) \leq \int_s^t \dot{\delta}_s(r)dr \leq \int_s^t \delta(r)dr.$$

Let

$$\gamma(t) \doteq \limsup_{h \rightarrow 0^+} \mathcal{R} \left(\frac{u(t+h) - u(t)}{h} \right).$$

By (2.13) then $\delta(t) \leq \gamma(t) \leq \delta(t)$. This proves that

$$\delta^+(t) \doteq \lim_{h \rightarrow 0^+} \mathcal{R} \left(\frac{u(t+h) - u(t)}{h} \right) = \delta(t) \quad \text{exists for a.e. } t \in (0, T).$$

Applying the previous argument to $t \mapsto u(T-t)$ and $v \mapsto \mathcal{R}(-v)$ in place of u and $\mathcal{R}$ respectively, follows that

$$\delta^-(t) = \lim_{h \rightarrow 0^-} \mathcal{R} \left(\frac{u(t+h) - u(t)}{h} \right) = \lim_{h \rightarrow 0^+} \mathcal{R} \left(\frac{u(t) - u(t-h)}{h} \right) \quad \text{exists for a.e. } t \in (0, T)$$

and satisfy the conditions

$$\mathcal{R}(u(t) - u(s)) \leq \int_s^t \delta^-(r) dr.$$

and $\delta^-(t) \leq m(t)$ for a.e. $t \in (0, T)$. This means that δ^+ and δ^- coincides both with the minimal $m \in L^1((0, T), (0, \infty))$ for which (2.12) holds. $\square$

Given a map $u: [0, T] \rightarrow V$ we define the total variation on $[a, b] \subset [0, T]$ as

$$\text{Var}_{\mathcal{R}}(u, [a, b]) = \sup \left\{ \sum_{k=1}^N \mathcal{R}(u(t_k)) - \mathcal{R}(u(t_{k-1})) \mid a = t_0 < \dots < t_N = b \right\}.$$

We say that $u \in \text{BV}_{\mathcal{R}}([0, T], V)$ if $\text{Var}_{\mathcal{R}}(u, [0, T]) < \infty$.

LEMMA 2.12. It holds that $\text{AC}([0, T], V) \subset \text{AC}_{\mathcal{R}}([0, T], V)$ and $\text{BV}([0, T], V) \subset \text{BV}_{\mathcal{R}}([0, T], V)$.

PROOF. Recall that there exists $M > 0$ such that $\mathcal{R}(v) \leq M\|v\|$. Let $u \in \text{AC}([0, T], V)$. Then

$$\mathcal{R}(u(t_1) - u(t_0)) \leq M\|u(t_1) - u(t_0)\| \leq \int_{t_0}^{t_1} Mm(s) ds.$$

This shows that $u \in \text{AC}_{\mathcal{R}}([0, T], V)$. Next let $u \in \text{BV}([0, T], V)$. Then

$$\text{Var}_{\mathcal{R}}(u, [0, T]) \leq M \text{Var}(u, [0, T]).$$

This implies that $u \in \text{BV}_{\mathcal{R}}([0, T], V)$. $\square$

3.5. A chain rule formula for the energy functional.

PROPOSITION 2.13. Let $u \in \text{AC}([0, T], V)$. Then $t \mapsto \mathcal{E}(u(t), t)$ is absolutely continuous in $[0, T]$ and

$$(2.14) \quad \frac{d}{dt} \mathcal{E}(u(t), t) = \langle \partial_u \mathcal{E}(u(t), t), \dot{u}(t) \rangle + \partial_t \mathcal{E}(u(t), t) \quad \text{for a.e. } t \in (0, T).$$

PROOF. *Step 1.* Since u is continuous then $K = u([0, T])$ is compact. Indeed let $u_n = u(t_n)$ be a sequence in K . Then up to a subsequence $t_n \rightarrow t$ and by continuity $u_n = u(t_n) \rightarrow u(t)$.

Step 2. Since $\mathcal{E} \in C^1(V \times [0, T])$ then by Proposition B.13 it is Lipschitz in $K \times [0, T]$. Let $L > 0$ be the Lipschitz constant.

Step 3. Let $0 \leq s < t \leq T$ and $m \in L^1((0, T), (0, \infty))$ as in Definition B.3. Then

$$|\mathcal{E}(u(t), t) - \mathcal{E}(u(s), s)| \leq L(\|u(t) - u(s)\| + |t - s|) \leq L \int_s^t (m(r) + r) dr.$$

This proves that $t \mapsto \mathcal{E}(u(t), t)$ is absolutely continuous in $[0, T]$.

Step 4. By Proposition B.4 it follows that u is a.e. differentiable in $(0, T)$. This implies formula (2.14). $\square$

3.6. Existence results in presence of positive viscosity.

PROPOSITION 2.14. Let $u_0 \in V$. Then there exists $u_\eta \in \text{AC}([0, T], V)$ such that $u_\eta(0) = u_0$ and

$$\partial_u \mathcal{E}(u_\eta(t), t) + \partial \Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for a.e. } t \in (0, T).$$

Moreover the identity

$$(2.15) \quad \mathcal{E}(u_\eta(t), t) + \int_s^t \left(\Phi_\eta(\dot{u}_\eta(r)) + \Phi_\eta^*(-\partial_u \mathcal{E}(u_\eta(r), r)) \right) dr = \mathcal{E}(u_\eta(s), s) + \int_s^t \partial_t \mathcal{E}(u_\eta(r), r) dr$$

holds for every $0 \leq s < t \leq T$.

PROOF. I still have to report the proof of existence. We prove the identity (2.15). By the properties of subdifferentials

$$\Phi_\eta(\dot{u}_\eta(t)) + \Phi_\eta^*(-\partial_t \mathcal{E}(u_\eta(t), t)) = -\langle \partial_u \mathcal{E}(u_\eta(t), t), \dot{u}_\eta(t) \rangle \quad \text{for a.e. } t \in (0, T).$$

By (2.14) we have

$$\Phi_\eta(\dot{u}_\eta(t)) + \Phi_\eta^*(-\partial_t \mathcal{E}(u_\eta(t), t)) = -\frac{d}{dt} \mathcal{E}(u_\eta(t), t) + \partial_t \mathcal{E}(u_\eta(t), t) \quad \text{for a.e. } t \in (0, T).$$

Integrating this identity in (s, t) we get the thesis. $\square$

3.7. Contact dissipation potential. The dissipation contact potential $p: V \times V^* \rightarrow [0, \infty)$ is

$$p(v, \xi) = \inf_{\eta > 0} \left(\Phi_\eta(v) + \Phi_\eta^*(\xi) \right) = \inf_{\eta > 0} \eta^{-1} \left(\Phi(\eta v) + \Phi^*(\xi) \right).$$

By (2.11)

$$\begin{aligned} p(v, \xi) &= \inf_{\eta > 0} \left(\mathcal{R}(v) + \eta^{-1} \phi(\eta \|v\|) + \eta^{-1} \min_{\zeta \in K} \phi^*(\|\xi - \zeta\|) \right) = \\ &= \mathcal{R}(v) + \inf_{\eta > 0} \left[\eta^{-1} \min_{\zeta \in K} (\phi(\eta \|v\|) + \phi^*(\|\xi - \zeta\|)) \right] = \\ &= \mathcal{R}(v) + \min_{\zeta \in K} \inf_{\eta > 0} \frac{\phi(\eta \|v\|) + \phi^*(\|\xi - \zeta\|)}{\eta} \end{aligned}$$

where last equality holds because the minimum point does not depends on η . But for all r and γ in $\mathbb{R}$

$$\inf_{\eta > 0} \frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} = \gamma r.$$

Indeed by Young–Fenchel inequality

$$\frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} \geq \gamma r$$

and the identity holds for η such that $\gamma = \phi'(\eta r)$. Then

$$(2.16) \quad p(v, \xi) = \mathcal{R}(v) + \|v\| \min_{\zeta \in K} \|\xi - \zeta\|.$$

Let $\mathfrak{f}: V \times V \times [0, T] \rightarrow [0, \infty)$ defined by

$$\mathfrak{f}(u, v, t) = p(v, -\partial_u \mathcal{E}(u, t)) = \mathcal{R}(v) + e(u, t) \|v\|$$

where $e: V \times [0, T] \rightarrow [0, \infty)$ is defined by

$$e(u, t) = \min_{\zeta \in K} \|\partial_u \mathcal{E}(u, t) + \zeta\|.$$

Notice that $e(u, t) = 0$ if and only if $-\partial_u \mathcal{E}(u, t) \in K$.

3.8. Parametrized balanced-viscosity solutions. Let $u_\eta \in AC([0, T], V)$ be the solution to the doubly nonlinear equation

$$\partial_u \mathcal{E}(u_\eta(t), t) + \partial \Phi_\eta(\dot{u}_\eta(t)) \ni 0 \quad \text{for } t \in (0, T)$$

The identity (2.15) implies

$$(2.17) \quad \mathcal{E}(u_\eta(t), t) + \int_s^t \mathfrak{f}(u_\eta(r), \dot{u}_\eta(r), r) dr \leq \mathcal{E}(u_\eta(s), s) + \int_s^t \partial_t \mathcal{E}(u_\eta(r), r) dr$$

for all $0 \leq s < t \leq T$.

LEMMA 2.15. There exists a constant $C > 0$ such that

$$\int_0^T \mathfrak{f}(u_\eta(t), \dot{u}_\eta(t), t) dt \leq C \quad \text{for all } \eta > 0.$$

PROOF. Since $\mathfrak{f} \geq 0$ by (E.3) then inequality (2.17) implies

$$\mathcal{E}(u_\eta(t), t) \leq \mathcal{E}(u_0, 0) + \int_0^t \partial_t \mathcal{E}(u(s), s) ds \leq \mathcal{E}(u_0, 0) + C \int_0^t (1 + \mathcal{E}(u_\eta(s), s)) ds.$$

By (E.3) and Gronwall inequality $-1 \leq \mathcal{E}(u_\eta(t), t) \leq C_1$ for all $t \in [0, T]$ and $\eta > 0$. Then (2.17) implies

$$\int_0^T \mathfrak{f}(u_\eta(t), \dot{u}_\eta(t), t) dt \leq 1 + \mathcal{E}(u_0, 0) + CT(1 + C_1).$$

This concludes the proof. $\square$

Let $s_\eta: [0, T] \rightarrow [0, S_\eta]$ defined by

$$(2.18) \quad s_\eta(t) = t + \int_0^t \mathfrak{f}(u_\eta(s), \dot{u}_\eta(s), s) ds \quad \text{and} \quad S_\eta = s_\eta(T).$$

Let also

$$t_\eta = s_\eta^{-1}: [0, S_\eta] \rightarrow [0, T] \quad \text{and} \quad \mathbf{u}_\eta = u_\eta(t_\eta): [0, S_\eta] \rightarrow V.$$

To write the time-rescaled energy identity we introduce the map $\mathfrak{F}_\eta: V \times [0, T] \times V \times (0, \infty) \rightarrow [0, \infty)$ defined by

$$\mathfrak{F}_\eta(u, t, v, a) = a\Phi_\eta(v/a) + \Phi_\eta^*(-\partial_u \mathcal{E}(u, t)) - a\partial_t \mathcal{E}(u, t) = \mathcal{R}(v) + \mathfrak{E}_\eta(u, t, v, a) - a\partial_t \mathcal{E}(u, t),$$

where $\mathfrak{E}_\eta: V \times [0, T] \times V \times (0, \infty) \rightarrow [0, \infty)$ is given by

$$\begin{aligned} \mathfrak{E}_\eta(u, t, v, a) &= a\mathcal{V}_\eta(v/a) + a\Phi_\eta^*(-\partial_u \mathcal{E}(u, t)) = a\mathcal{V}_\eta(v/a) + \frac{a}{\eta} \min_{\zeta \in V^*} \phi^*(\|\partial_u \mathcal{E}(u, t) + \zeta\|) = \\ &= a\mathcal{V}_\eta(v/a) + \frac{a}{\eta} \phi^*\left(\min_{\zeta \in V^*} \|\partial_u \mathcal{E}(u, t) + \zeta\|\right) = a\mathcal{V}_\eta(v/a) + \frac{a}{\eta} \phi^*(e(u, t)). \end{aligned}$$

Then (2.15) becomes

$$\mathcal{E}(\mathbf{u}_\eta(t), t_\eta(t)) + \int_s^t \mathfrak{F}_\eta(\mathbf{u}_\eta(r), t_\eta(r), \dot{\mathbf{u}}_\eta(r), \dot{t}_\eta(r)) dr = \mathcal{E}(\mathbf{u}_\eta(s), t_\eta(s)) \quad \text{for all } 0 \leq s < t \leq S_\eta.$$

Identity (2.18) implies

$$1 = \dot{t}_\eta(s) + \mathfrak{f}(\mathbf{u}_\eta(s), \dot{\mathbf{u}}_\eta(s), t_\eta(s)) \quad \text{for a.e. } s \in (0, S_\eta).$$

Now let $\mathfrak{F}: V \times [0, T] \times V \times [0, \infty) \rightarrow [0, \infty]$ defined by

$$\mathfrak{F}(u, t, v, a) = \mathcal{R}(v) + \mathfrak{E}(u, t, v, a) - a\partial_t \mathcal{E}(u, t)$$

where $\mathfrak{E}: V \times [0, T] \times V \times [0, \infty) \rightarrow [0, \infty]$

$$\mathfrak{E}(u, t, v, a) = \begin{cases} \iota_K(-\partial_u \mathcal{E}(u, t)) & \text{if } a > 0 \\ e(u, t)\|v\| & \text{if } a = 0. \end{cases}$$

DEFINITION 2.16. We say that a pair $(\mathbf{u}, \mathbf{t}): [0, S] \rightarrow V \times [0, T]$ is an admissible parametrized curve if

- (i) $\mathbf{t}$ is non decreasing and absolutely continuous, $\mathbf{u} \in \text{AC}_{\mathcal{R}}([0, S], D_E)$ for some $E > 0$
- (ii) $\mathbf{u}$ locally absolutely continuous (with respect to $\|\cdot\|$) in the open set

$$G = \{s \in [0, S] \mid e(\mathbf{u}(s), \mathbf{t}(s)) > 0\} = \{s \in [0, S] \mid -\partial_u \mathcal{E}(\mathbf{u}(s), \mathbf{t}(s)) \notin K\}$$

- (iii) the estimate

$$\int_0^S \mathfrak{f}[\mathbf{u}, \dot{\mathbf{u}}, \mathbf{t}](s) ds = \int_0^S \mathcal{R}[\dot{\mathbf{u}}](s) ds + \int_G e(\mathbf{u}(s), \mathbf{t}(s)) \|\dot{\mathbf{u}}(s)\| ds < \infty$$

holds.

For every admissible parametrized curve and for a.e. $s \in [0, S]$ is well defined

$$\mathfrak{F}[\mathbf{u}, \mathbf{t}, \dot{\mathbf{u}}, \dot{\mathbf{t}}](s) = \begin{cases} \mathcal{R}[\dot{\mathbf{u}}](s) + \iota_K(-\partial_u \mathcal{E}(\mathbf{u}(s), \mathbf{t}(s))) - \dot{\mathbf{t}}(s)\partial_t \mathcal{E}(\mathbf{u}(s), \mathbf{t}(s)) & \text{if } \dot{\mathbf{t}}(s) > 0 \\ \mathcal{R}[\dot{\mathbf{u}}](s) + e(\mathbf{u}(s), \mathbf{t}(s)) \|\dot{\mathbf{u}}(s)\| & \text{if } \dot{\mathbf{t}}(s) = 0. \end{cases}$$

where is implicit that if $s \notin G$ then $e(\mathbf{u}(s), \mathbf{t}(s)) \|\dot{\mathbf{u}}(s)\| = 0$.

We denote with $\mathcal{A}([0, S], V \times [0, T])$ the set of all admissible parametrized curves and we say that $(\mathbf{u}, \mathbf{t})$ is

- (i) non degenerate if $\dot{\mathbf{t}}(s) + \mathcal{R}[\dot{\mathbf{u}}](s) > 0$ for a.e. $s \in [0, S]$
- (ii) surjective if $\mathbf{t}(0) = 0$ and $\mathbf{t}(S) = T$
- (iii) $\mathbf{m}$ -normalized with $\mathbf{m} \in L^\infty((0, S), (0, \infty))$ if $(\mathbf{u}, \mathbf{t})$ fulfills

$$\dot{\mathbf{t}}(s) + \mathcal{R}[\dot{\mathbf{u}}](s) + e(\mathbf{u}(s), \mathbf{t}(s)) \|\dot{\mathbf{u}}(s)\| = \mathbf{m}(s) \quad \text{for a.e. } s \in (0, S)$$

DEFINITION 2.17. A parametrized balanced-viscosity solution is a non degenerate and surjective curve $(\mathbf{u}, \mathbf{t}) \in \mathcal{A}([0, S], V \times [0, T])$ satisfying

$$\mathcal{E}(\mathbf{u}(t), \mathbf{t}(t)) + \int_s^t \mathfrak{F}[\mathbf{u}, \mathbf{t}, \dot{\mathbf{u}}, \dot{\mathbf{t}}](r) dr = \mathcal{E}(\mathbf{u}(s), \mathbf{t}(s)) \quad \text{for all } 0 \leq s < t \leq S.$$

LEMMA 2.18. Let $t_n \in AC([0, S], [0, T])$ non decreasing and surjective. Let $E > 0$ and $u_n \in AC([0, S], D_E)$. Assume that there exists $L > 0$ such that

$$(2.19) \quad \dot{t}_n(s) + \mathfrak{f}(u_n(s), \dot{u}_n(s), t_n(s)) \leq L \quad \text{for a.e. } s \in (0, S)$$

Then up to a extracting a subsequence, (u_n, t_n) converges uniformly on $[0, S]$ to some $(u, t) \in \mathcal{A}([0, S], D_E \times [0, T])$. Moreover

$$\dot{t}(s) + \mathfrak{f}[u, \dot{u}, t](s) \leq L \quad \text{for a.e. } s \in (0, S)$$

and the following lower semicontinuity properties holds

$$\begin{aligned} \liminf_{n \rightarrow \infty} \int_s^t \mathcal{R}(\dot{u}_n(r)) dr &\geq \int_s^t \mathcal{R}[\dot{u}](r) dr \\ \liminf_{n \rightarrow \infty} \int_s^t e(u_n(r), t_n(r)) \|\dot{u}_n(r)\| dr &\geq \int_s^t e(u(r), t(r)) \|\dot{u}(r)\| dr \\ \liminf_{n \rightarrow \infty} \int_s^t \mathfrak{G}_{\eta_n}(u_n(r), t_n(r), \dot{u}_n(r), \dot{t}_n(r)) dr &\geq \int_s^t \mathfrak{G}(u(r), t(r), \dot{u}(r), \dot{t}(r)) dr \end{aligned}$$

for every $0 \leq s < t \leq S$ and $\eta_n \in (0, T)$ with $\eta_n \rightarrow 0$.

PROOF. *Step 1.* From the fact that $t_n(0) = 0$ and that $\dot{t}_n$ are uniformly bounded (by (2.19)), it follows that t_n are uniformly Lipschitz continuous. Then by Ascoli–Arzelà up to extracting a subsequence, t_n converges uniformly to some $t \in C^{0,1}([0, T])$. VERIFICARE CHE t SODDISFA LE PROPRIETÀ RICHIESTE.

Step 2. Since $\mathcal{R}$ is continuous and D_E is weakly compact we can prove that for all $\delta > 0$ there exists $M_\delta > 0$ such that

$$|v - w| \leq \delta + M_\delta \mathcal{R}(v - w).$$

Indeed by contradiction assume that there is some $\delta > 0$ such that for all n there exist v_n and w_n with

$$|v_n - w_n| \geq \delta + n \mathcal{R}(v - w).$$

Since D_E is weakly compact, we can assume that $v_n \rightharpoonup v$ and $w_n \rightharpoonup w$. Up to extracting a subsequence we can also assume that $v_n \rightarrow v$ and $w_n \rightarrow w$ in H . QUI VA SISTEMATA LA SITUAZIONE. IL MOTIVO È CHE SERVIREBBE CONVERGENZA FORTE NELLA (SEMI)NORMA IN CUI $\mathcal{R}$ È CONTINUA. Then

$$|v - w| \geq \delta + \limsup_{n \rightarrow \infty} n \mathcal{R}(v_n - w_n).$$

This is a contradiction because necessarily $v \neq w$ so that $\mathcal{R}(v - w) > 0$ and in this case

$$\limsup_{n \rightarrow \infty} n \mathcal{R}(v_n - w_n) = \infty.$$

Let $\omega: [0, \infty) \rightarrow [0, \infty)$ defined by $\omega(r) = \inf_{\delta > 0} (\delta + M_\delta r)$. Notice that ω is non decreasing and $\lim_{r \rightarrow 0} \omega(r) = 0$.

Step 3. By Jensen inequality and assumption (2.19)

$$\mathcal{R}(u_n(t) - u_n(s)) = |t - s| \mathcal{R}\left(\frac{1}{|t - s|} \int_s^t \dot{u}_n(r) dr\right) \leq \int_s^t \mathcal{R}(\dot{u}_n(r)) dr \leq L|t - s|.$$

Then

$$|u_n(t) - u_n(s)| \leq \omega(\mathcal{R}(u_n(t) - u_n(s))) \leq \omega(L|t - s|).$$

This implies that u_n are equicontinuous with respect to $|\cdot|$. Then up to extracting a subsequence

$$\lim_{n \rightarrow \infty} \sup_{s \in [0, S]} |u(s) - u_n(s)| = 0$$

for some $u \in AC_{|\cdot|}([0, T], V)$. It follows that

□

THEOREM 2.19 (existence of parametrized solutions). Let $u_\eta \in AC([0, T], V)$ be solutions to the doubly nonlinear equation (2.9) with fixed initial condition $u_0 \in V$. Consider non decreasing and surjective absolutely continuous time rescalings $t_\eta: [0, S] \rightarrow [0, T]$ and let $u_\eta = u_\eta(t_\eta)$. Suppose that as $\eta \rightarrow 0$

$$(2.20) \quad m_\eta = \dot{t}_\eta + \mathfrak{f}(u_\eta, \dot{u}_\eta, t_\eta) \xrightarrow{*} m \quad \text{in } L^\infty((0, S))$$

for some $m \in L^\infty((0, S), (0, \infty))$. Then there exists a sequence $\eta_n \rightarrow 0$ such that (u_{η_n}, t_{η_n}) converges uniformly on $[0, S]$ to a parametrized balanced viscosity solution $(u, t) \in \mathcal{A}([0, S], V \times [0, T])$. Moreover (u, t) is m -normalized.

PROOF. Arguing as in the proof of Lemma 2.15 one finds that there exists $E > 0$ such that $\mathbf{u}_\eta(s) \in D_E$ for all $\eta > 0$ and $s \in [0, S]$. Since (2.20) holds, there exists $L > 0$ such that $\|\dot{\mathbf{t}}_\eta + \mathbf{f}(\mathbf{u}_\eta, \dot{\mathbf{u}}_\eta, \mathbf{t}_\eta)\|_\infty \leq L$. Then the hypotheses of Lemma 2.18 are satisfied and we can find a sequence $\eta_n \rightarrow 0$ such that $(\mathbf{u}_{\eta_n}, \mathbf{t}_{\eta_n})$ converges uniformly on $[0, S]$ to some $(\mathbf{u}, \mathbf{t}) \in \mathcal{A}([0, S], V \times [0, T])$. Since $\mathcal{E} \in C^1(V \times [0, T])$ then

$$(2.21) \quad \mathcal{E}(\mathbf{u}_{\eta_n}(s), \mathbf{t}_{\eta_n}(s)) \rightarrow \mathcal{E}(\mathbf{u}(s), \mathbf{t}(s)) \quad \text{for all } s \in [0, S].$$

Now, since $\dot{\mathbf{t}}_\eta$ are bounded in $L^\infty((0, S))$, up to extracting a subsequence it holds that $\dot{\mathbf{t}}_{\eta_n} \xrightarrow{*} \dot{\mathbf{t}}$ in $L^\infty((0, T))$. In particular $\dot{\mathbf{t}}_{\eta_n} \rightharpoonup \dot{\mathbf{t}}$ in $L^1((0, T))$. Then by Lemma 2.20

$$(2.22) \quad \liminf_{n \rightarrow \infty} \int_s^t \partial_t \mathcal{E}(\mathbf{u}_{\eta_n}(r), \mathbf{t}_{\eta_n}(r)) dr \geq \int_s^t \partial_t \mathcal{E}(\mathbf{u}(r), \mathbf{t}(r)) dr.$$

Now taking the inferior limit as $n \rightarrow \infty$ of the energy identity

$$\mathcal{E}(\mathbf{u}_{\eta_n}(t), \mathbf{t}_{\eta_n}(t)) + \int_s^t \mathfrak{F}_{\eta_n}(\mathbf{u}_{\eta_n}(r), \mathbf{t}_{\eta_n}(r), \dot{\mathbf{u}}_{\eta_n}(r), \dot{\mathbf{t}}_{\eta_n}(r)) dr \leq \mathcal{E}(\mathbf{u}_{\eta_n}(s), \mathbf{t}_{\eta_n}(s)) \quad \text{for all } 0 \leq s < t \leq T$$

taking into account (2.21), (2.22) and the lower semicontinuity estimates of Lemma 2.18

$$\mathcal{E}(\mathbf{u}(t), \mathbf{t}(t)) + \int_s^t \mathfrak{F}[\mathbf{u}, \mathbf{t}, \dot{\mathbf{u}}, \dot{\mathbf{t}}](r) dr \leq \mathcal{E}(\mathbf{u}(s), \mathbf{t}(s)) + \int_s^t \partial_t \mathcal{E}(\mathbf{u}(r), \mathbf{t}(r)) dr$$

for all $0 \leq s < t \leq T$. ADESSO MANCA DA OTTENERE L'ALTRA STIMA (E VOLENDO MOSTRARE CHE SODDISFA LA GIUSTA NORMALIZZAZIONE). $\square$

LEMMA 2.20. Let $f_n, f, m_n, m: (0, T) \rightarrow [0, \infty]$ be measurable functions that satisfy

$$\begin{aligned} \liminf_{n \rightarrow \infty} f_n(s) &\geq f(s) \quad \text{for a.e. } s \in (0, T) \\ m_n &\rightharpoonup m \quad \text{in } L^1((0, T)). \end{aligned}$$

Then

$$\liminf_{n \rightarrow \infty} \int_0^T f_n(s) m_n(s) ds \geq \int_0^T f(s) m(s) ds.$$

PROOF. Let $g_k(s) = \inf_{n \geq k} \min\{f_n(s), n\}$. Notice that $g_k \in L^\infty((0, T))$. Fix k and notice that for all $n \geq k$

$$\int_0^T f_n(s) m_n(s) ds \geq \int_0^T g_k(s) m_n(s) ds.$$

Taking the inferior limit as $n \rightarrow \infty$

$$\liminf_{n \rightarrow \infty} \int_0^T f_n(s) m_n(s) ds \geq \int_0^T g_k(s) m(s) ds.$$

Now g_k is an increasing sequence and

$$\lim_{k \rightarrow \infty} g_k(s) = \lim_{k \rightarrow \infty} \inf_{n \geq k} \min\{f_n(s), n\} = \liminf_{n \rightarrow \infty} f_n(s) \geq f(s) \quad \text{for a.e. } s \in (0, T).$$

The passing to the limit $k \rightarrow \infty$ and using monotone convergence the proof is concluded. $\square$

3.9. An example. Let $V = L^2(\Omega)$ with $\Omega \subset \mathbb{R}^d$ open and bounded. Let

$$\mathcal{R}(v) = \int_\Omega |v| dx \quad \text{and} \quad \mathcal{V}(v) = \frac{1}{2} \int_\Omega |v|^2 dx.$$

They satisfy the assumptions (D). In particular the continuity of $\mathcal{R}$ follows from Hölder inequality

$$|\mathcal{R}(v) - \mathcal{R}(w)| \leq \int_\Omega ||v| - |w|| dx \leq \int_\Omega |v - w| dx \leq C \|v - w\|_2.$$

It holds that K is the closed unit ball in $L^\infty(\Omega)$. Indeed if $\xi \in V^* = L^2(\Omega)$ satisfies $\|\xi\|_\infty \leq 1$ then

$$\mathcal{R}^*(\xi) = \sup_{v \in V} \int_\Omega (\xi v - |v|) dx \leq \sup_{v \in V} \int_\Omega (\xi v - |v|) dx \leq \sup_{v \in V} \int_\Omega (|\xi| - 1) |v| dx \leq 0$$

and $\mathcal{R}^*(\xi) \geq 0$ trivially. If $\|\xi\|_\infty > 1$ then there exists $\epsilon > 0$ and $w \in L^1(\Omega)$ such that $\langle \xi, w \rangle \geq 1 + \epsilon$. Since $L^2(\Omega)$ is dense in $L^1(\Omega)$ then for all $\delta > 0$ there exist $v \in L^2(\Omega)$ such that $\|w - v\| \leq \delta$. Then for all $\lambda > 0$

$$\mathcal{R}^*(\xi) \geq \langle \xi, \lambda v \rangle - \|\lambda v\|_1 \geq \lambda (\langle \xi, w \rangle - \|w\|_1 - \delta(1 + \|\xi\|)) \geq \lambda (\epsilon - \delta(1 + \|\xi\|)).$$

Choosing δ such that $\epsilon - \delta(1 + \|\xi\|) > 0$ and letting $\lambda \rightarrow \infty$ we get $\mathcal{R}(\xi) = \infty$. Since $\mathcal{V}^* = \frac{1}{2}\|\cdot\|_2^2$ and by (2.10)

$$\Phi_\epsilon^*(\xi) = \frac{1}{2\eta} \min_{\zeta \in K} \|\xi - \zeta\|_2^2 = \frac{1}{2\eta} \int_{\Omega} \min_{|\zeta| \leq 1} |\xi - \zeta|^2 dx.$$

By (2.16) it follows

$$\mathfrak{p}(v, \xi) = \|v\|_1 + \|v\|_2 \min_{\zeta \in K} \|\xi - \zeta\|_2.$$

Part 1

One dimensional model of a plastic cylindrical bar

CHAPTER 3

Introduction

Model at small deformations

In this chapter we develop and analyze a small deformations model of a bar whose sections can only translate and change their size. In particular shear and bending are not allowed. We split our discussion into three parts. The first section is devoted to the development of the model, the second to an existence and uniqueness results. The last section is devoted to the explicit solution of two boundary value problems describing the tensile testing of the bar. In both the cases we show that the small deformation model is not able to predict the necking phenomenon.

1. Formulation

1.1. Motivation from the multidimensional kinematics. We start describing the kinematics of the bar from a multidimensional perspective. This will serve as a motivation for the development of the purely one dimensional model discussed in the forthcoming sections.

Let $\Omega = (0, L) \times \Sigma$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in Ω as (x, y) with $x \in (0, L)$ and $y \in \Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix}$$

where u_1 and u_2 are functions from $(0, L)$ to $\mathbb{R}$. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u'_1 & 0 \\ u'_2 y & u_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix}$$

where $p: (0, L) \rightarrow \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u'_1 - p & 0 \\ u'_2 y & (u_2 + \frac{p}{d-1}) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u'_1 - p & \frac{u'_2}{2} y^T \\ \frac{u'_2}{2} y & (u_2 + \frac{p}{d-1}) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u'_2}{2} y^T \\ \frac{u'_2}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u'_1 - p$ and $e_2 = u_2 + (d-1)^{-1}p$. Assuming an isotropic elastic response the elastic free energy density is given by

$$\psi_e = \frac{1}{2} \kappa (\text{tr } E_e)^2 + \mu |\text{dev } E_e|^2$$

where $\kappa \geq 0$ and $\mu \geq 0$ are the bulk and shear modulus respectively. We notice that

$$\text{tr } E_e = e_1 + (d-1)e_2 \quad \text{and} \quad \text{dev } E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u'_2}{2} y^T \\ \frac{u'_2}{2} y & -\frac{1}{d}(e_1 - e_2)I \end{pmatrix}.$$

Then the elastic free energy density can be written as

$$\psi_e = \frac{1}{2} \kappa (e_1 + (d-1)e_2)^2 + \frac{d-1}{d} \mu (e_1 - e_2)^2 + \frac{1}{2} |y|^2 \mu |u'_2|^2.$$

We conclude that the elastic free energy per unit axial length is

$$(4.1) \quad \int_{\Sigma} \psi_e dy = \frac{1}{2} A \kappa (e_1 + (d-1)e_2)^2 + \frac{d-1}{d} A \mu (e_1 - e_2)^2 + \frac{1}{2} J \mu |u'_2|^2$$

where

$$A = \int_{\Sigma} dy \quad \text{and} \quad J = \int_{\Sigma} |y|^2 dy$$

are the polar and the polar second order moment of area.

1.2. Kinematics. We now develop a purely one dimensional model of the bar taking inspiration from the multidimensional kinematics just discussed.

Let $L > 0$ be the length of the cylindrical bar. The macroscopic configuration of the bar is described by $u_i: (0, L) \rightarrow \mathbb{R}$ for $i \in \{1, 2\}$. Here u_1 is the longitudinal displacement of the sections in the axis direction and u_2 is the quantity describing the transversal expansion of the sections. The configuration of the microscopic structure of the bar is described by the plastic distortion $p: (0, L) \rightarrow \mathbb{R}$. We define $e_1 = u_1' - p$ and $e_2 = u_2 + (d - 1)^{-1}p$ the elastic distortions.

The space Z of the configurations $z = (u, p)$ will be clearly defined subsequently. The virtual velocities will be denoted as $\tilde{w} = (\tilde{v}, \tilde{q})$

1.3. Statics. To describe the coupling between the macroscopic configuration u and the microscopic configuration p we require that the internal generalized force is the form

$$\langle \mathcal{J}, \tilde{w} \rangle = \int_0^L \zeta \tilde{v}_2' dx + \int_0^L (\sigma_1(\tilde{v}_1' - \tilde{q}) + \sigma_2(\tilde{v}_2 + (d - 1)^{-1}\tilde{q})) dx + \int_0^L \tau \tilde{q} dx.$$

The first integral describes a purely macroscopic response.. Here $\zeta: (0, L) \rightarrow \mathbb{R}$ is the transversal stress. The second integral describes the elastic response that couples the macroscopic and the microscopic configurations. Here $\sigma_1: (0, L) \rightarrow \mathbb{R}$ is the longitudinal elastic stress and $\sigma_2: (0, L) \rightarrow \mathbb{R}$ the longitudinal elastic force. The last integral describes the microscopic response and $\tau: (0, L) \rightarrow \mathbb{R}$ is the microscopic force.

The external generalized force is required to be of the form

$$\langle \mathcal{X}, \tilde{w} \rangle = \int_0^L f \cdot \tilde{v} dx + h(L) \cdot \tilde{v}(L) - h(0) \cdot \tilde{v}(0).$$

Here $f: (0, L) \rightarrow \mathbb{R}^2$ is an external force distributed along the bar and $h: \{0, L\} \rightarrow \mathbb{R}^2$ is an external force acting at the ends of the bar. The first component of both f and h is referred to the axial direction, while the second component is referred to the longitudinal direction.

PROPOSITION 4.1. The local form of the equilibrium condition is

$$(4.2a) \quad \sigma_1' + f_1 = 0 \quad \text{in } (0, L)$$

$$(4.2b) \quad h_1(x) = \sigma_1(x) \quad \text{for } x \in \{0, L\}$$

$$(4.2c) \quad -\sigma_2 + \zeta' + f_2 = 0 \quad \text{in } (0, L)$$

$$(4.2d) \quad h_2(x) = \zeta(x) \quad \text{for } x \in \{0, L\}$$

$$(4.2e) \quad \sigma_1 - (d - 1)^{-1}\sigma_2 - \tau = 0 \quad \text{in } (0, L)$$

PROOF. First we choose $\tilde{v}_2 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_1$ arbitrary. Then we get

$$-\int_0^L \sigma_1' \tilde{v}_1 dx + \sigma_1(L) \tilde{v}_1(L) - \sigma_1(0) \tilde{v}_1(0) = \int_0^L f_1 \tilde{v}_1 dx + h_1(L) \tilde{v}_1(L) - h_1(0) \tilde{v}_1(0).$$

We get (4.2a) and (4.2b). Next we choose $\tilde{v}_1 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_2$ arbitrary. Then

$$-\int_0^L \zeta' \tilde{v}_2 dx + \zeta(L) \tilde{v}_2(L) - \zeta(0) \tilde{v}_2(0) + \int_0^L \sigma_2 \tilde{v}_2 dx = \int_0^L f_2 \tilde{v}_2 dx + h_2(L) \tilde{v}_2(L) - h_2(0) \tilde{v}_2(0).$$

We get (4.2c) and (4.2d). Then we choose $\tilde{v} = 0$ and $\tilde{q}$ arbitrary obtaining

$$\int_0^L (\sigma_1 - (d - 1)^{-1}\sigma_2 - \tau) \tilde{q} dx = 0.$$

This implies (4.2e). □

1.4. Constitutive relations. Taking inspiration from (4.1), we choose the free energy for unit axial length

$$(4.3) \quad \psi = \frac{1}{2}\kappa(e_1 + (d-1)e_2)^2 + \frac{1}{2}\mu(e_1 - e_2)^2 + \frac{1}{2}l^2\mu|u'_2|^2 + \frac{1}{2}Bp^2.$$

where κ is a bulk modulus, μ is a shear modulus, l is length parameter related to the geometry of the section and B is the hardening parameter. We also choose the dissipation potential for unit axial length

$$(4.4) \quad \phi = R(\dot{p})$$

where $R: \mathbb{R} \rightarrow [0, \infty]$ is convex and satisfy $R(0) = 0$. A possible choice could be $R(\dot{p}) = Y|\dot{p}|$ where $Y > 0$ is the yield strength. Consistently with these choices and (1.3), we impose the following constitutive relations

$$\begin{aligned} \zeta &= \mu l^2 u'_2 \\ \sigma_1 &= \kappa(e_1 + (d-1)e_2) + \mu(e_1 - e_2) \\ \sigma_2 &= (d-1)\kappa(e_1 + (d-1)e_2) - \mu(e_1 - e_2) \\ \tau &\in Bp + \partial R(\dot{p}). \end{aligned}$$

We notice that introducing these constitutive relations into (4.2e) we get

$$\frac{d}{d-1}\mu(e_1 - e_2) - Bp \in \partial R(\dot{p}).$$

We refer to

$$m = \frac{d}{d-1}\mu(e_1 - e_2)$$

as the Mandel stress. It is the elastic force acting on the microscopic configuration and we notice that it depends on the shear modulus only.

2. Existence and uniqueness result

In this section we discuss the existence and uniqueness of solutions of evolutions. For simplicity we will assume homogeneous Dirichlet boundary conditions $u(x) = 0$ for $x \in \{0, L\}$.

We define the spaces $U = H^1((0, L), \mathbb{R} \times \mathbb{R})$ and $P = L^2((0, L))$. Let also $Z = U \times P$ the space of configurations of the system. We will write the generic configuration as $z = (u, p) \in Z$ with $u \in U$ and $p \in P$ and the generic virtual rate as $w = (v, q) \in Z$ with $v \in U$ and $q \in P$.

The evolution problem associated with the model described in this chapter is of the form of an evolutionary variational inequality that has been studied in Section 1. According to (4.3), the free energy $\Psi: Z \rightarrow \mathbb{R}$ is defined by

$$\Psi(z) = \int_0^L \left[\frac{1}{2}\kappa(u'_1 + (d-1)u_2)^2 + \frac{1}{2}\mu \left(u'_1 - u_2 - \frac{d}{d-1}p \right)^2 + \frac{1}{2}l^2\mu|u'_2|^2 + \frac{1}{2}Bp^2 \right] dx$$

Its Fréchet differential is $\mathcal{A} = \partial\Psi: Z \rightarrow Z^*$ given by

$$\begin{aligned} \langle \mathcal{A}z, w \rangle &= \int_0^L l^2\mu u'_2 v'_2 dx + \int_0^L \left[\kappa(u'_1 + (d-1)u_2) + \mu \left(u'_1 - u_2 - \frac{d}{d-1}p \right) \right] (v'_1 - q) dx + \\ &\quad + \int_0^L \left[(d-1)\kappa(u'_1 + (d-1)u_2) - \mu \left(u'_1 - u_2 - \frac{d}{d-1}p \right) \right] (v_2 + (d-1)^{-1}p) dx + \\ &\quad + \int_0^L Bpq dx. \end{aligned}$$

According to (4.4) the dissipation potential $\mathcal{R}: Z \rightarrow \mathbb{R} \cup \{\infty\}$ is defined by

$$\mathcal{P}(w) = \int_0^L R(q) dx.$$

The external generalized force is $\mathcal{X}: [0, T] \rightarrow Z^*$,

$$\langle \mathcal{X}(t), w \rangle = \int_0^T f(t) \cdot v dx.$$

Now to prove the existence of solution we need only to satisfy the assumptions of Theorem 2.2. This will be done under the following assumptions:

(G) $L > 0$ and $d = 2$ or $d = 3$,

- (ψ) κ, μ, l and B are positive constants,
 (R) $R: \mathbb{R} \rightarrow \mathbb{R}$ is given by $R(q) = Y|q|$ for some $Y > 0$,
 (f) $f \in H^1((0, T), U^*)$ and $f(0) = 0$.

THEOREM 4.2. Assume hypotheses (G), (ψ), (R) and (f). Then there exists a unique solution to the evolutionary problem associated with .

PROOF. We verify the hypothesis of Theorem 2.2.

Step 1. It is trivial to check that $\mathcal{A}$ is linear and continuous. To prove coercivity we show that there exists $\alpha > 0$ such that $\langle \mathcal{A}z, z \rangle \geq \alpha \|z\|_Z^2$. If we call $e_1 = u'_1 - p$ and $e_2 = u_2 + (d-1)^{-1}p$ we have by applying Young inequality repeatedly

$$\begin{aligned} \langle \mathcal{A}z, z \rangle &= 2 \int_0^L \psi dx = \kappa \|e_1 + (d-1)e_2\|_2^2 + \mu \|e_1 - e_2\|_2^2 + l^2 \mu \|u'_2\|_2^2 + B \|p\|_2^2 = \\ &= \kappa \|u'_1 + (d-1)u_2\|_2^2 + \mu \left\| u'_1 - u_2 - \frac{d}{d-1} p \right\|_2^2 + l^2 \mu \|u'_2\|_2^2 + B \|p\|_2^2 \geq \\ &\geq (1-\theta) \kappa \|u'_1\|_2^2 + \left(1 - \frac{1}{\theta}\right) (d-1)^2 \kappa \|u_2\|_2^2 + (1-\tau) \mu \|u'_1 - u_2\|_2^2 + \\ &\quad l^2 \mu \|u'_2\|_2^2 + \left[\left(1 - \frac{1}{\tau}\right) \left(\frac{d}{d-1}\right)^2 \mu + B \right] \|p\|_2^2 \\ &\geq [(1-\theta) \kappa + (1-\eta)(1-\tau) \mu] \|u'_1\|_2^2 + \\ &\quad + \left[\left(1 - \frac{1}{\theta}\right) (d-1)^2 \kappa + \left(1 - \frac{1}{\eta}\right) (1-\tau) \mu \right] \|u_2\|_2^2 + l^2 \mu \|u'_2\|_2^2 + \\ &\quad + \left[\left(1 - \frac{1}{\tau}\right) \left(\frac{d}{d-1}\right)^2 \mu + B \right] \|p\|_2^2 \end{aligned}$$

for all θ, τ and η in $(0, \infty)$. We restrict to θ, τ and η in $(0, 1)$. Then by Poincarè inequality there exist $C_1 > 0$ and $C_2 > 0$ such that

$$\begin{aligned} \langle \mathcal{A}z, z \rangle &\geq C_1 [(1-\theta) \kappa + (1-\eta)(1-\tau) \mu] \|u_1\|_{1,2}^2 + \\ &\quad + \left[C_2 l^2 \mu + \left(1 - \frac{1}{\theta}\right) (d-1)^2 \kappa + \left(1 - \frac{1}{\eta}\right) (1-\tau) \mu \right] \|u_2\|_{1,2}^2 + \\ &\quad + \left[\left(1 - \frac{1}{\tau}\right) \left(\frac{d}{d-1}\right)^2 \mu + B \right] \|p\|_2^2 \end{aligned}$$

Now we choose θ, τ and η sufficiently close to 1 so that the coefficients multiplying $\|u_1\|_{1,2}^2, \|u_2\|_{1,2}^2$ and $\|p\|_2^2$ are positive and we get the desired coercivity.

Step 2. Since $\Phi(w) = Y\|q\|_1$ we see immediately that Φ is convex, proper, continuous and 1-homogeneous.

Step 3. Since $f \in H^1((0, T), U^*)$ and $f(0) = 0$ then $\mathcal{X} \in H^1((0, T), Z^*)$ and $\mathcal{X}(0) = 0$. $\square$

3. Analytical solution of the tensile test problem

In this section we obtain explicit solutions of the tensile test problem, meaning that an axial tension force is applied at the ends of the bar. We will consider two different variants of the problem: in the first variant the end section are left free, while in the second variant and the end section are clamped. We will see that the behavior of the solutions in this two situations is qualitatively similar, and in particular necking does not occur.

3.1. Traction with free end sections. Suppose that $f(x, t) = 0$, $h_1(L, t) = h_1(0, t) = F(t)$, and $h_2(L, t) = h_2(0, t) = 0$ for all $x \in (0, L)$ and $t \in [0, T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. From equation (4.2a) and (4.2b) we have that $\sigma_1 = F$ on all $(0, L)$. Then

$$\kappa(e_1 + (d-1)e_2) = F - \mu(e_1 - e_2) \quad \text{and} \quad e_1 = \frac{F - (d-1)\kappa e_2 + \mu e_2}{\kappa + \mu}.$$

Moreover

$$e_1 - e_2 = \frac{F - d\kappa e_2}{\kappa + \mu}.$$

Then

$$\begin{aligned}\sigma_2 &= (d-1)(F - \mu(e_1 - e_2)) - \mu(e_1 - e_2) = (d-1)F - d\mu(e_1 - e_2) = \\ &= (d-1)F - d\frac{\mu}{\kappa + \mu}F + d^2\frac{\kappa\mu}{\kappa + \mu}e_2 = \frac{(d-1)\kappa - \mu}{\kappa + \mu}F + d^2\frac{\kappa\mu}{\kappa + \mu}e_2 = \\ &= \frac{(d-1)\kappa - \mu}{\kappa + \mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa + \mu}p + d^2\frac{\kappa\mu}{\kappa + \mu}u_2.\end{aligned}$$

Now from equation (4.2c) we have

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa + \mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa + \mu}p + d^2\frac{\kappa\mu}{\kappa + \mu}u_2.$$

The solution is given by

$$u_2 = -\frac{p}{d-1} - \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

since conditions $h_2(L, t) = h_2(0, t)$ are satisfied because $u_2' = 0$. Notice that

$$e_2 = u_2 + \frac{p}{d-1} = -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

Now the Mandel stress is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa + \mu}(F - d\kappa e_2) = F.$$

When $F \leq Y$ then the response is purely elastic and $p = 0$. When $F > Y$ then $\dot{p} > 0$ and

$$F - Bp = Y.$$

This implies $p = B^{-1}(F - Y)$. Assuming $(d-1)\kappa - \mu > 0$ the sections constricts in traction.

3.2. Traction with clamped end sections.

Setup of the problem. Suppose that $f(x, t) = 0$, $h_1(L, t) = h_1(0, t) = F(t)$, and $u_2(L, t) = u_2(0, t) = 0$ for all $x \in (0, L)$ and $t \in [0, T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. By the same computations of Section 3.1 we find the equation

$$(4.5) \quad l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa + \mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa + \mu}p + d^2\frac{\kappa\mu}{\kappa + \mu}u_2.$$

Solution in the elastic range. In the elastic range $p = 0$ everywhere in $(0, L)$. In this case a particular solution is given by

$$x \mapsto -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F$$

and a solution of the homogeneous equation which is also symmetric with respect to $x = L/2$ is

$$x \mapsto \cosh\left(d\sqrt{\frac{\kappa}{\kappa + \mu}}\frac{x - \frac{L}{2}}{l}\right).$$

Then taking into account the boundary conditions we find the solution

$$u_2 = e_2 = \frac{(d-1)\kappa - \mu}{d^2\kappa\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa + \mu}}\frac{x - \frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa + \mu}}\frac{L}{2l}\right)} - 1 \right) F.$$

When $(d-1)\kappa - \mu > 0$ the sections constricts in traction. The middle section is the most affected by the constriction. The Mandel stress in the elastic range is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa + \mu}(F - d\kappa e_2) = \frac{d}{d-1}\frac{\mu}{\kappa + \mu} \left(1 - \frac{(d-1)\kappa - \mu}{d\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa + \mu}}\frac{x - \frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa + \mu}}\frac{L}{2l}\right)} - 1 \right) \right) F.$$

Its maximum is reached in the middle section so we conclude that the Mandel stress reach yield strength here first.

Solution in the plastic range. From the above considerations we are let to seek for a solution where $p > 0$ in the interval $(L/2 - \epsilon, L/2 + \epsilon)$ where $\epsilon \in [0, L/2]$ is a function of time.

REMARK 4.3. Since the solution is symmetric with respect to $x = L/2$ from now on we focus on the $(0, L/2)$ half of the bar only.

On $(0, L/2 - \epsilon)$ the solution is given by $p = 0$ and u_2 of the form

$$(4.6) \quad u_2 = \frac{(d-1)\kappa - \mu}{d^2\kappa\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{x - \frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l}\right)} - 1 \right) F + C_1 \sinh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} x\right)$$

where $C_1 \in \mathbb{R}$. On $(L/2 - \epsilon, L/2)$ we have

$$m - Bp = Y$$

$$\frac{d}{d-1} \frac{\mu}{\kappa + \mu} (F - d\kappa u_2 - \frac{d\kappa}{d-1} p) - Bp = Y.$$

Then we find

$$p = \frac{\frac{d}{d-1} \frac{\mu}{\kappa + \mu} F - \frac{d^2}{d-1} \frac{\kappa\mu}{\kappa + \mu} u_2 - Y}{\frac{d^2}{(d-1)^2} \frac{\mu\kappa}{\kappa + \mu} + B}.$$

Substituting that identity into equation (4.5) we have obtain a differential equation in y of the form

$$y'' = ay + b$$

where a and b depends on time only. A solution which is symmetric with respect to $x = L/2$ is of the form

$$(4.7) \quad y = -\frac{b}{a} + C_2 \cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right).$$

REMARK 4.4. From now on, since most of the resultant expressions are too long we do not report them.

We now look for a solution $u_2 \in C^2([0, L])$. When $\epsilon = L/2$ the solution is simply given by

$$u_2 = \frac{b}{a} \left(\frac{\cosh\left(\sqrt{a}\left(x - \frac{L}{2}\right)\right)}{\cosh\left(\sqrt{a}\frac{L}{2}\right)} - 1 \right)$$

When $\epsilon \in (0, L/2)$ we consider (4.6) on $(0, L/2 - \epsilon)$ and (4.7) on $(L/2 - \epsilon, L/2)$ and we impose the continuity of u_2 and u_2' at $x = L/2 - \epsilon$. Solving the resultant linear system we find the constants C_1 and C_2 .

Determination of ϵ . We are left with the determination of $\epsilon \in [0, L/2]$ as a function of time. To this aim we consider the Mandel stress computed the elastic range at $x = L/2$. We call F_Y the value of F for which that Mandel stress reach Y . We find

$$F_Y = \frac{d-1}{d} \frac{\kappa + \mu}{\mu} \left(-\frac{(d-1)\kappa - \mu}{d\mu} \left(\frac{1}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l}\right)} - 1 \right) + 1 \right)^{-1} Y.$$

For $F \leq F_Y$ we have $\epsilon = 0$. To determine ϵ when $F > F_Y$ we consider the equation $m(L/2 - \epsilon) = Y$. That equation cannot be solved analytically in ϵ . Instead we can find explicitly F as a function $f =: [0, \epsilon_0) \rightarrow \mathbb{R}$ of ϵ . Here $[0, \epsilon_0)$ is the maximal interval where f is well defined. After a meticulous analysis of the analytical expression of f we find that $\epsilon_0 > L/2$. This means that when the load F reach the value

$$f(L/2) = \frac{d-1}{d} \frac{\kappa + \mu}{\mu} Y,$$

the plastic distortion satisfies $p > 0$ in the entire interval $(0, L)$. We conclude that

$$\epsilon = \begin{cases} 0 & \text{if } F \leq F_Y \\ f^{-1}(F) & \text{if } F_Y < F \leq f(L/2) \\ L/2 & \text{if } F > f(L/2). \end{cases}$$

Absence of necking. From the analytical expression one can see that for $F > f(L/2)$ the solution u_2 depends linearly on F . In particular as $F \rightarrow \infty$ we have that $u_2 \rightarrow -\infty$ uniformly on all compact sets contained in $(0, L)$. We conclude that the small deformation model is not able to predict necking. Indeed the shrinkage of the cross section seems to be not localized on a small region around the middle of the bar but to be distributed along its entire length.

The limit of thin bar. To get further insight on the solution we look at the limit $l \rightarrow 0^+$.

PROPOSITION 4.5. In the limit $l \rightarrow 0^+$ the solution u_2 that we have found converges pointwise on $(0, L) \times [0, T]$ to the homogeneous solution of Section 3.1. This convergence is also uniform on compact subsets of $(0, L) \times [0, T]$.

PROOF. It is a direct computation. □

This results confirm the absence of necking in the small deformation model. Indeed it says that a thin bar deforms almost uniformly.

Model at large deformations

As we showed in the previous chapter, the one dimensional small deformations model of a bar is not able to predict necking. This leads us to refine the one dimensional model allowing large deformations.

1. Formulation

1.1. Motivation from the multidimensional kinematics. We start with the description of the kinematics of the bar from a multidimensional perspective. The results obtained here will serve as a motivational basis for the development of the one dimensional model. Let $\Omega = (0, L) \times \Sigma \subset \mathbb{R}^d$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ the reference configuration of the cylindrical bar. The generic material point will be denoted as $(X, Y) \in \Omega$ with $X \in (0, L)$ and $Y \in \Sigma$. We consider deformations of the form

$$(5.1) \quad (X, Y) \mapsto \begin{pmatrix} \chi_1(X) \\ \chi_2(X)Y \end{pmatrix}$$

where $\chi_1: (0, L) \rightarrow \mathbb{R}$ satisfy $\chi_1' > 0$ and $\chi_2: (0, L) \rightarrow (0, \infty)$. The corresponding deformation gradient is

$$F = \begin{pmatrix} \chi_1' & 0 \\ \chi_2'Y & \chi_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$(5.2) \quad F_p = \begin{pmatrix} P & 0 \\ 0 & P^{-\frac{1}{d-1}} I \end{pmatrix}$$

where $P: (0, L) \rightarrow (0, \infty)$. This choice implies plastic incompressibility. The elastic distortion is then given by the Kröner–Lee multiplicative decomposition

$$F_e = FF_p^{-1} = \begin{pmatrix} \chi_1' P^{-1} & 0 \\ \chi_2' P^{-1} Y & \chi_2 P^{\frac{1}{d-1}} I \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ GY & E_2 I \end{pmatrix},$$

where we have defined $E_1 = \chi_1' P^{-1}$, $E_2 = \chi_2 P^{\frac{1}{d-1}}$ and $G = \chi_2' P^{-1}$. In order to get some inspiration for developing constitutive relations for the one dimensional model, we look how the neo-Hookean elastic free energy density looks like when the deformation is of the form (5.1) and the plastic distortion is as in (5.2). Since the first principal invariant of $C_e = F_e^T F_e$ is

$$I_e = \text{tr}(C_e) = E_1^2 + G^2|Y|^2 + E_2^2$$

and the determinant of F_e is

$$J_e = \det F_e = E_1 E_2,$$

then the neo-Hookean elastic free energy density is given by

$$(5.3) \quad \psi_e = C \left(\frac{I_e}{J_e} - 2 \right) + w(J_e) = C \left(\frac{E_1^2 + G^2|Y|^2 + E_2^2}{E_1 E_2} - 2 \right) + w(E_1 E_2).$$

1.2. Kinematics. Let $L > 0$ the length of the cylindrical bar. The macroscopic configuration of the bar is described by $\chi_1: (0, L) \rightarrow \mathbb{R}$ with $\chi_1' > 0$ and $\chi_2: (0, L) \rightarrow (0, \infty)$. Here χ_1 describes the positions of the sections and χ_2 is the transversal deformation factor of the sections. The configuration of the microscopic structure of the bar, that is the plastic distortion, is described by $P: (0, L) \rightarrow (0, \infty)$. As suggested by the multidimensional kinematics, we define the elastic distortions as $E_1 = \chi_1' P^{-1}$, $E_2 = \chi_2 P^{\frac{1}{d-1}}$ and $G = \chi_2' P^{-1}$.

The configuration space Z of the system are thus the elements of the form $z = (\chi, P): (0, L) \rightarrow \mathbb{R} \times (0, \infty) \times (0, \infty)$. The virtual rates are denoted as $\tilde{w} = (\tilde{v}, \tilde{Q}): (0, L) \rightarrow \mathbb{R} \times (0, \infty) \times (0, \infty)$. The right functional setting will be discussed in Section 2 where the existence theorem is proved.

1.3. Statics. We assume that the internal generalized force is of the form

$$\langle \mathcal{J}, \tilde{w} \rangle = \int_0^L \left(\zeta \tilde{G} + \sigma_1 \tilde{E}_1 + \sigma_2 \tilde{E}_2 \right) dX + \int_0^L \left(\tau \tilde{Q} + \kappa \tilde{Q}' \right) dX,$$

Here $\zeta: (0, L) \rightarrow \mathbb{R}$ is the transversal elastic stress, $\sigma_1: (0, L) \rightarrow \mathbb{R}$ is the longitudinal elastic stress and $\sigma_2: (0, L) \rightarrow \mathbb{R}$ is the transversal elastic force while $\tau: (0, L) \rightarrow \mathbb{R}$ is the microscopic force and $\kappa: (0, L) \rightarrow \mathbb{R}$ the microscopic stress. Moreover the virtual rates $\tilde{E}_1$, $\tilde{E}_2$ and $\tilde{G}$ are defined by the relations

$$\begin{aligned} G^{-1} \tilde{G} &= (\chi_2')^{-1} \tilde{v}_2' - P^{-1} \tilde{Q} \\ E_1^{-1} \tilde{E}_1 &= (\chi_1')^{-1} \tilde{v}_1' - P^{-1} \tilde{Q} \\ E_2^{-1} \tilde{E}_2 &= \chi_2^{-1} \tilde{v}_2 + (d-1)^{-1} P^{-1} \tilde{Q}, \end{aligned}$$

Then the internal generalized force can be written more explicitly as

$$\begin{aligned} \langle \mathcal{J}, \tilde{w} \rangle &= \int_0^L G \zeta \left((\chi_2')^{-1} \tilde{v}_2' - P^{-1} \tilde{Q} \right) dX + \\ &+ \int_0^L \left[E_1 \sigma_1 \left((\chi_1')^{-1} \tilde{v}_1' - P^{-1} \tilde{Q} \right) + E_2 \sigma_2 \left(\chi_2^{-1} \tilde{v}_2 + (d-1)^{-1} P^{-1} \tilde{Q} \right) \right] dX + \\ &+ \int_0^L \left[\tau \tilde{Q} + \kappa \tilde{Q}' \right] dX. \end{aligned}$$

REMARK 5.1. In the small strain model of Chapter 4 we did not included the microscopic stress κ . However, we are now including it because we expect that in the large deformation regime, it is necessary to obtain an existence result in Sobolev spaces. This is suggested by the existence theorem for three-dimensional elastoplasticity at large strains presented in [27].

The external generalized force is chosen of the form

$$\tilde{X} = \int_0^L f \cdot \tilde{v} dx + h(L) \cdot \tilde{v}(L) - h(0) \cdot \tilde{v}(0).$$

As in the small strain model, $f: (0, L) \rightarrow \mathbb{R}^2$ is an external force distributed along the bar and $h: \{0, L\} \rightarrow \mathbb{R}^2$ is an external force acting at the ends of the bar. The first component of both f and h is referred to the axial direction, while the second component is referred to the longitudinal direction.

PROPOSITION 5.2. The local form of the equilibrium condition is

$$\begin{aligned} (P^{-1} \sigma_1)' + f_1 &= 0 \quad \text{in } (0, L) \\ h_1(X) &= P(X)^{-1} \sigma_1(X) \quad \text{for } X \in \{0, L\} \\ -P^{\frac{1}{d-1}} \sigma_2 + (P^{-1} \zeta)' + f_2 &= 0 \quad \text{in } (0, L) \\ h_2(X) &= P(X)^{-1} \zeta(X) \quad \text{for } X \in \{0, L\} \\ \chi_1' P^{-2} \sigma_1 - (d-1)^{-1} \chi_2 P^{\frac{1}{d-1}-1} \sigma_2 + \chi_2' P^{-2} \zeta - \tau + \kappa' &= 0 \quad \text{for } X \in \{0, L\} \\ \kappa(X) &= 0 \quad \text{for } X \in \{0, L\} \end{aligned}$$

PROOF. Rearranging the terms we have

$$\begin{aligned} \langle \mathcal{J}, \tilde{w} \rangle &= \int_0^L \left[P^{-1} \sigma_1 \tilde{v}_1' + P^{\frac{1}{d-1}} \sigma_2 \tilde{v}_2 + P^{-1} \zeta \tilde{v}_2' \right] dX + \\ &+ \int_0^L \left[\left(\tau - \chi_1' P^{-2} \sigma_1 + (d-1)^{-1} \chi_2 P^{\frac{1}{d-1}-1} \sigma_2 - \chi_2' P^{-2} \zeta \right) \tilde{Q} + \kappa \tilde{Q}' \right] dX. \end{aligned}$$

By integration by parts and application of the fundamental lemma of the calculus of variation we get the thesis. $\square$

We call Mandel stress the quantity

$$(5.5) \quad m = \chi_1' P^{-2} \sigma_1 - (d-1)^{-1} \chi_2 P^{\frac{1}{d-1}-1} \sigma_2 + \chi_2' P^{-2} \zeta.$$

It is the elastic force felt by the microscopic configuration.

1.4. Constitutive relations. We choose a free energy for unit axial length of the form

$$\psi = W(E_1, E_2, G) + H(P, P')$$

where $W: (0, \infty) \times (0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ is elastic free energy density and $H: (0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ is the hardening free energy density. The plastic response is obtained by introducing a dissipation potential per unit axial length of the form

$$\phi = R(P, \dot{P})$$

where $R: (0, \infty) \times \mathbb{R} \rightarrow [0, \infty]$ is convex in $\dot{P}$ and satisfies $R(0) = 0$. Consistently with that choice of ψ and ϕ we impose the following constitutive relations

$$\begin{aligned} \sigma_i &= \frac{\partial W}{\partial E_i} \quad \text{and} \quad \zeta = \frac{\partial W}{\partial G} \\ \tau &\in \frac{\partial H}{\partial P} + P^{-1} \partial_{\dot{P}} R \quad \text{and} \quad \kappa = \frac{\partial H}{\partial P'}. \end{aligned}$$

PROPOSITION 5.3. The constitutive relations (5.6) satisfy the dissipation inequality.

PROOF. Let $\tau = \frac{\partial H}{\partial P} + P^{-1} \xi$ with $\xi \in \partial R(\dot{P} P^{-1})$. Then the dissipation per unit axial length is given by

$$\zeta \dot{G} + \sigma_2 \dot{E}_2 + \sigma_1 \dot{E}_1 + \tau \dot{P} + \kappa \dot{P}' - \dot{\psi} = \xi P^{-1} \dot{P}.$$

By definition of subdifferential we have $0 = R(0) \geq R(P^{-1} \dot{P}) - \xi P^{-1} \dot{P}$ so that $\xi P^{-1} \dot{P} \geq R(P^{-1} \dot{P}) \geq 0$. This implies that the dissipation is nonnegative. $\square$

We will see particular examples of constitutive relations in the next section, after we have clarified which assumptions are required for the model to be well-posed.

2. Existence result

In this section we prove the existence of energetic solutions. A self-contained introduction to this solution concept, including an explanation of the notation employed in this section, can be found in Section 2 of Appendix 2. The aim of this section is to verify the hypotheses required to apply Theorem 2.8.

2.1. Formulation of the initial boundary value problem. We now describe how the model formulated in the previous section, can be treated within the framework of energetic solutions.

Dirichlet conditions. Suppose that at time $t \in [0, T]$ the following Dirichlet conditions are imposed

$$(5.7) \quad \chi_i(X) = g_i(X, t) \quad \text{for } X \in \Gamma_D^i \text{ and } i \in \{1, 2\},$$

where $\Gamma_D^i \subset \{0, L\}$ and $g_i: (0, L) \times [0, T] \rightarrow \mathbb{R}$ satisfy $g_1' > 0$ and $g_2 > 0$. To avoid free translations of the bar, it is required that $\Gamma_D^1 \neq \emptyset$, but Γ_D^2 can be empty instead. Conditions (5.7) can be converted into time-independent conditions by expressing χ_i as

$$\chi_1(X) = g_1(y_1(X), t) \quad \text{and} \quad \chi_2(X) = g_2(y_1(X), t) y_2(X),$$

with $y_i: (0, L) \rightarrow \mathbb{R}$, $y_1' > 0$ and $y_2 > 0$, and requiring that

$$y_1(X) = X \quad \text{for } X \in \Gamma_D^1 \quad \text{and} \quad y_2(X) = 1 \quad \text{for } X \in \Gamma_D^2.$$

Having made that change of variables, the unknown of the problem are y and P . We notice that the elastic distortions E_1, E_2 and G are given by

$$E_1 = g_1'(y_1, t) y_1' P^{-1}, \quad E_2 = g_2(y_1, t) y_2 P \quad \text{and} \quad G = g_2'(y_1, t) y_1' y_2 P^{-1} + g_2(y_1, t) y_2' P^{-1}.$$

Function spaces. Let $Y = W^{1, q_Y}((0, L), \mathbb{R}^2)$ with $q_Y \in (1, \infty)$ be the space of y and $Z = L^{q_H}((0, L)) \cap W^{1, r_H}((0, L))$ with $q_H \in (1, \infty)$, $r_H \in (1, q_H]$ be the space of P . Let $Q = Y \times Z$. The Dirichlet conditions are imposed requiring that y, P belong respectively to the admissible spaces

$$\mathcal{Y} = \{y \in Y \mid y_1(X) = X \text{ for } X \in \Gamma_D^1 \text{ and } y_2(X) = 1 \text{ for } X \in \Gamma_D^2\} \quad \text{and} \quad \mathcal{Z} = Z.$$

Let $\mathcal{Q} = \mathcal{Y} \times \mathcal{Z}$. It is evident that $\mathcal{Q}$ is a weakly closed subset of Q .

Energy functional and dissipation distance. To complete the formulation of the initial boundary value problem, we need to specify the energy functional $\mathcal{E}: \mathcal{Q} \times [0, T] \rightarrow (-\infty, \infty]$ and the dissipation distance $\mathcal{D}: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty]$. The energy functional is defined by

$$\begin{aligned} \mathcal{E}(y, P, t) = & \int_0^L W(g'_1(y_1, t)y_1'P^{-1}, g_2(y_1, t)y_2P, g'_2(y_1, t)y_1'y_2P^{-1} + g_2(y_1, t)y_2'P^{-1}) dX + \\ & + \int_0^L H(P, P') dX - \int_0^L (f_1(t)g_1(y_1, t) + f_2(t)g_2(y_1, t)y_2) dX. \end{aligned}$$

The dissipation distance can be expressed in term of a dissipation distance density $D: \mathbb{R}^2 \rightarrow [0, \infty]$ as

$$\mathcal{D}(P_0, P_1) = \int_0^L D(P_0, P_1) dX.$$

According to Remark 2.4, D can be defined from the dissipation potential density R by

$$(5.8) \quad D(P_0, P_1) = \inf \left\{ \int_0^1 R(P(t), \dot{P}(t)) dt \mid P \in C^1([0, 1]) \text{ and } P(i) = P_i \text{ for } i \in \{0, 1\} \right\}.$$

2.2. Assumptions and preliminary results. We now discuss the assumptions under which we will prove the validity of hypotheses (E1)-(E4), (D1)-(D2) and (C1)-(C2) of Section 2 in Appendix 2. Subsequently, these assumptions will be implicitly supposed to be valid and will not be recalled.

REMARK 5.4 (on notation). Sometimes the notation $G = E_3$ and $E = (E_1, E_2, E_3)$ will be used. Given $E, N \in \mathbb{R}^3$, then $NE \in \mathbb{R}^3$ is the component-wise multiplication between E and N , i.e., $(NE)_i = N_i E_i$ for $i \in \{1, 2, 3\}$.

The assumption on the Dirichlet data is

$$(G) \quad g_1 \in C^2(\mathbb{R} \times [0, T]), g_2 \in C^2(\mathbb{R} \times [0, T], (0, \infty)) \text{ and there exists } 0 < \gamma < \Gamma < \infty \text{ for } i \in \{1, 2\} \text{ such that } \gamma < g'_1 < \Gamma, \gamma < g_2 < \Gamma \text{ and } |g'_2| \leq \Gamma.$$

Only to prove Proposition 5.11 we will assume also the following additional requirement

$$(\hat{G}) \quad g'_2 = 0.$$

We do not know whether the proof works without $(\hat{G})$. Anyway assuming $(\hat{G})$, it is not restrictive for our purposes.

The elastic free energy density is supposed to satisfy the following assumptions

(W1) there exists $\mathbb{W}: \mathbb{R}^4 \rightarrow (-\infty, \infty]$ continuous such that

$$W(E_1, E_2, G) = \mathbb{W}(E_1, E_2, G, E_1 E_2)$$

for all $E_1, E_2, G \in \mathbb{R}$ and $\mathbb{W}$ is convex in $(E_1, G, E_1 E_2)$,

(W2) there exist $C_W > 0$ and $q_W \in (1, \infty)$ such that $W(E_1, E_2, G) \geq C_W(-1 + |E_1|^{q_W} + |E_2|^{q_W} + |G|^{q_W})$ for all $E_1, E_2, G \in \mathbb{R}$,

(W3) $W(E_1, E_2, G) = \infty$ if and only if $E_1 \leq 0$ or $E_2 \leq 0$,

(W4) W is differentiable in $\text{dom } W$ and there exist $C_0^W \in \mathbb{R}$, $C_1^W > 0$ and a modulus of continuity ω^W such that

(i) for all $E \in \text{dom } W$ and $i \in \{1, 2, 3\}$ it holds $\left| E_i \frac{\partial W}{\partial E_i}(E) \right| \leq C_1^W (C_0^W + W(E))$,

(ii) let $K(E) = E \cdot \frac{\partial W}{\partial E}(E)$, then $|K(NE) - K(E)| \leq \omega^W(|N - 1|)(W(E) + C_0^W)$ for all $N \in (0, \infty)^3$.

Assumption (W1) is a sort policonvexity requirement necessary to ensure lower semicontinuity, (W2) is a growth condition, (W3) is an incompentratibility condition for the elastic distortion and (W4) is a further regularity assumption which is needed to get suitable a priori estimates. Later on we will use the following consequence of (W4).

LEMMA 5.5. Fix $0 < \delta < 1$ small enough. Then there exists $C > 0$ such that

$$W(NE) + \max_{i \in \{1, 2, 3\}} \left| E_i \frac{\partial W}{\partial E_i}(NE) \right| \leq C(W(E) + C_0^W)$$

for all $E \in (0, \infty) \times (0, \infty) \times \mathbb{R}$ and $N \in (1 - \delta, 1 + \delta)^3$. Here $NE \in \mathbb{R}^3$ denotes the component-wise multiplication, i.e., $(NE)_i = N_i E_i$ for all $i \in \{1, 2, 3\}$.

PROOF. *Step 1.* For all $i \in \{1, 2, 3\}$, by (W4, i)

$$\begin{aligned} \left| E_i \frac{\partial W}{\partial E_i}(NE) \right| &= |N_i|^{-1} \left| N_i E_i \frac{\partial W}{\partial E_i}(NE) \right| \leq |N_i|^{-1} C_1^W (C_0^W + W(NE)) \leq \\ &\leq \frac{C_1^W}{1-\delta} (C_0^W + W(NE)). \end{aligned}$$

Step 2. We have that

$$|W(NE) - W(E)| = \left| \int_0^1 \frac{d}{d\lambda} W(\tilde{N}(\lambda)E) d\lambda \right|,$$

where $\tilde{N}(\lambda) = (1-\lambda) + \lambda N$, i.e., $\tilde{N}_i(\lambda) = (1-\lambda) + \lambda N_i \in (1-\delta, 1+\delta)$ for all $i \in \{1, 2, 3\}$. Then

$$\begin{aligned} |W(NE) - W(E)| &= \left| \int_0^1 \frac{\partial W}{\partial E}(\tilde{N}(\lambda)E) \cdot (N-1) d\lambda \right| \leq \sum_{i=1}^3 \int_0^1 \left| \tilde{N}_i(\lambda) E_i \frac{\partial W}{\partial E_i}(\tilde{N}(\lambda)E) \right| \frac{|N_i-1|}{\tilde{N}_i(\lambda)} d\lambda \leq \\ &\leq \frac{\delta}{1-\delta} C_1^W \int_0^1 (W(\tilde{N}(\lambda)E) + C_0^W) d\lambda. \end{aligned}$$

Step 3. Let $\theta(E) = \sup_{N \in (1-\delta, 1+\delta)^3} W(NE)$. From Step 2

$$|W(NE) - W(E)| \leq \theta(E) - W(E) \leq \frac{\delta}{1-\delta} C_1^W (\theta(E) + C_0^W).$$

This implies

$$\left(1 - \frac{\delta}{1-\delta} C_1^W\right) \theta(E) \leq W(E) + \frac{\delta}{1-\delta} C_1^W C_0^W.$$

If δ is small enough then

$$\theta(E) \leq \frac{W(E) + \frac{\delta}{1-\delta} C_1^W C_0^W}{1 - \frac{\delta}{1-\delta} C_1^W}.$$

Step 4. From Step 3

$$|W(NE) - W(E)| \leq \frac{\delta}{1-\delta} C_1^W \left(\frac{W(E) + \frac{\delta}{1-\delta} C_1^W C_0^W}{1 - \frac{\delta}{1-\delta} C_1^W} + C_0^W \right) \leq \left(\frac{1}{\frac{\delta}{1-\delta} C_1^W} - 1 \right)^{-1} (W(E) + C_0^W).$$

Step 5. From Step 1

$$\begin{aligned} \max_{i \in \{1, 2, 3\}} \left| E_i \frac{\partial W}{\partial E_i}(NE) \right| &\leq \frac{C_1^W}{1-\delta} (C_0^W + \theta(E)) \leq \frac{C_1^W}{1-\delta} \left(\frac{W(E) + \frac{\delta}{1-\delta} C_1^W C_0^W}{1 - \frac{\delta}{1-\delta} C_1^W} + C_0^W \right) = \\ &= \frac{C_1^W}{1-\delta(1+C_1^W)} (W(E) + C_0^W). \end{aligned}$$

Step 6. Summing together the results of Step 4 and Step 5 we get the thesis. $\square$

The plastic free energy density satisfies

(H1) $H: \mathbb{R}^2 \rightarrow (-\infty, \infty]$ is continuous and $H(P, \cdot)$ is convex for all $P \in \mathbb{R}$,

(H2) there exist $C_H > 0$ and $q_H \in (1, \infty)$, $r_H \in (1, q_H]$ such that $H(P, P') \geq C_H(-1 + |P|^{q_H} + |P'|^{r_H})$ for all $P, P' \in \mathbb{R}$,

(H3) there exists $\Pi > 0$ such that $H(P, P') = \infty$ if and only if $P \leq \Pi$.

The convexity assumption (H1) is required for lower semicontinuity, (H2) is a growth condition ensuring that the hardening and the plastic gradient regularization is sufficiently strong. Assumption (H3) is a strong incompressibility condition for the plastic distortion.

To handle correctly the nonlinear coupling between elastic and plastic distortions, we require the following compatibility condition on the Hölder exponents

$$(\ddot{O}) \quad \frac{1}{q_Y} = \frac{1}{q_W} + \frac{1}{q_H}.$$

The requirements on the dissipation potential density are

(d1) $D: \mathbb{R}^2 \rightarrow [0, \infty]$ is lower semicontinuous,

(d2) D is a quasidistance, in the sense that $D(P_0, P_1) = 0$ if and only if $P_0 = P_1$ and $D(P_0, P_2) \leq D(P_0, P_1) + D(P_1, P_2)$ for all $P_0, P_1, P_2 \in \mathbb{R}$,

(d3) D is finite and continuous on $[\Pi, \infty) \times [\Pi, \infty)$, where Π is the constant appearing in (H3).

2.3. Verification the properties of the energy functional. We start with the following lemma which is necessary to handle the nonlinear coupling between elastic and plastic distortion.

LEMMA 5.6. Let $a, b > 0$. Then for all $\delta > 0$ and $\rho > 1$

$$\frac{a}{b} \geq \rho \delta^{\frac{\rho-1}{\rho}} a^{\frac{1}{\rho}} - (\rho-1) \delta b^{\frac{1}{\rho-1}}$$

PROOF. By Young inequality, if $A, B > 0$ then

$$AB \leq \frac{A^\rho}{\rho} + \frac{B^{\rho^*}}{\rho^*}$$

and

$$A^{\frac{1}{\rho}} B^{\frac{1}{\rho^*}} \leq \frac{A}{\rho} + \frac{B}{\rho^*}.$$

It follows that

$$A \geq \rho A^{\frac{1}{\rho}} B^{\frac{1}{\rho^*}} - \frac{\rho}{\rho^*} B.$$

If we put $A = a/b$ and B such that

$$B^{\frac{1}{\rho^*}} = \delta^{\frac{1}{\rho^*}} b^{\frac{1}{\rho}}$$

the thesis follows. $\square$

We now prove that (E2) is satisfied. We start with proving the boundedness of sublevels.

PROPOSITION 5.7 (Boundedness of energy sublevels). The sublevels of $\mathcal{E}(\cdot, t)$ are bounded in Q for all $t \in [0, T]$.

PROOF. From (W1) and (H1) it follows

$$(5.9) \quad \mathcal{E}(y, P, t) \geq C \left(-1 + \|g'_1(y_1, t) y'_1 P^{-1}\|_{q_W}^{q_W} + \|g_2(y_1, t) y_2 P\|_{q_W}^{q_W} + \|g'_2(y_1, t) y'_1 y_2 P^{-1} + g_2(y_1, t) y'_2 P^{-1}\|_{q_W}^{q_W} + \|P\|_{q_H}^{q_H} + \|P'\|_{r_H}^{r_H} \right).$$

By Hölder inequality

$$\|y'_1\|_{q_Y} = \|y'_1 P^{-1} P\|_{q_Y} \leq \|y'_1 P^{-1}\|_{q_W} \|P\|_{q_H}$$

so that using Lemma 5.6 with $\rho = q_W/q_Y$ one gets

$$\|y'_1 P^{-1}\|_{q_W}^{q_W} \geq \frac{\|y'_1\|_{q_Y}^{q_W}}{\|P\|_{q_H}^{q_W}} \geq \rho \delta^{\frac{\rho-1}{\rho}} \|y'_1\|_{q_Y}^{q_Y} - (\rho-1) \delta \|P\|_{q_H}^{q_H}$$

for all $\delta > 0$. We have used that $q_W/(\rho-1) = q_H$. Moreover

$$\|y'_1 P^{-1}\|_{q_W} = \|g'_1(y_1, t)^{-1} g'_1(y_1, t) y'_1 P^{-1}\|_{q_W} \leq \gamma^{-1} \|g'_1(y_1, t) y'_1 P^{-1}\|_{q_W}.$$

This implies that

$$(5.10) \quad \|g'_1(y_1, t) y'_1 P^{-1}\|_{q_W}^{q_W} \geq \gamma^{q_W} \rho \delta^{\frac{\rho-1}{\rho}} \|y'_1\|_{q_Y}^{q_Y} - \gamma^{q_W} (\rho-1) \delta \|P\|_{q_H}^{q_H}.$$

Similarly by Hölder and Young inequalities, for all $\delta > 0$

$$\|y_2\|_{q_Y}^{q_Y} = \|y_2 P P^{-1}\|_{q_Y}^{q_Y} \leq \|y_2 P\|_{q_W}^{q_Y} \|P^{-1}\|_{q_H}^{q_Y} \leq \frac{1}{\rho \delta^\rho} \|y_2 P\|_{q_W}^{q_W} + \frac{\delta^{\rho^*}}{\rho^*} \|P^{-1}\|_{q_H}^{q_H} \leq \frac{1}{\rho \delta^\rho} \|y_2 P\|_{q_W}^{q_W} + \frac{\delta^{\rho^*}}{\rho^*} L \Pi^{-q_H}$$

where now $\rho = 1 + q_W/q_H$. Last inequality holds since $P > \Pi$ a.e. on the sublevels of $\mathcal{E}$. Then

$$(5.11) \quad \|g_2(y_1, t) y_2 P\|_{q_W}^{q_W} \geq \gamma^{q_W} \rho \delta^\rho \|y_2\|_{q_Y}^q - \gamma^{q_W} \frac{\rho}{\rho^*} \delta^{\rho+\rho^*} \Pi^{-q_H}.$$

Substituting (5.10) and (5.11) into (5.9) with δ sufficiently small, one gets that on sublevels of $\mathcal{E}$ the norms $\|y'_1\|_{q_Y}$, $\|y\|_{q_Y}$, $\|P\|_{q_H}$ and $\|P'\|_{r_H}$ are bounded. By Poincaré inequality it follows that also $\|y_1\|_{1, q_Y}$ is bounded. It remains to prove that $\|y'_2\|_{q_Y}$ is bounded. Let

$$G = g'_2(y_1, t) y'_1 y_2 P^{-1} + g_2(y_1, t) y'_2 P^{-1}.$$

Then

$$y'_2 = \frac{GP - g'_2(y_1, t) y'_1 y_2}{g_2(y_1, t)}.$$

It follows that

$$\begin{aligned} \|y_2'\|_{q_Y} &\leq \frac{1}{\gamma} \|GP\|_{q_Y} + \frac{1}{\gamma} \|g_2'(y_1, t)y_1'y_2\|_{q_Y} \leq \frac{1}{\gamma} \|G\|_{q_W} \|P\|_{q_H} + \frac{\Gamma}{\gamma} \|y_1'y_2\|_{q_Y} \leq \\ &\leq \frac{1}{\gamma} \|G\|_{q_W} \|P\|_{q_H} + \frac{\Gamma}{\gamma} \|y_1'\|_{q_Y} \|y_2\|_{\infty}. \end{aligned}$$

Since $L^\infty((0, T))$ is continuously embedded in $W^{1,q_Y}((0, L))$, it follows that for all $\epsilon > 0$ there exists $C > 0$ such that

$$\|y_2\|_{\infty} \leq C(\|y_2\|_{q_Y} + \epsilon \|y_2'\|_{q_Y}) \quad \text{for all } y_2 \in W^{1,q_Y}((0, L)).$$

On a sublevel of $\mathcal{E}$ the quantities $\|G\|_{q_W}$, $\|P\|_{q_H}$, $\|y_1'\|_{q_Y}$ and $\|y_2\|_{q_Y}$ are bounded, so that there exists $M > 0$ such that

$$\|y_2'\|_{q_Y} \leq M(1 + \epsilon \|y_2'\|_{q_Y}).$$

Choosing ϵ small enough, it follows that $\|y_2'\|_{q_Y}$ is bounded on the sublevels of $\mathcal{E}$. This concludes the proof. $\square$

We now move to the proof of the weak lower semicontinuity of $\mathcal{E}$. We first need the following technical result.

PROPOSITION 5.8 (convergence of elastic distortion). Let $y^k, y \in W^{1,q_Y}((0, L), \mathbb{R} \times \mathbb{R})$ and $P_k, P \in C^0([0, L])$ such that

$$\begin{aligned} y^k &\rightharpoonup y \quad \text{in } W^{1,q_Y}((0, L)) \\ P_k &\rightarrow P \quad \text{in } C^0([0, L]). \end{aligned}$$

Assume also that $P_k, P \geq \Pi > 0$ a.e. in $(0, L)$. Let also $\gamma_1^k, \gamma_1, \gamma_2^k, \gamma_2, \rho_2^k, \rho_2 \in C^0([0, L])$ such that

$$\begin{aligned} \gamma_1^k &\rightarrow \gamma_1 \quad \text{in } C^0([0, L]) \\ \gamma_2^k &\rightarrow \gamma_2 \quad \text{in } C^0([0, L]) \\ \rho_2^k &\rightarrow \rho_2 \quad \text{in } C^0([0, L]) \end{aligned}$$

Then, up to extracting a subsequence,

$$\begin{aligned} \gamma_1^k (y_1^k)' P_k^{-1} &\rightharpoonup \gamma_1 y_1' P^{-1} && \text{in } L^1((0, L)) \\ \gamma_2^k y_2^k P_k &\rightarrow \gamma_2 y_2 P && \text{in } C^0([0, L]) \\ \rho_2^k (y_1^k)' y_2^k P_k^{-1} + \gamma_2^k (y_2^k)' P_k^{-1} &\rightharpoonup \rho_2 y_1' y_2 P^{-1} + \gamma_2 y_2' P^{-1} && \text{in } L^1((0, L)) \\ \gamma_1^k (y_1^k)' \gamma_2^k y_2^k &\rightharpoonup \gamma_1 y_1' \gamma_2 y_2 && \text{in } L^1((0, L)) \end{aligned}$$

PROOF. First notice that $P_k^{-1} \rightarrow P^{-1}$ in $L^\infty((0, L))$. Indeed

$$|P_k^{-1} - P^{-1}| = \frac{|P_k - P|}{|P_k P|} \leq \frac{|P_k - P|}{\Pi^2}.$$

Notice also that, since $W^{1,q}((0, L))$ is compactly embedded in $C^0([0, L])$, up to extracting a subsequence $y_2^k \rightarrow y_2$ in $C^0([0, L])$. Now the results follows from Lemma 5.9 below. $\square$

LEMMA 5.9. Let $f_k, f \in L^q((0, L))$ with $q \in (1, \infty)$, $g_k, g \in L^\infty((0, L))$ and $h_k, h \in C^0([0, L])$ such that

$$\begin{aligned} f_k &\rightharpoonup f \quad \text{in } L^q((0, L)) \\ g_k &\rightarrow g \quad \text{in } L^\infty((0, L)) \\ h_k &\rightarrow h \quad \text{in } C^0([0, L]). \end{aligned}$$

Then $f_k g_k h_k \rightharpoonup f g h$ in $L^1((0, L))$.

PROOF. Let $\phi \in L^\infty((0, L))$. Then

$$\begin{aligned} \left| \int_0^L (f_k g_k h_k - f g h) \phi dX \right| &\leq \\ &\leq \left| \int_0^L (g_k - g) f_k h_k \phi dX \right| + \left| \int_0^L g f_k (h_k - h) \phi dX \right| + \left| \int_0^L g (f_k - f) h \phi dX \right|. \end{aligned}$$

Since f_k are bounded in $L^q((0, L))$, h_k are bounded in $C^0([0, L])$, then $f_k h_k \phi$ and $g f_k \phi$ are bounded in $L^1((0, L))$. Moreover $g h \phi \in L^q((0, L))$. Then the thesis follows. $\square$

We are now ready to get the weakly lower semicontinuity of $\mathcal{E}$.

PROPOSITION 5.10 (lower semicontinuity of the energy). For all $t \in [0, T]$ the map $\mathcal{E}(\cdot, t)$ is lower semicontinuous.

PROOF. Let $(y^k, P_k) \rightharpoonup (y, P)$ in $Y \times Z$. We have to prove that

$$\mathcal{E}(y, P, t) \leq \liminf_{k \rightarrow \infty} \mathcal{E}(y_k, P_k, t).$$

We take a (not relabeled) subsequence such that $\mathcal{E}(y_k, P_k, t)$ has a limit. Then we need to show that

$$(5.12) \quad \mathcal{E}(y, P, t) \leq \lim_{k \rightarrow \infty} \mathcal{E}(y_k, P_k, t).$$

If $\mathcal{E}(y_k, P_k, t) \rightarrow \infty$ we have nothing to prove. Then we assume that instead it converges to a finite limit. In this case we can assume that $\mathcal{E}(y_k, P_k, t) < \infty$ for all k . This implies that $P_k > \Pi$ a.e. in $(0, L)$, so that also $P \geq \Pi$ a.e. in $(0, L)$.

By Rellich–Kondrachov theorem, up to extracting a subsequence, $y_i^k \rightarrow y_i$ for $i \in \{1, 2\}$ and $P_k \rightarrow P$ in $C^0([0, L])$. Moreover $g'_1(y_1^k, t) \rightarrow g'_1(y_1, t)$, $g_2(y_1^k, t) \rightarrow g_2(y_1, t)$ and $g'_2(y_1^k, t) \rightarrow g'_2(y_1, t)$ in $C^0([0, L])$. Indeed if $\gamma \in C^1(\mathbb{R})$ then

$$(5.13) \quad \begin{aligned} |\gamma(y_1^k) - \gamma(y_1)| &= \left| \int_0^1 \frac{d}{d\lambda} \gamma((1-\lambda)y_1^k + \lambda y_1) d\lambda \right| = \left| \int_0^1 \gamma'((1-\lambda)y_1^k + \lambda y_1) (y_1 - y_1^k) d\lambda \right| \leq \\ &\leq \|y_1 - y_1^k\|_\infty \int_0^1 |\gamma'((1-\lambda)y_1^k + \lambda y_1)| d\lambda. \end{aligned}$$

Since y_1^k converges to y_1 uniformly, then $y_1^k(t)$ takes values in a bounded set as $k \in \mathbb{N}$ and $t \in [0, T]$. Then the continuity of γ' implies that the last integral in (5.13). We have proved that $\gamma(y_1^k) \rightarrow \gamma(y_1)$ in $C^0([0, T])$, and this provides the desired results.

Applying Proposition 5.8, it follows that, up to extracting a subsequence,

$$\begin{aligned} g'_1(y_1^k, t)(y_1^k)' P_k^{-1} &\rightharpoonup g'_1(y_1, t) y_1' P^{-1} && \text{in } L^1((0, L)) \\ g_2(y_1^k, t) y_2^k P_k &\rightarrow g_2(y_1, t) y_2 P && \text{in } C^0([0, L]) \\ g'_2(y_1^k, t)(y_1^k)' y_2^k P_k^{-1} + g_2(y_1^k, t)(y_2^k)' P_k^{-1} &\rightharpoonup g'_2(y_1, t) y_1' y_2 P^{-1} + g_2(y_1, t) y_2' P^{-1} && \text{in } L^1((0, L)) \\ g'_1(y_1^k, t)(y_1^k)' g_2(y_1^k, t) y_2^k &\rightharpoonup g'_1(y_1, t) y_1' g_2(y_1, t) y_2 && \text{in } L^1((0, L)). \end{aligned}$$

By these convergence results, together with $P_k \rightarrow P$ in $C^0([0, L])$ and $P'_k \rightharpoonup P'$ in $L^{r_H}((0, T))$, and by the convexity assumptions (W0) and (H0), we can apply Theorem B.25 which leads to (5.12). This concludes the proof. $\square$

Until now we have established (E1) and (E2). We now prove (E3) and (E4).

PROPOSITION 5.11. Then there exists $\mathcal{D} \subset \mathcal{Q}$ such that $\text{dom } \mathcal{E}(\cdot, t) = \mathcal{D}$ for all $t \in [0, T]$. Moreover

- (i) for all $q \in \mathcal{D}$ the function $\mathcal{E}(q, \cdot)$ is $C^1([0, T])$,
- (ii) there exist C_1^E and C_0^E such that $|\partial_t \mathcal{E}(q, t)| \leq C_1^E (C_0^E + \mathcal{E}(q, t))$ for all $q \in \mathcal{Q}$ and $t \in [0, T]$,
- (iii) there exists a modulus of continuity ω^E such that for all $q \in \mathcal{D}$ and $t, s \in [0, T]$,

$$|\partial_t \mathcal{E}(q, t) - \partial_t \mathcal{E}(q, s)| \leq \omega^E(|t - s|) \min\{\mathcal{E}(q, t) + C_0^W, \mathcal{E}(q, s) + C_0^W\}.$$

PROOF. Since $g'_1 > 0$ and $g_2 > 0$, it is clear that

$$\mathcal{D} = \{(y, P) \in Y \times Z \mid y'_1, y_2, P > 0 \text{ a.e. on } (0, L)\}.$$

Proof of (i) Let $E_1 = g'_1(y_1, t) y_1' P^{-1}$, $E_2 = g_2(y_1, t) y_2 P$ and $G = g_2(y_1, t) y_2' P^{-1}$ as functions of $(y, y', P, t) \in \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \times t \in [0, T]$. Let $q = (y, P) \in \mathcal{D}$. Then $(E_1, E_2, G) \in \text{dom } W$ a.e. on $(0, L)$ and everywhere on $[0, T]$. Then $t \mapsto W(E_1, E_2, G)$ is differentiable in $[0, T]$ and a.e. in $(0, L)$ and

$$\frac{d}{dt} W(E_1, E_2, G) = E_1 \frac{\partial W}{\partial E_1}(E_1, E_2, G) \frac{\dot{g}'_1(y_1, t)}{g'_1(y_1, t)} + E_2 \frac{\partial W}{\partial E_2}(E_1, E_2, G) \frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)} + G \frac{\partial W}{\partial G}(E_1, E_2, G) \frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)}$$

there. We have to prove that for all $t \in [0, T]$, there exists

$$(5.14) \quad \partial_t \mathcal{E}(q, t) = \lim_{\epsilon \rightarrow 0} \frac{\mathcal{E}(q, t + \epsilon) - \mathcal{E}(q, t)}{\epsilon} = \int_0^L \frac{d}{dt} W(E_1, E_2, G) dx.$$

By Lagrange theorem there exists $\eta \in (0, 1)$ depending on X, t and ϵ , such that

$$\frac{\mathcal{E}(q, t + \epsilon) - \mathcal{E}(q, t)}{\epsilon} = \int_0^L W(g'_1(y_1, t + \eta\epsilon)y'_1P^{-1}, g_2(y_1, t + \eta\epsilon)y_2P, g_2(y_1, t + \eta\epsilon)y'_2P^{-1})dX.$$

For ϵ small, $g'_1(y_1, t + \eta\epsilon)/g'_1(y_1, t)$ and $g_2(y_1, t + \eta\epsilon)/g_2(y_1, t)$ are close to 1 uniformly on $(0, L) \times [0, T]$. Thus, for ϵ small enough, we can apply Lemma 5.5 and we get

$$\begin{aligned} -C_0^W &\leq W(g'_1(y_1, t + \eta\epsilon)y'_1P^{-1}, g_2(y_1, t + \eta\epsilon)y_2P, g_2(y_1, t + \eta\epsilon)y'_2P^{-1}) \leq \\ &\leq C(W(g'_1(y_1, t)y'_1P^{-1}, g_2(y_1, t)y_2P, g_2(y_1, t)y'_2P^{-1}) + C_0^W) \in L^1((0, L)). \end{aligned}$$

Then, by dominated convergence theorem, we can pass to the limit $\epsilon \rightarrow 0$ and we get (5.14).

Proof of (ii). Taking into account the fact that $\frac{\dot{g}'_1}{g'_1}$ and $\frac{\dot{g}_2}{g_2}$ are bounded in $\mathbb{R} \times [0, T]$, assumption (W4, i) yields

$$|\partial_t \mathcal{E}(y, P, t)| \leq 3C_1^W \max \left\{ \left\| \frac{\dot{g}'_1}{g'_1} \right\|_\infty, \left\| \frac{\dot{g}_2}{g_2} \right\|_\infty \right\} \int_\Omega (C_0^W + W(E_1, E_2, G))dX.$$

This implies the existence of $C_0^E \in \mathbb{R}$ and $C_1^E > 0$ such that $|\partial_t \mathcal{E}(q, t)| \leq C_1^E(C_0^E + \mathcal{E}(q, t))$ for all $(q, t) \in \mathcal{D} \times [0, T]$.

Proof of (iii). Let

$$K_i = E_i \frac{\partial W}{\partial E_i} \quad \text{for } i \in \{1, 2, 3\} \quad \text{and} \quad V = \left(\frac{\dot{g}'_1(y_1, t)}{g'_1(y_1, t)}, \frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)}, \frac{\dot{g}_2(y_1, t)}{g_2(y_1, t)} \right)$$

as functions of $(X, t) \in (0, L) \times [0, T]$. Then

$$\frac{d}{dt} W(E) = K \cdot V.$$

Let $E \in \mathbb{R}$ and $q \in \mathcal{D}$ such that $\sup_{t \in [0, T]} \mathcal{E}(q, t)$ and let $t, s \in [0, T]$. By the regularity properties (G) of g , there exists modulus of continuity ω^V and ω^G independent of q such that

$$|V(t) - V(s)| \leq \omega^V(|t - s|) \quad \text{and} \quad \max \left\{ \frac{g'_1(y_1, s)}{g'_1(y_1, t)}, \frac{g_2(y_1, s)}{g_2(y_1, t)} \right\} \leq \omega^V(|t - s|)$$

on $(0, L)$. Moreover $|K| \leq 3C_1^W(W(E) + C_0^W)$ and there exists $M > 0$ independent of q , such that $\|V\|_\infty \leq M$. Then by (W2, i) and (W2, ii),

$$\begin{aligned} |\partial_t \mathcal{E}(q, t) - \partial_t \mathcal{E}(q, s)| &\leq \int_0^L |K(t) \cdot V(t) - K(s) \cdot V(s)|dX \leq \\ &\leq \int_0^L |K(t) - K(s)| |V(s)|dX + \int_0^L |K(t)| |V(t) - V(s)|dX \leq \\ &\leq (L\omega^W(\omega^G(|t - s|))M + 3C_1^W\omega^V(|t - s|)) \int_0^L (W(E(t)) + C_0^W)dX \\ &\leq (L\omega^W(\omega^G(|t - s|))M + 3C_1^W\omega^V(|t - s|)) (\mathcal{E}(q, t) + C_0^W). \end{aligned}$$

This concludes the proof. $\square$

2.4. Verification the properties of the dissipation distance. Notice that (d1) implies that $\mathcal{D}$ is well defined. Property (D1) follows immediately from (d2). We now prove (D2).

PROPOSITION 5.12. Let $P_k \rightarrow P$ and $\hat{P}_k \rightarrow \hat{P}$ in Z . Then $\mathcal{D}(P, \hat{P}) \leq \liminf_{k \rightarrow \infty} \mathcal{D}(P_k, \hat{P}_k)$.

PROOF. Let $P_k, P, \hat{P}_k, \hat{P} \in Z$ such that

$$P_k \rightarrow P \quad \text{and} \quad \hat{P}_k \rightarrow \hat{P} \quad \text{in } Z.$$

By Proposition B.2 it follows that

$$P_k \rightarrow P \quad \text{and} \quad \hat{P}_k \rightarrow \hat{P} \quad \text{in } C^0([0, L]).$$

Now

$$\liminf_{k \rightarrow \infty} \mathcal{D}(P_k, \hat{P}_k) = \liminf_{k \rightarrow \infty} \int_0^L D(P_k, \hat{P}_k)dX = \lim_{k \rightarrow \infty} \inf_{j \geq k} \int_0^L D(P_j, \hat{P}_j)dX \geq \lim_{k \rightarrow \infty} \int_0^L \inf_{j \geq k} D(P_j, \hat{P}_j)dX.$$

By monotone convergence the last limit can be took inside the integral yielding

$$\liminf_{k \rightarrow \infty} \mathcal{D}(P_k, \hat{P}_k) \geq \int_0^L \liminf_{k \rightarrow \infty} D(P_k, \hat{P}_k) dX \geq \int_0^L D(P, \hat{P}) dX.$$

The last inequality holds since D is lower semicontinuous. This concludes the proof. $\square$

2.5. Verification of the compatibility conditions. We now verify (C1) and (C2). In view of Proposition 2.9 we need to verify (PC1) and (PC2). Notice that (PC2) has been established in Proposition 5.11 (iii). Thus we need to verify (PC1) only. A stronger version of (PC1) is the following proposition.

PROPOSITION 5.13. Let $(q_k, t_k), (q, t) \in Q \times [0, T]$ such that $(q_k, t_k) \rightarrow (q, t)$ in $Q \times [0, T]$ and $q_k \in \mathcal{D}$. Then for all $\hat{q} \in Q$,

$$\limsup_{k \rightarrow \infty} (\mathcal{E}(\hat{q}, t_k) + \mathcal{D}(q_k, \hat{q})) \leq \mathcal{E}(\hat{q}, t) + \mathcal{D}(q, \hat{q}).$$

PROOF. Let $(q_k, t_k) \rightarrow (q, t)$ in $Q \times [0, T]$. If $\hat{q} \notin \mathcal{D}$ there is nothing to prove since $\mathcal{E}(\hat{q}, t) = \infty$. Then assume that $\hat{q} \in \mathcal{D}$. By Proposition (5.11) the map $\mathcal{E}(\hat{q}, \cdot)$ is continuous. Then $\mathcal{E}(\hat{q}, t_k) \rightarrow \mathcal{E}(\hat{q}, t)$ and it remains to prove only that

$$\limsup_{k \rightarrow \infty} \mathcal{D}(q_k, \hat{q}) \leq \mathcal{D}(q, \hat{q}).$$

Recall that this is the same as

$$\limsup_{k \rightarrow \infty} \mathcal{D}(P_k, \hat{P}) \leq \mathcal{D}(P, \hat{P}).$$

Since $\hat{q} \in \mathcal{D}$, then (H3) implies that $\hat{P} > \Pi$ a.e. in $(0, L)$, so that $\hat{P} \geq \Pi$ in $[0, L]$. Since also $q_k \in \mathcal{D}$ then, for the same reason, also $P_k \geq \Pi$ in $[0, L]$. Notice also that by B.2, $P_k \rightarrow P$ in $C^0([0, T])$. Then actually

$$\lim_{k \rightarrow \infty} \mathcal{D}(P_k, \hat{P}) = \lim_{k \rightarrow \infty} \int_0^L D(P_k, \hat{P}) dX = \mathcal{D}(P, \hat{P}),$$

where last inequality holds because D is finite and continuous on $[\Pi, \infty) \times [\Pi, \infty)$ by (d3). This concludes the proof. $\square$

2.6. Conclusion of the existence proof. The application of Theorem 2.8 leads to the desired existence result.

THEOREM 5.14. Let $q_0 = (y_0, P_0) \in \mathcal{Q}$ such that $\mathcal{E}(q_0, 0) < \infty$. Then there exists an energetic solution $q = (y, P): [0, T] \rightarrow \mathcal{Q}$ such that $q(0) = q_0$.

2.7. Particular constitutive relations that satisfy the hypotheses of the existence theorem. We now discuss particular constitutive relations that satisfies the hypotheses required to apply Theorem 5.14.

Taking inspiration from the neo-Hookean elastic free energy density (5.3), we consider

$$W(E_1, E_2, G) = \begin{cases} \mathbb{W}(E_1, E_2, G, E_1 E_2) & \text{if } E_1 > 0 \text{ and } E_2 > 0 \\ \infty & \text{otherwise,} \end{cases}$$

with

$$\mathbb{W}(E_1, E_2, G, J) = A \frac{|E_1 - 1|^p + |E_2 - 1|^p + |lG|^p}{J^\alpha} + B|J - 1|^r$$

where $A, B, l > 0, r > 1, 0 < \alpha \leq p - 1$ and $p > 1 + \frac{\alpha}{r}$. We now check that this particular W satisfies (W1), (W2), (W3) and (W4) of previous section. We start with the check of the assumptions (W1) and (W2). Hypotheses (W1) is a consequence of the following lemma.

LEMMA 5.15. Let $w: (0, \infty) \rightarrow \mathbb{R}$ strictly convex and let $g: (0, \infty) \times (0, \infty) \rightarrow \mathbb{R}$ given by $g(r, y) = \frac{r^p}{y^\alpha} + w(y)$ for some $p > 1$ and $0 < \alpha \leq p - 1$. Let also $n \in \mathbb{N}$ and $f: \mathbb{R}^n \times (0, \infty) \rightarrow \mathbb{R}$ given by $f(x, y) = g(|x|, y)$. Then f is strictly convex in $\mathbb{R}^n \times (0, \infty)$.

PROOF. We first prove that g is strictly convex. The hessian matrix of g is given by

$$\nabla^2 g(r, y) = \begin{pmatrix} p(p-1) \frac{r^{p-2}}{y^\alpha} & -p\alpha \frac{r^{p-1}}{y^{\alpha+1}} \\ -p\alpha \frac{r^{p-1}}{y^{\alpha+1}} & \alpha(\alpha+1) \frac{r^p}{y^{\alpha+2}} + w''(y) \end{pmatrix}.$$

Then

$$\begin{aligned}\det \nabla^2 g(r, p) &= p(p-1) \frac{r^{p-2}}{y^\alpha} \left(\alpha(\alpha+1) \frac{r^p}{y^{\alpha+2}} + w''(y) \right) - p^2 \alpha^2 \frac{r^{2p-2}}{y^{\alpha+2}} = \\ &= p\alpha(p-\alpha-1) \frac{r^{2p-2}}{y^{\alpha+2}} + p(p-1) \frac{r^{p-2}}{y^\alpha} w''(y) > 0.\end{aligned}$$

and

$$\operatorname{tr} \nabla^2 g(r, y) = p(p-1) \frac{r^{p-2}}{y^\alpha} + \alpha(\alpha+1) \frac{r^p}{y^{\alpha+2}} + w''(y) > 0.$$

This proves that $\nabla^2 g$ is everywhere positive-definite. This implies that g is strictly convex. Then for any $(x_0, y_0), (x_1, y_1) \in \mathbb{R}^n \times (0, \infty)$ and $0 < \lambda < 1$ it holds

$$\begin{aligned}f((1-\lambda)(x_0, y_0) + \lambda(x_1, y_1)) &= \frac{|(1-\lambda)x_0 + \lambda x_1|^p}{y^\alpha} + w(y) \leq \\ &\leq \frac{[(1-\lambda)|x_0| + \lambda|x_1|]^p}{[(1-\lambda)y_0 + \lambda y_1]^\alpha} + w((1-\lambda)y_0 + \lambda y_1) = \\ &= g((1-\lambda)(|x_0|, y_0) + \lambda(|x_1|, y_1)) < (1-\lambda)g(|x_0|, y_0) + \lambda g(|x_1|, y_1) = \\ &= (1-\lambda)f(x_0, y_0) + \lambda f(x_1, y_1).\end{aligned}$$

This is exactly the thesis. $\square$

We now prove the growth condition (W2) for $q_W = p/(1 + \frac{\alpha}{r}) > 1$. Lemma 5.6 yields for all $\delta > 0$ and $\rho > 1$ that

$$\begin{aligned}\mathbb{W}(E_1, E_2, G, J) &\geq A\rho\delta^{\frac{\rho-1}{\rho}} |E_1 - 1|^{\frac{p}{\rho}} + A\rho\delta^{\frac{\rho-1}{\rho}} |E_2 - 1|^{\frac{p}{\rho}} + A\rho\delta^{\frac{\rho-1}{\rho}} |IG|^{\frac{p}{\rho}} - \\ &\quad - (2A+1)(\rho-1)\delta J^{\frac{\alpha}{\rho-1}} + B|J-1|^r.\end{aligned}$$

Choosing ρ such that $\frac{\alpha}{\rho-1} = r$ we get $\frac{p}{\rho} = q_H$. Moreover choosing $\delta > 0$ sufficiency small we have

$$\inf_{J>0} \left[-(2A+1)(\rho-1)\delta J^r + B|J-1|^r \right] > -\infty.$$

This is implied immediately from the following lemma.

LEMMA 5.16. Let $r \geq 1$. Then there exists $C > 0$ such that $|x-1|^r \geq C|x|^r - 1$ for all $x \in \mathbb{R}$.

PROOF. By the equivalence of the norms there exists $K > 0$ such that

$$(|x-1|^r + 1)^{1/r} \geq K(|x-1| + 1) \geq K|x|.$$

Now the thesis follows immediately. $\square$

A possible choice of the hardening free energy density is

$$H(P, P') = \begin{cases} C(P-1)^q + \frac{D}{P-\Pi} + E|IP'|^q & \text{if } P > \Pi \\ \infty & \text{otherwise.} \end{cases}$$

where $q > 1$, $0 < \Pi < 1$ and $C, D, E > 0$. It trivially satisfies hypotheses (H1), (H2) and (H3). The dissipation potential density can be chosen as

$$R(P, \dot{P}) = \begin{cases} Y \frac{|\dot{P}|}{P} & \text{if } P \geq \Pi \\ \infty & \text{otherwise,} \end{cases}$$

where $Y > 0$ is the yield strength. The corresponding dissipation distance density $D: \mathbb{R} \rightarrow [0, \infty]$ can be calculated by mean of formula (5.8). We find that

$$(5.15) \quad D(P_0, P_1) = \begin{cases} Y |\log P_1 - \log P_0| & \text{if } P_0, P_1 \geq p \\ \infty & \text{otherwise.} \end{cases}$$

Indeed notice that for each $P \in C^1([0, 1], (0, \infty))$ connecting P_0 with P_1 , it holds

$$\int_0^1 R(P(t), \dot{P}(t)) dt \geq Y \int_0^1 \left| \frac{\dot{P}(t)}{P(t)} \right| dt \geq Y \left| \int_0^1 \frac{\dot{P}(t)}{P(t)} dt \right| = Y |\log P_1 - \log P_0|.$$

This implies $D(P_0, P_1) \geq Y |\log P_1 - \log P_0|$. If $P_i < \Pi$ for some $i \in \{0, 1\}$, then every C^1 curve P connecting P_0 with P_1 , has $\int_0^1 R(P(t), \dot{P}(t)) dt = \infty$. If instead $P_0, P_1 \geq \Pi$ then we can take $P(t) = P_0^t P_1^{1-t} \geq \Pi$ and in this case

$$\int_0^1 R(P(t), \dot{P}(t)) dt = Y \int_0^1 \left| \frac{\dot{P}(t)}{P(t)} \right| dt = Y |\log P_1 - \log P_0|.$$

This implies $D(P_0, P_1) = Y |\log P_1 - \log P_0|$. Now it is easy to see that D in (5.15) satisfies the assumptions (d1), (d2) and (d3).

3. Qualitative analysis of the tensile test problem

We consider the following boundary value problem describing the tensile test: the bar is not subject to any distributed load, that is $f = 0$. Moreover the end sections are left free to slide in the transversal direction, this means that $h_2(X, t) = 0$ for $X \in \{0, L\}$ and $t \in [0, T]$ and $\Gamma_D^2 = \emptyset$. The longitudinal position of the end sections is assigned, this means that $\Gamma_D^1 = \{0, L\}$. In particular we consider the boundary conditions $\chi_1(X, t) = \eta(t)X$ for $X \in \{0, L\}$ and $t \in [0, T]$, where $\eta \in C^\infty([0, T], (0, \infty))$ is the imposed elongation. We suppose that $\eta(0) = 1$, $\eta \geq 1$ and η is nondecreasing. To handle the boundary conditions we $g: \mathbb{R}^2 \times [0, T] \rightarrow \mathbb{R}^2$ by $g(y, t) = (\eta(t)y_1, y_2)$ so that $\chi(X, t) = \eta(t)y_1(X, t)$ and $\chi_2(X, t) = y_2(X, t)$. Then the unknown of the problem are $y: (0, L) \times [0, T] \rightarrow \mathbb{R}^2$ and $P: (0, L) \times [0, T] \rightarrow \mathbb{R}$ with the constrain $y_1(X, t) = X$ for $X \in \{0, L\}$.

We will see that a trivial strong solution to this problem is the spatially homogeneous one that is $\bar{y}_1(X, t) = X$ for all $X \in (0, L)$ and $t \in [0, T]$, $\bar{y}_1(\cdot, t)$ and $\bar{P}(\cdot, t)$ constant for all $t \in [0, T]$. However we expect that this trivial solution is unstable when η is sufficiently large, in the sense that it does not satisfy the global stability condition

$$\mathcal{E}(\bar{y}(t), \bar{P}(t), t) \leq \mathcal{E}(\hat{y}, \hat{P}, t) + \mathcal{D}(\bar{P}(t), \hat{P}) \quad \text{for all } (\hat{y}, \hat{P}) \in \mathcal{Q}.$$

We first discuss the stability when the elastic response only is considered. In this case we are able to obtain some explicit results. Then we will show our efforts to extend the analysis also in the elastoplastic case. In this second case we cannot obtain explicit results, however we can gain some insight about the phenomenology of necking.

REMARK 5.17. Most of the stability analysis below is only formal. Some additional effort should be made in order to get fully rigorous results. The mathematical theory that should be employed to reach a full justification of the procedure below is that of bifurcation theory in Banach spaces. Some results regarding this theory and application to instability phenomena in nonlinear elasticity can be found in our review [2].

3.1. Purely elastic model. In the purely elastic model, equations (5.4) reduces to

$$\begin{aligned} \sigma_1' &= 0 \quad \text{in } (0, L) \\ -\sigma_2 + \zeta' &= 0 \quad \text{in } (0, L) \\ \zeta(X) &= 0 \quad \text{for } X \in \{0, L\}. \end{aligned}$$

From the first equation it follows that $\sigma_1 = F$, where F is a constant, representing the axial reaction force needed to impose the boundary conditions on χ_1 . The constitutive relations yields $\sigma_1 = W_1(E_1, E_2, G)$, $\sigma_2 = W_2(E_1, E_2, G)$ and $\zeta = W_3(E_1, E_2, G)$, where $E_1 = \chi_1'$, $E_2 = \chi_2$ and $G = \chi_2'$. We have denoted the partial derivatives of W with respect to its arguments E_1 , E_2 and G with the subscripts 1, 2 and 3 respectively. Inserting these constitutive relations into (5.16), we get

$$\begin{aligned} \frac{d}{dX} W_1(\chi_1', \chi_2, \chi_2') &= 0 \quad \text{in } (0, L) \\ -W_2(\chi_1', \chi_2, \chi_2') + \frac{d}{dX} W_3(\chi_1', \chi_2, \chi_2') &= 0 \quad \text{in } (0, L) \\ W_3(\chi_1'(X), \chi_2(X), \chi_2'(X)) &= 0 \quad \text{for } X \in \{0, L\}. \end{aligned}$$

We first look for homogeneous solutions of the form $\bar{\chi}_1(X, t) = \eta(t)X$, $\bar{\chi}_2(X, t) = \bar{E}_2(t)$ where $\bar{E}_2: (0, T) \rightarrow (0, \infty)$ is the only unknown. Notice that $\bar{E}_1(t) = \bar{\chi}_1(X, t) = \eta(t)$ and $\bar{G}(t) = \bar{\chi}_2'(X, t) = 0$ for all $X \in (0, L)$ and $t \in [0, T]$. To find $\bar{E}_2$ we need only to solve the nonlinear equation

$$W_2(\eta(t), \bar{E}_2(t), 0) = 0.$$

This yields $\bar{E}_2$ as a function of η .

We now turn to the study of the stability of the homogeneous solution $(\bar{\chi}, \bar{P})$. To this aim we should check whether the second variation of the energy becomes indefinite for η large enough. This can be done

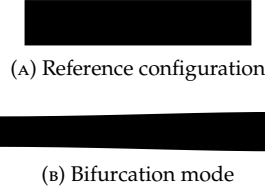


FIGURE 5.1. Bifurcation mode of the elastic bar under traction.

as follows: we insert a perturbation of the homogeneous solution $\chi_1 = \bar{\chi}_1 + v_1$ and $\chi_2 = \bar{\chi}_2 + v_2$ into problem (5.17), then we linearize the resulting problem around the homogeneous solution and we look for nontrivial solutions. The linearized problem reads as follows

$$\begin{aligned} \bar{W}_{11}v_1'' + \bar{W}_{12}v_2' + \bar{W}_{13}v_2'' &= 0 \quad \text{in } (0, L) \\ -\bar{W}_{12}v_1' - \bar{W}_{22}v_2 - \bar{W}_{23}v_2' + \\ + \bar{W}_{13}v_1'' + \bar{W}_{23}v_2' + \bar{W}_{33}v_2'' &= 0 \quad \text{in } (0, L) \end{aligned}$$

together with the boundary conditions

$$\begin{aligned} v_1(X) &= 0 \quad \text{for } X \in \{0, L\} \\ v_2'(X) &= 0 \quad \text{for } X \in \{0, L\}. \end{aligned}$$

We have denoted with a superimposed bar, the evaluation at $(\eta, \bar{E}_2, 0)$. From mechanical ground, it is reasonable to assume that $\sigma_i = W_i$ for $i \in \{1, 2\}$ is even in G and $\zeta = W_3$ is odd in G . Then $\bar{W}_{i3} = 0$ for $i \in \{1, 2\}$. Moreover we assume that $\bar{W}_{33} > 0$, $\bar{W}_{11} > 0$, $\bar{W}_{22} > 0$. Then (5.18) simplifies to

$$\begin{aligned} \bar{W}_{11}v_1'' + \bar{W}_{12}v_2' &= 0 \quad \text{in } (0, L) \\ \bar{W}_{33}v_2'' - \bar{W}_{12}v_1' - \bar{W}_{22}v_2 &= 0 \quad \text{in } (0, L). \end{aligned}$$

The first equation yields

$$(5.20) \quad v_1' = -\frac{\bar{W}_{12}}{\bar{W}_{11}}v_2 + K_1$$

for some constant $K_1 \in \mathbb{R}$. Substituting into the second equation we obtain

$$\bar{W}_{33}v_2'' + \left(\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22} \right) v_2 = \bar{W}_{12}K_1 \quad \text{in } (0, L).$$

Let

$$\lambda = \frac{\frac{(\bar{W}_{12})^2}{\bar{W}_{11}} - \bar{W}_{22}}{\bar{W}_{33}}.$$

Taking into account the boundary condition $v_2'(X) = 0$ for $X \in \{0, L\}$, we find that a nontrivial solution exists if and only if $\lambda = \frac{k^2\pi^2}{L^2}$ for $k \in \{1, 2, \dots\}$ and is of the form

$$v_2(X) = \frac{\bar{W}_{12}K_1}{\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22}} + K_2 \cos(\sqrt{\lambda}X).$$

Here $K_2 \in \mathbb{R}$ is a constant. Inserting in (5.20) we get that

$$v_1(X) = \left(-\frac{\bar{W}_{12}}{\bar{W}_{11}} \frac{\bar{W}_{12}K_1}{\frac{\bar{W}_{12}^2}{\bar{W}_{11}} - \bar{W}_{22}} + 1 \right) K_1 X - \frac{\bar{W}_{12}}{\bar{W}_{11}} \frac{K_2}{\sqrt{\lambda}} \sin(\sqrt{\lambda}X) + K_3$$

with $K_3 \in \mathbb{R}$. Boundary condition imposes that $K_1 = 0$ and $K_3 = 0$. From these considerations we conclude that the homogeneous solution becomes unstable when $\lambda = \pi^2/L^2$ for the first time. The shape of the instability is showed schematically in Figure 5.1

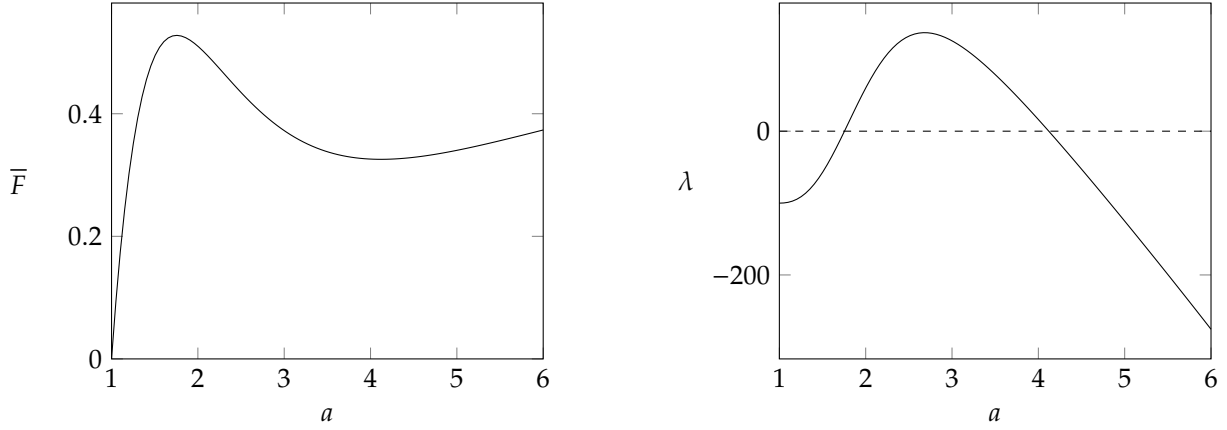


FIGURE 5.2. Graphs of $\bar{F}$ and λ as functions of the imposed elongation η for the homogeneous deformation in the purely elastic model with the choice of the value of the parameters in 5.21.

Now let $\bar{F}(t) = W_1(\eta(t), \bar{E}_2(t), 0)$ be the axial force along the homogeneous solution. We now show that the instability can occur only after that F reaches its maximum. Indeed let us view $\bar{F}$ as a function of η , so that

$$\frac{d\bar{F}}{d\eta} = \bar{W}_{11} + \bar{W}_{12} \frac{d\bar{E}_2}{d\eta}.$$

On the other end we know that $\bar{W}_2 = W_2(\eta, \bar{E}_2, 0) = 0$ so that, by differentiating with respect to η ,

$$\bar{W}_{12} + \bar{W}_{22} \frac{d\bar{E}_2}{d\eta} = 0 \quad \text{that implies} \quad \frac{d\bar{E}_2}{d\eta} = -\frac{\bar{W}_{12}}{\bar{W}_{22}}.$$

Then

$$\frac{d\bar{F}}{d\eta} = \bar{W}_{11} - \frac{\bar{W}_{12}^2}{\bar{W}_{22}} = -\bar{W}_{33} \frac{\bar{W}_{11}}{\bar{W}_{22}} \lambda.$$

Now, at the first stages of deformation the force increases with elongation so $d\bar{F}/d\eta > 0$ that implies $\lambda < 0$ and no instability can occur. It $\bar{F}$ has a (local) maximum then λ become positive and instability occur if it reach the value π^2/L^2 . This clearly happen if the bar is sufficiently long.

Consider now the particular constitutive model described in Section 2.7. If we choose

$$(5.21) \quad A = 1, B = 10^{-4}, p = 2, \alpha = 1 \text{ and } r = 2,$$

we get the graphs of $\bar{F}$ and λ as functions of η reported in Figure 5.2. Notice that F has a maximum and λ became positive so that instability occur if L is sufficiently large.

REMARK 5.18. We notice that the particular choice of the parameters in (5.21) is not mechanically reasonable for metals, and was motivated only by the need to show that the mechanical model that we have developed in this chapter can exhibit instability phenomena. The inaccuracy of (5.21) stems to the fact that, since B is small compared to A , the nearly-compressibility behavior of metals is not correctly reproduced. Moreover (5.21) implies that cross-sections growth in traction, this is clearly wrong for metals.

3.2. Elastoplastic model. We start with the computation of the trivial solution of the form $\bar{y}_1(X, t) = \eta(t)X$, $\bar{y}_2(X, t) = \bar{y}_2(t)$ and $\bar{P}(X, t) = \bar{P}(t)$ for all $X \in (0, L)$ and $t \in [0, T]$. In the elastic range, $\bar{y}_2(t)$ can be found as we showed previously while $\bar{P} = 1$. When the Madel stress (5.5) overcomes the yield strength Y , then $\dot{\bar{P}} > 0$ and the plastic force is everywhere given by $\partial_{\dot{P}} R(P, \dot{P}) = Y/P$. Then the balance of microforces coupled with the balance in the transversal direction allows to compute $\bar{y}_2$ and $\bar{P}$.

To study the stability of the homogeneous solution we should to check if

$$(5.22) \quad \mathcal{E}(\bar{y}(t), \bar{P}(t), t) \leq \mathcal{E}(\hat{y}, \hat{P}, t) + \mathcal{D}(\bar{P}(t), \hat{P}) \quad \text{for all } (\hat{y}, \hat{P}) \in \mathcal{Q}.$$

This since $\mathcal{D}$ is a quasidistance, and in particular $\mathcal{D}(P_0, P_1) \geq 0$ and equality holds if and only if $P_0 = P_1$, the stability condition is equivalent to the requirement that $(\bar{y}(t), \bar{P}(t))$ is a global minimizer of $(y, P) \mapsto \mathcal{E}(y, P, t) + \mathcal{D}(\bar{P}(t), P)$ in $\mathcal{Y} \times \mathcal{X}$. We expect that (5.22) is violated when η is sufficiently large. We were

not able to get conclusive results in this sense, so we now propose an heuristic consideration that suggest that instability in the form of necking is likely to occur if suitable constitutive relation for the hardening free energy density are chosen.

First of all we assume that instability does not occur in the elastic range. This is certainly true in the case of metals since yielding starts when the geometric nonlinearities have still a negligible effect. So fix $t \in [0, T]$ and suppose that the yield point has been already reached at that time. Let

$$\bar{F}(t) = \bar{P}(t)^{-1} W_1(\eta(t) \bar{P}(t)^{-1}, \bar{y}_2(t) \bar{P}(t), \bar{y}_2'(t) \bar{P}(t)^{-1})$$

be the reaction tensile force acting on the ends of the bar along the trivial solution. Consider now the function $J: (0, \infty)^3 \rightarrow \mathbb{R} \cup \{\infty\}$ given by

$$J_t(\rho, y_2, P) = W(\rho P^{-1}, y_2 P, 0) + H(P, 0) + D(\bar{P}(t), P) - \bar{F}(t) \rho.$$

The quantity $J_t(\rho, y_2, P) - J_t(\eta(t), \bar{y}_2(t), \bar{P}(t))$ is the energy per unit length that must be supplied to the system to bring it from the configuration $\bar{y}_1(t), \bar{y}_2(t)$ and $\bar{P}(t)$ to the configuration $y_1(X) = \rho X, y_2(X) = y_2$ and $P(X) = P$ if the ends are now left free to move and the tensile force is kept equal to $\bar{F}(t)$. We now want to study if $(\eta, \bar{y}_2(t), \bar{P}(t))$ is a (local) minimizer of J . If not, this would signify that the trivial solution $(\bar{y}, \bar{P})$ is not the most energetically favored if the same tensile force is applied and the ends are made free to move. Thus assume now the particular constitutive relations proposed in Section 2.7. In this case J_t is not differentiable at $P = \bar{P}(t)$ but $J_t \in C^1(H_t^+)$ and $J_t \in C^1(H_t^-)$, where

$$H_t^+ = \{(\rho, y_2, P) \in (0, \infty)^3 \mid P \geq \bar{P}(t)\} \quad \text{and} \quad H_t^- = \{(\rho, y_2, P) \in (0, \infty)^3 \mid P < \bar{P}(t)\}.$$

We easily find that

$$\nabla J_t(\eta, \bar{y}_2(t), \bar{P}(t)^-) = \left(0, 0, -\frac{2Y}{\bar{P}(t)}\right) \quad \text{and} \quad \nabla J_t(\eta, \bar{y}_2(t), \bar{P}(t)^+) = (0, 0, 0).$$

Then, to study whether $(\eta(t), \bar{y}_2(t), \bar{P}(t))$ is a local minimum depends on the eigenvalues of the hessian matrix

$$h(t) = \nabla^2 J_t(\eta(t), \bar{y}_2(t), \bar{P}(t)^+).$$

Let

$$a(t) = h_{11}(t), \quad b(t) = \begin{pmatrix} h_{22}(t) & h_{23}(t) \\ h_{32}(t) & h_{33}(t) \end{pmatrix} \quad \text{and} \quad c(t) = \begin{pmatrix} h_{21}(t) \\ h_{31}(t) \end{pmatrix}$$

so that

$$h(t) = \begin{pmatrix} a(t) & c(t)^T \\ c(t) & b(t) \end{pmatrix}.$$

LEMMA 5.19. It holds that

$$\frac{dF}{d\eta}(t) := \frac{dF}{dt}(t) \left(\frac{d\eta}{dt}(t) \right)^{-1} = \frac{\det h(t)}{\det b(t)}.$$

PROOF. Notice that

$$F(s) = \frac{\partial J_t}{\partial \rho}(\eta(s), \bar{y}_2(s), \bar{P}(s)) + \bar{F}(t).$$

Then, differentiating with respect to s , computing for $s = t$ and dividing by $\frac{d\eta}{dt}(t)$, we get

$$(5.23) \quad \frac{dF}{d\eta} = \frac{\partial J_t}{\partial \rho \partial \rho}(\eta, \bar{y}_2, \bar{P}) + \frac{\partial J_t}{\partial \rho \partial y_2}(\eta, \bar{y}_2, \bar{P}) \frac{d\bar{y}_2}{d\eta} + \frac{\partial J_t}{\partial \rho \partial P}(\eta, \bar{y}_2, \bar{P}) \frac{d\bar{P}}{d\eta}.$$

On the other hand, the balance in the transversal direction and the balance of microforces can be written respectively as

$$\frac{\partial J_t}{\partial y_2}(\eta, \bar{y}_2, \bar{P}) = 0 \quad \text{and} \quad \frac{\partial J_t}{\partial P}(\eta, \bar{y}_2, \bar{P}) = 0.$$

Differentiating with respect to η we get

$$\begin{cases} \frac{\partial J_t}{\partial \rho \partial y_2}(\eta, \bar{y}_2, \bar{P}) + \frac{\partial J_t}{\partial y_2 \partial y_2}(\eta, \bar{y}_2, \bar{P}) \frac{d\bar{y}_2}{d\eta} + \frac{\partial J_t}{\partial y_2 \partial P}(\eta, \bar{y}_2, \bar{P}) \frac{d\bar{P}}{d\eta} = 0 \\ \frac{\partial J_t}{\partial \rho \partial P}(\eta, \bar{y}_2, \bar{P}) + \frac{\partial J_t}{\partial y_2 \partial P}(\eta, \bar{y}_2, \bar{P}) \frac{d\bar{y}_2}{d\eta} + \frac{\partial J_t}{\partial P \partial P}(\eta, \bar{y}_2, \bar{P}) \frac{d\bar{P}}{d\eta} = 0. \end{cases}$$

Then

$$\left(\frac{\frac{d\bar{P}}{d\eta}}{\frac{dP}{d\eta}} \right) = -b^{-1}c.$$

Substituting into (5.23) we get

$$\frac{dF}{d\eta} = a - c \cdot b^{-1}c.$$

But since

$$\det h = \det b (a - c \cdot b^{-1}c)$$

the thesis follows. $\square$

It is easy to see that the constitutive model described in Section 2.7 implies that $\det b(t) > 0$. Then the sign of $dF/d\eta$ coincides with the sign of $\det h$. This suggest that h is positive definite when F increases and became indefinite if F reaches a maximum. This is indeed confirmed by the numerical computation of the trivial solution.

We notice that the constitutive equation of the hardening free energy proposed in Section 2.7 gives a monotone relation between F and η if C is large enough, see the dashed graph in Figure 5.3. According to the considerations above we expect that hardening does not occur in this case. When $C > 0$ is sufficiency reduces a non monotone relation is obtained, see the solid graph in Figure 5.3.

REMARK 5.20. A possible refinement of the hardening free energy function is

$$(5.24) \quad H(P, P') = \begin{cases} C(P^q + \frac{q}{\delta}e^{-\delta(P-1)}) + \frac{D}{P-\Pi} + E|P'|^q & \text{if } P > p \\ \infty & \text{otherwise,} \end{cases}$$

where $C, D, E, \delta > 0$ and $q > 1$. This constitutive relation still satisfies the assumptions (H1), (H2) and (H3). The expression (5.24) is obtained as a modification of the Armstrong–Frederick kinematical hardening model. According to the Armstrong–Frederick model, the backstress β (that in our model correspond to the conservative microforce $\beta = \frac{\partial H}{\partial P}$) is required to satisfy the condition

$$(5.25) \quad \dot{\beta} = \delta(C\dot{P} - |\dot{P}|\beta)$$

for some $\delta, C > 0$, see [40]. In tension $\dot{P} \geq 0$ and (5.25) becomes

$$\delta\dot{P} = \frac{\dot{\beta}}{C - \beta} = -\frac{d}{dt} \log(C - \beta).$$

This suggest the relation

$$(5.26) \quad \log \frac{C - \beta}{C} = -\delta(P - 1) \quad \text{so that} \quad \beta = C(1 - e^{-\delta(P-1)}).$$

The main feature of this relation is that β is bounded from above by C and $P \rightarrow C^-$ as $P \rightarrow \infty$. Now (5.26) corresponds to the energy density

$$H = C \left(P + \frac{1}{\delta} e^{-\delta(P-1)} \right).$$

Formula (5.24) is a modification of this as the linear growth in P has been substituted with a superlinear growth in order to satisfy (H2). The superlinear growth makes β unbounded as $\beta \rightarrow \infty$, but the main features of the original Armstrong–Frederick model are preserved.

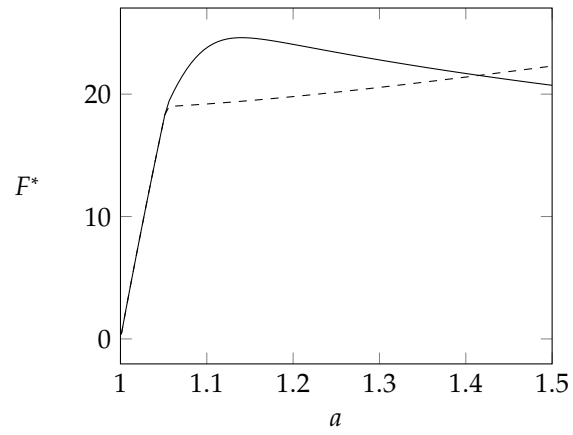


FIGURE 5.3. Graphs of $\bar{F}$ as functions of the elongation η along the homogeneous solution in the plastic model.

Part 2

A model describing pressure sensitive plastic materials

CHAPTER 6

Introduction

Small deformation model

1. Formulation

Scrivere una introduzione

1.1. Kinematics. Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic displacement is a map $u: \Omega \rightarrow \mathbb{R}^d$. We set $E = \text{sym grad } u$ and $M = \text{grad}^2 u$. We subdivide $\partial\Omega$ in two disjoint parts Γ_D and Γ_N . On Γ_D we impose the kinematical constrain $u = \bar{u}$. The configuration of the microscopic structure is described by the plastic distortion $H_p: \Omega \rightarrow \text{Lin}$. The skew symmetric part $W_p = \text{skew } H_p$ describes the rotation of the microscopic structure and for simplicity we assume that it vanish. The symmetric part is the plastic strain $E_p = \text{sym } H_p: \Omega \rightarrow \text{Sym}$ which describes the deformation of microscopic structure. We can regard E_p as the elastically relaxed strain of the body so we define the elastic strain as $E_e = E - E_p$. The volumetric deformation of the microscopic structure is described by $\lambda_p = \text{tr } E_p$ while the deviatoric deformation by $U_p = \text{dev } E_p$. We call $M_p = \text{grad } U_p$ and we set $\mathbb{U}_p = (U_p, l M_p) \in \text{DevPla}$ where $l > 0$ is an internal length scale. Here $\text{DevPla} = \text{Sym} \otimes (\text{Sym} \otimes \mathbb{R})$ is the vector space of the deviatoric plastic strains and their gradients. In what follows we will denote with $\circ$ the natural scalar product on DevPla . We assume that the material is not dilatant (this is a simplification and should be generalized to describe the dilatant behavior observed in most materials exhibiting a pressure dependent yield strength) so we add the constrain $\lambda_p = 0$. To impose this constrain we add the Lagrange multiplier $p: \Omega \rightarrow \mathbb{R}$ with the meaning of a plastic pressure. The constrain can be written in weak form as

$$\int_{\Omega} \lambda_p \tilde{q} dx = 0 \quad \text{for all } \tilde{q}: \Omega \rightarrow \mathbb{R}.$$

1.2. Statics. We denote with $\tilde{v}$, $\tilde{V}_p$ and $\tilde{\mu}_p$ the virtual rates of u , U_p and λ respectively. The virtual rate of E_e is $\tilde{L}_e = \text{grad } \tilde{v} - \tilde{\rho} - \frac{\tilde{\mu}}{d} I$. Due to the presence of the Dirichlet condition we require that $\tilde{v}$ vanish on Γ_D . Instead we do not require that $\tilde{\mu}_p = 0$ because the constrain $\lambda_p = 0$ is imposed by mean of the Lagrange multiplier p . To describe the coupling between the macroscopic deformation and the microscopic configuration we choose an internal generalized force of the form

$$\tilde{J} = \int_{\Omega} \left(T : \tilde{L} + K : \text{grad}^2 \tilde{v} \right) dx + \int_{\Omega} T_e : \tilde{L}_e dx + \int_{\Omega} \left(T_p : \tilde{\rho} + K_p : \text{grad } \tilde{\rho} \right) dx.$$

The first integral is related to the macroscopic response. Here $T \in \text{Sym}$ is the macrostress and $K \in \text{Lin} \otimes \mathbb{R}$ is the macrohyperstress. The presence of the second order of $\tilde{v}$ is due to the need of a sufficient regularization in view of the existence theorem. The second integral describe the coupling between the macroscopic and the microscopic response and $T_e \in \text{Sym}$ is the elastic stress. The third integral is associated with the microscopic response. Here $T_p \in \text{DevSym}$ is the microforce and $K_p \in \text{DevSym} \otimes \mathbb{R}$ is the microstress. In Figure 7.1 we report a picture that represents schematically the meaning of the internal generalized force that we have chosen. We define $\tilde{\mathbb{V}}_p = (\tilde{V}_p, l \text{grad } \tilde{V}_p)$ and $\mathbb{T}_p = (T_p, l^{-1} K_p)$ so that

$$T_p : \tilde{V}_p + K_p : \text{grad } \tilde{V}_p = \mathbb{T}_p \circ \tilde{\mathbb{V}}_p.$$

The action of the plastic pressure is obtained by considering the reaction virtual power

$$\tilde{\mathcal{R}} = - \int_{\Omega} p \tilde{\mu}_p dx.$$

We notice that this power vanish on virtual rates that satisfy the constrain that is $\tilde{\mu}_p = 0$. The external actions of the volume force f on Ω and the traction h on Γ_N are included choosing the external generalized forces

$$\tilde{\mathcal{X}} = \int_{\Omega} f \cdot \tilde{v} dx + \int_{\Gamma_N} h \cdot \tilde{v} dx.$$

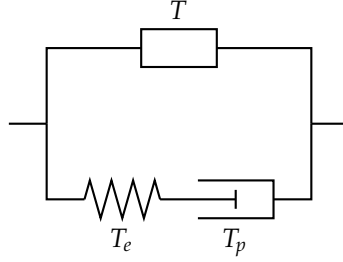


FIGURE 7.1. Rheological model describing the choice of the internal virtual power $\tilde{W}$. The box in upper branch represents the (generic) macroscopic response. In the lower branch, the spring represents the elastic response and the piston represents the microscopic response. We notice that the microscopic response besides the usual dissipative plastic microforce force can include conservative microforces to model hardening. A more detailed discussion on constitutive relations is in Section 1.3.

For simplicity we assume that no external torque is applied to the body and that there is no external action on the microscopic configuration. These external actions could be added to describe more accurately the boundary conditions imposed on the body.

LEMMA 7.1. The local form of the equilibrium condition is

$$\begin{aligned} \operatorname{div}(T + T_e - \operatorname{div} K) + f &= 0 \quad \text{in } \Omega \\ (T + T_e - \operatorname{div} K)n - \operatorname{div}_P((Kn)P) &= h \quad \text{in } \Gamma_N \cap \partial^{d-1}\Omega \\ \{(Kn)\tau\} &= 0 \quad \text{in } \partial^{d-2}\Omega \\ -\operatorname{dev} T_e + T_p - \operatorname{div} K_p &= 0 \quad \text{in } \Omega \\ p &= -\frac{1}{d} \operatorname{tr} T_e \quad \text{in } \Omega \\ K_p n &= 0 \quad \text{in } \partial^{d-1}\Omega. \end{aligned}$$

PROOF. By integration by parts we can write

$$\begin{aligned} \int_{\Omega} \left(T : \operatorname{grad} \tilde{v} + K : \operatorname{grad}^2 \tilde{v} \right) dx &= \\ &= - \int_{\Omega} \operatorname{div}(T - \operatorname{div} K) \cdot \tilde{v} dx + \int_{\partial^{d-1}\Omega} [(T - \operatorname{div} K)n \cdot \tilde{v} + Kn : \operatorname{grad} \tilde{v}] dx. \end{aligned}$$

Now denoting with $P = I - n \otimes n$ the projector on $\partial^{d-1}\Omega$ we have

$$Kn : \operatorname{grad} \tilde{v} = Kn : \left(\frac{\partial \tilde{v}}{\partial n} \otimes n + \operatorname{grad}_P \tilde{v} \right) = (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} + Kn : \operatorname{grad}_P \tilde{v}$$

and

$$Kn : \operatorname{grad}_P \tilde{v} = (Kn)P : \operatorname{grad}_P \tilde{v} = \operatorname{div}_P(P(Kn)^T \tilde{v}) - \operatorname{div}_P((Kn)P) \cdot \tilde{v}.$$

But by integration by parts

$$\int_{\partial^{d-1}\Omega} \operatorname{div}_P(P(Kn)^T \tilde{v}) dx = \int_{\partial^{d-2}\Omega} \{P(Kn)^T \tilde{v} \cdot \tau\} dx = \int_{\partial^{d-2}\Omega} \{(Kn)\tau\} \cdot \tilde{v} dx.$$

Here $\{\cdot\}$ denotes sum of its argument evaluated on both sides of $\partial^{d-2}\Omega$ and τ denotes the unit normal vector to $\partial^{d-2}\Omega$ and outer tangent to $\partial^{d-1}\Omega$. We have obtained

$$\begin{aligned} \int_{\Omega} \left(T : \operatorname{grad} \tilde{v} + K : \operatorname{grad}^2 \tilde{v} \right) dx &= \\ &= - \int_{\Omega} \operatorname{div}(T - \operatorname{div} K) \cdot \tilde{v} dx + \int_{\partial^{d-1}\Omega} [(T - \operatorname{div} K)n - \operatorname{div}_P((Kn)P)] \cdot \tilde{v} dx + \\ &\quad + \int_{\partial^{d-1}\Omega} (Kn)n \cdot \frac{\partial \tilde{v}}{\partial n} dx + \int_{\partial^{d-2}\Omega} \{(Kn)\tau\} \cdot \tilde{v} dx. \end{aligned}$$

Moreover the remaining terms of the virtual power can be written as

$$- \int_{\Omega} \operatorname{div} T_e : \tilde{v} dx + \int_{\partial^{d-1}\Omega} T_e n \cdot \tilde{v} dx + \int_{\Omega} (T_p - \operatorname{dev} T_e - \operatorname{div} K_p) : \tilde{V}_p dx + \\ + \int_{\partial^{d-1}\Omega} K_p n : \tilde{V}_p dx - \int_{\Omega} \frac{1}{d} \operatorname{tr} T_e \tilde{\mu} dx.$$

Now the results follows from the fundamental lemma of the calculus of variations. $\square$

1.3. Constitutive relations. We assume a constitutive relation for the free energy density of the form

$$\psi = \frac{1}{2} \mathfrak{C} E_e : E_e + \frac{1}{2} \mathfrak{M} M : M + \frac{1}{2} \mathfrak{B} U_p \circ U_p.$$

with $\mathfrak{C}$, $\mathfrak{M}$ and $\mathfrak{B}$ a symmetric and positive definite linear operators on Sym , $\operatorname{Lin} \otimes \mathbb{R}$ and DevPla respectively. Moreover we choose a dissipation potential density of the form

$$\phi = \frac{1}{2} \mathfrak{D} \dot{E} : \dot{E} + \frac{1}{2} \mathfrak{E} \dot{M} : \dot{M} + \frac{1}{2} \mathfrak{F} \dot{U}_p \circ \dot{U}_p + Y(p) G(\dot{U}_p).$$

Here $\mathfrak{D}$, $\mathfrak{E}$, $\mathfrak{F}$ are symmetric and positive definite linear operators on Sym , $\operatorname{Lin} \otimes \mathbb{R}$ and DevPla respectively. Moreover $G : \operatorname{DevSym} \rightarrow [0, \infty]$ is the convex and subdifferentiable plastic dissipation potential satisfying $G(0) = 0$ and $Y : \mathbb{R} \rightarrow [0, \infty)$ is the pressure dependent yield strength. Consistently with these choices of the free energy density and of the dissipation potential density we impose the following constitutive relations. The macroscopic response is described by

$$T = \mathfrak{D} \dot{E} \quad \text{and} \quad K = \mathfrak{M} + \mathfrak{E} \dot{M}.$$

with $\mathfrak{D}$ and $\mathfrak{E}$ symmetric and positive definite linear operators on Sym and $\operatorname{Lin} \otimes \mathbb{R}$ respectively. The elastic response is given by

$$T_e = \mathfrak{C} E_e$$

and the viscoplastic response by

$$T_p \in \mathfrak{B} U_p + \mathfrak{F} \dot{U}_p + Y(p) \partial_{\dot{U}_p} G(\dot{U}_p)$$

where $\mathfrak{F}$ is a symmetric positive definite tensor on DevPla , $G : \operatorname{DevSym} \rightarrow \mathbb{R}$ is the plastic dissipation potential satisfying $\partial_{\dot{U}_p} G(\dot{U}_p) : \dot{U}_p \geq 0$ and $Y : \mathbb{R} \rightarrow [0, \infty)$ is the pressure dependent yield strength.

REMARK 7.2. In the above expression we are viewing $Y(p) \partial_{\dot{U}_p} G(\dot{U}_p) \in \operatorname{DevSym}$ as an element of DevPla by considering DevSym as a subspace of DevPla .

1.4. Refinements of the model. To include the dilatancy in the model we could do the following. We put the unilateral constrain $\lambda_p \geq F(U_p)$, meaning that when the microscopic structure suffer shear distortions it must suffer also an expansion. This constrain is still imposed through the Lagrange multiplier p representing the microscopic pressure. To make the expansion irreversible (once the microscopic structure expanded it can not be compacted again) we add a microscopic force ξ conjugate to $\tilde{\mu}_p$ in the internal virtual power and in the dissipation potential density we add $\iota_{[0, \infty)}(\dot{\xi})$. In this way ξ opposes to the compaction of the microscopic structure.

2. Mathematical analysis

In what follows we write simply U and λ in place of U_p and λ_p respectively. Moreover for simplicity we assume $\Gamma_D = \partial\Omega$ and $\bar{u} = 0$ since the extension to the general case is trivial.

2.1. Hypotheses. To get the global existence result we assume the following hypotheses.

- (S) (a) $\Omega \subset \mathbb{R}^d$ with $d \in \{2, 3\}$ is open, bounded and Lipschitz
- (b) Γ_D and Γ_N are relatively open disjoint subsets of $\partial\Omega$
- (Y) (a) $Y \in C^{0,1}(\mathbb{R}, [0, \infty))$
- (G) (a) $G \in C^{1,1}(\operatorname{DevSym}, \mathbb{R})$ is convex
- (b) $\partial G \in L^\infty(\operatorname{DevSym}, \operatorname{DevPla})$
- (c) $\partial G(\dot{U}) : \dot{U} \geq 0$ for all $\dot{U} \in \operatorname{DevSym}$
- (L) (a) $f \in C^1([0, T], L^2(\Omega)^d)$
- (b) $h \in C^1([0, T], L^2(\Gamma_N)^d)$
- (E) (a) $\mathfrak{C}$ and $\mathfrak{D}$ are in $\operatorname{Sym} \otimes \operatorname{Sym}$, $\mathfrak{B}$ and $\mathfrak{F}$ are in $\operatorname{DevPla} \otimes \operatorname{DevPla}$, $\mathfrak{M}$ and $\mathfrak{E}$ are in $(\operatorname{Lin} \otimes \mathbb{R}) \otimes (\operatorname{Lin} \otimes \mathbb{R})$

- (b) there exist $\gamma > 0$ and $\delta > 0$ such that $\mathfrak{C}E : E \geq \gamma|E|^2$ and $\mathfrak{D}E : E \geq \delta|E|^2$ for all $E \in \text{Sym}$, there exist $\beta > 0$ and ϕ such that $\mathfrak{B}\mathbb{U} \circ \mathbb{U} \geq \beta|\mathbb{U}|^2$ and $\mathfrak{F}\mathbb{U} \circ \mathbb{U} \geq \phi|\mathbb{U}|^2$, there exists $\alpha > 0$ and $\epsilon > 0$ such that $\mathfrak{A}M : M \geq \alpha|\text{sym } M|^2$ and $\mathfrak{C}M : M \geq \epsilon|\text{sym } M|^2$ for all $M \in \text{Lin} \otimes \mathbb{R}$ where sym denote the symmetrization of the first two indices

2.2. Function spaces. We now introduce the function spaces that we use to formulate the problem. Let

$$W = \left\{ z = (u, U, \lambda) \left| \begin{array}{l} u \in H^2(\Omega)^d \text{ and } u = 0 \text{ on } \Gamma_D \\ U \in H^1(\Omega, \text{DevSym}) \\ \lambda \in L^2(\Omega) \end{array} \right. \right\} \quad \text{and} \quad \mathcal{Q} = \{ p \in L^2(\Omega) \}.$$

The rates of $z = (u, U, \lambda) \in W$ will be denoted by $\dot{z} = (\dot{u}, \dot{U}, \dot{\lambda}) \in W$. The variations of $\dot{z} = (\dot{u}, \dot{\xi}, \dot{\lambda}) \in W$ and $p \in \mathcal{Q}$ will be denoted as $w = (v, V, \mu) \in W$ and $q \in \mathcal{Q}$. The test functions will be denoted as $\tilde{w} = (\tilde{v}, \tilde{V}, \tilde{\mu}) \in W$.

2.3. Formulation of the problem. We define $\mathcal{F} : W \times \mathcal{Q} \times Z \times [0, T] \rightarrow Z' \times \mathcal{Q}'$ by

$$\begin{aligned} \langle \mathcal{F}(p, y, \dot{y}, t), (\tilde{z}, \tilde{q}) \rangle &= \int_{\Omega} \mathfrak{C} \left(Eu - U - \frac{\lambda}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ &+ \int_{\Omega} (\mathfrak{A} \text{grad}^2 u + \mathfrak{C} \text{grad}^2 \dot{u}) : \text{grad}^2 \tilde{v} dx + \int_{\Omega} \mathfrak{D} E \dot{u} : E \tilde{v} dx + \\ &+ \int_{\Omega} [(\mathfrak{B}\mathbb{U} + \mathfrak{F}\dot{\mathbb{U}}) \circ \tilde{\mathbb{V}} + Y(p) \partial G(\dot{U}) : \tilde{V}] dx - \\ &\quad - \int_{\Omega} p \tilde{\mu} dx - \int_{\Omega} \lambda \tilde{q} dx - \\ &\quad - \int_{\Omega} f(t) \cdot \tilde{v} dx - \int_{\Gamma_N} h(t) \cdot \tilde{v} dx. \end{aligned}$$

LEMMA 7.3. The map $\mathcal{F}$ is well defined.

PROOF. The linear terms are trivially well defined so we need to check only

$$\int_{\Omega} Y(p) \partial G(\dot{U}) : \tilde{V} dx.$$

But

$$\int_{\Omega} |Y(p)| |\partial G(\dot{U})| |\tilde{V}| dx \leq \|\partial G\|_{\infty} \int_{\Omega} (Y(0) + \text{Lip}(Y)|p|) |\tilde{V}| dx \lesssim (1 + \|p\|_2) \|\tilde{V}\|_{1,2}.$$

This concludes the proof. $\square$

Now we can formulate the evolution problem as follows.

PROBLEM 7.4. Find $z \in H^1((0, T), W)$ and $p \in L^2([0, T], \mathcal{Q})$ such that $z(0) = 0$ and

$$\mathcal{F}(p(t), y(t), \dot{y}(t), t) = 0$$

for a.e. $t \in (0, T)$.

2.4. Global existence. This section is devoted to prove the following existence of a global in time solution to Problem 7.4.

THEOREM 7.5. Assume the hypotheses of Section 2.1. Then there exists $z \in H^1((0, T), W)$ and $p \in C^0([0, T], \mathcal{Q})$ solution to Problem 7.4.

The proof is based on the Rothe method so we first prove the existence to a discrete in time version of the problem and then we pass to the vanishing time step limit.

Existence of the discrete in time solution. Let $N \in \mathbb{N}$. We define $\tau_N = N^{-1}T$ and $t_n^N = n\tau_N$ for $n \in \{0, \dots, N\}$. We consider the discrete in time version of Problem 7.4.

PROBLEM 7.6 (Discrete problem). Let $y_0^N = 0 \in \mathcal{X}$. Find $y_n^N \in \mathcal{X}$ and $p_n^N \in \mathcal{Q}$ for $n \in \{1, \dots, N\}$ such that

$$(7.1) \quad \mathcal{F} \left(\frac{y_n^N - y_{n-1}^N}{\tau_N}, p_n^N, y_n^N, t_n^N \right) = 0$$

for all $n \in \{1, \dots, N\}$.

We first prove the existence of a solution to Problem 7.6. Reasoning by induction we reduce to find $y_n^N \in \mathcal{X}$ and $p_n^N \in \mathcal{Q}$ that satisfy (7.1) given $y_{n-1}^N \in \mathcal{X}$. We immediately observe that in view of the constrain $\lambda_n^N = 0$ for all n . To simplify the notation we call $y = y_{n-1}^N$, $\dot{y} = \tau^{-1}(y_n^N - y_{n-1}^N)$ and $t = t_n^N$. Then $y_n^N = y + \tau \dot{y}$. It is enough to prove that there exists $\dot{y} \in \mathcal{X}$ and $p \in \mathcal{Q}$ such that

$$(7.2) \quad \mathcal{F}(\dot{y}, p, y + h\dot{y}, t) = 0.$$

We define the operator $\mathcal{A}: \mathcal{Q} \times \mathcal{X} \rightarrow \mathcal{X}'$ by

$$\begin{aligned} \langle \mathcal{A}(p, \dot{y}), \tilde{z} \rangle &= \int_{\Omega} \tau \mathfrak{C} \left(E\dot{u} - \dot{U} - \frac{\dot{\lambda}}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \int_{\Omega} \mathfrak{D} E\dot{u} : E\tilde{v} dx + \\ &+ \int_{\Omega} (\mathfrak{C} + \tau \mathfrak{A}) \text{grad}^2 \dot{u} : \text{grad}^2 \tilde{v} dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B}) \dot{U} \circ \tilde{V} + Y(p) \partial G(\dot{U}) : \tilde{V}] dx \end{aligned}$$

the linear operator $\mathcal{B}: \mathcal{Q} \rightarrow \mathcal{X}'$ by

$$\langle \mathcal{B}p, \tilde{z} \rangle = - \int_{\Omega} p \tilde{\mu} dx$$

and the operator $\mathcal{C}: \mathcal{X} \rightarrow \mathcal{Q}'$ by

$$\langle \mathcal{C}(\dot{y}), \tilde{q} \rangle = \int_{\Omega} (\lambda + \tau \dot{\lambda}) \tilde{q} dx = 0.$$

Moreover we introduce the linear form $l \in \mathcal{X}'$

$$\begin{aligned} \langle l, \tilde{z} \rangle &= \int_{\Omega} f(t) \cdot \tilde{v} dx + \int_{\Gamma_N} h(t) \cdot \tilde{v} dx - \\ &- \int_{\Omega} \mathfrak{C} \left(Eu - U - \frac{\lambda}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx - \\ &- \int_{\Omega} \mathfrak{A} \text{grad}^2 u : \text{grad}^2 \tilde{v} dx - \int_{\Omega} \mathfrak{B} U \circ \tilde{V} dx. \end{aligned}$$

Now equation (7.2) can be written as

$$\begin{cases} \mathcal{A}(p, \dot{y}) + \mathcal{B}p = l(t) \\ \mathcal{C}(\dot{y}) = 0. \end{cases}$$

We have already noticed that $\lambda = 0$ and $\dot{\lambda} = 0$. We call

$$\mathcal{N} = \left\{ \dot{y} = (\dot{u}, \dot{U}, 0) \mid \begin{array}{l} \dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega, \text{DevSym}) \end{array} \right\}.$$

Now we study the first equation in (7.2). We first study the solvability of that equation in $\dot{y} \in \mathcal{N}$ given $p \in \mathcal{Q}$. If we test the first equation with $\tilde{z} \in \mathcal{N}$ we get

$$\langle \mathcal{A}(p, \dot{y}), \tilde{z} \rangle = \langle l, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \mathcal{N}.$$

Now we define the space

$$\mathcal{X} = \left\{ \xi = (\dot{u}, \dot{U}) \mid \begin{array}{l} \dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega, \text{DevSym}) \end{array} \right\}$$

and we can consider the natural bijection $\mathcal{J}$ from $\mathcal{X}$ to $\mathcal{N}$. In particular if $\xi = (\dot{u}, \dot{U})$ we define

$$\mathcal{J}(\xi) = (\dot{u}, \dot{U}, 0).$$

Now we introduce the operator $\mathcal{T}_p$ from $\mathcal{X}$ to $\mathcal{X}'$ as follows. Given $\xi = (\dot{u}, \dot{U})$ and $\tilde{\xi} = (\tilde{v}, \tilde{V})$ in $\mathcal{X}$ we set

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle &= \langle \mathcal{A}(p, \mathcal{J}(\xi)), \mathcal{J}(\tilde{\xi}) \rangle = \int_{\Omega} \tau \mathfrak{C} (E\dot{u} - \dot{U}) : (E\tilde{v} - \tilde{V}) dx + \int_{\Omega} \mathfrak{D} E\dot{u} : E\tilde{v} dx + \\ &+ \int_{\Omega} (\mathfrak{C} + \tau \mathfrak{A}) \text{grad}^2 \dot{u} : \text{grad}^2 \tilde{v} dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B}) \dot{U} \circ \tilde{V} + Y(p) \partial G(\dot{U}) : \tilde{V}] dx. \end{aligned}$$

Now for each $p \in \mathcal{Q}$ we are reduced to find $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle = \langle l, \mathcal{J}(\tilde{\xi}) \rangle \quad \text{for all } \tilde{\xi} \in \mathcal{X}.$$

We want to apply the Browder–Minty theorem (Theorem B.20). We need to establish the boundedness, continuity, coercivity and monotonicity of $\mathcal{T}_p$. This done in the following lemmas.

LEMMA 7.7. The operator $\mathcal{T}_p: \mathcal{X} \rightarrow \mathcal{X}'$ is bounded and continuous.

PROOF. We have

$$|\langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle| \lesssim (\|E\dot{u}\|_2 + \|\dot{U}\|_2) (\|E\tilde{v}\|_2 + \|\tilde{V}\|_2) + \|\text{grad}^2 \dot{u}\|_2 \|\text{grad}^2 \tilde{v}\|_2 + \|\dot{U}\|_2 \|\tilde{V}\|_2 + (1 + \|p\|_2) \|\tilde{V}\|_2.$$

and this proves that $\mathcal{T}_p$ is bounded. Now it is enough to prove the continuity of the nonlinear term only. Let $\xi_n \rightarrow \xi$ in $\mathcal{X}$. Then $\dot{U}_n \rightarrow \dot{U}$ in L^2 . We have

$$\left| \int_{\Omega} Y(p)(\partial G(\dot{U}_n) - \partial G(\dot{U})) \circ \tilde{V} dx \right| \leq \|Y(p)(\partial G(\dot{U}_n) - \partial G(\dot{U}))\|_2 \|\tilde{V}\|_2$$

and

$$\|Y(p)(\partial G(\dot{U}_n) - \partial G(\dot{U}))\|_2^2 = \int_{\Omega} Y(p)^2 (\partial G(\dot{U}_n) - \partial G(\dot{U}))^2 dx \rightarrow 0.$$

Indeed for every subsequence there exists a further subsequence such that $\dot{U}_n \rightarrow \dot{U}$ in L^2 and we can apply the dominated convergence theorem to pass to the limit. $\square$

LEMMA 7.8. The operator $\mathcal{T}_p: \mathcal{X} \rightarrow \mathcal{X}'$ is coercive uniformly in p .

PROOF. For any $\xi \in \mathcal{X}$ we have

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \xi \rangle &= \int_{\Omega} \tau \mathfrak{C} (E\dot{u} - \dot{U}) : (E\dot{u} - \dot{U}) dx + \int_{\Omega} \mathfrak{D} E\dot{u} : E\dot{u} dx + \\ &+ \int_{\Omega} (\mathfrak{C} + \tau \mathfrak{A}) \text{grad}^2 \dot{u} : \text{grad}^2 \dot{u} dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B}) \dot{U} \circ \dot{U} + Y(p) \partial G(\dot{U}) \circ \dot{U}] dx. \end{aligned}$$

Using the hypotheses (E, b) and (G, b) and using that $\dot{U} : I = 0$ since $\dot{U} \in \text{Dev}$ we have

$$\langle \mathcal{T}_p(\xi), \xi \rangle \geq \tau \gamma \|E\dot{u} - \dot{U}\|_2^2 + \tau \alpha \|\text{grad} E\dot{u}\|_2^2 + (\phi + \tau \beta) \|\dot{U}\|_2^2.$$

By the Young inequality

$$\|E\dot{u} - \dot{U}\|_2^2 \geq (1 - \theta) \|E\dot{u}\|_2^2 + \left(1 - \frac{1}{\theta}\right) \|\dot{U}\|_2^2$$

for any $\theta > 0$. Then

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \xi \rangle &\geq (1 - \theta) \tau \gamma \|E\dot{u}\|_2^2 + \tau \alpha \|\text{grad} E\dot{u}\|_2^2 + \\ &+ \left(1 - \frac{1}{\theta}\right) \tau \gamma \|\dot{U}\|_2^2 + (\phi + \tau \beta) \|\dot{U}\|_2^2. \end{aligned}$$

We choose $\theta < 1$ sufficiently close to 1 such there exists a constant $C_1 > 0$ with the property that

$$\left(1 - \frac{1}{\theta}\right) \tau \gamma \|\dot{U}\|_2^2 + (\phi + \tau \beta) \|\dot{U}\|_2^2 \geq C_1 \|\dot{U}\|_2^2 \geq C_2 \|\dot{U}\|_{1,2}^2.$$

The thesis follows by application of Korn inequality. $\square$

LEMMA 7.9. The operator $\mathcal{T}_p: \mathcal{X} \rightarrow \mathcal{X}'$ is strongly monotone uniformly in p , i.e., there exists $M > 0$ such that

$$\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle \geq M \|\xi_2 - \xi_1\|_{\mathcal{X}}^2$$

for all $p \in \mathcal{Q}$ and ξ_1, ξ_2 in $\mathcal{X}$.

PROOF. For all ξ_1 and ξ_2 in $\mathcal{X}$ we have

$$\begin{aligned} &\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle = \\ &= \int_{\Omega} \tau \mathfrak{C} (E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)) : (E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)) dx + \\ &\quad + \int_{\Omega} \mathfrak{D} (E(\dot{u}_2 - \dot{u}_1)) : (E(\dot{u}_2 - \dot{u}_1)) dx + \\ &\quad + \int_{\Omega} (\mathfrak{C} + \tau \mathfrak{A}) (\text{grad}^2 \dot{u}_2 - \text{grad}^2 \dot{u}_1) : (\text{grad}^2 \dot{u}_2 - \text{grad}^2 \dot{u}_1) dx + \\ &+ \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B}) (\dot{U}_2 - \dot{U}_1) \circ (\dot{U}_2 - \dot{U}_1) + Y(p) (\partial G(\dot{U}_2) - \partial G(\dot{U}_1)) : (\dot{U}_2 - \dot{U}_1)] dx \geq \\ &\geq (\delta + \tau \gamma) \|E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)\|_2^2 + (\phi + \tau \beta) \|\dot{U}_2 - \dot{U}_1\|_2^2. \end{aligned}$$

We have used that since G is convex then ∂G is monotone. Reasoning as in the proof of Lemma 7.8 we have

$$\begin{aligned} \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle &\geq (1 - \theta)(\delta + \tau\gamma) \|E(\dot{u}_2 - \dot{u}_1)\|_2^2 + (\alpha + \tau\epsilon) \|\text{grad } E(\dot{u}_2 - \dot{u}_1)\|_2^2 + \\ &\quad + \left(1 - \frac{1}{\theta}\right) (\delta + \tau\gamma) \|\dot{U}_2 - \dot{U}_1\|_2^2 + \\ &\quad + (\phi + \tau\beta) \|\dot{U}_2 - \dot{U}_1\|_2^2. \end{aligned}$$

Choosing $0 < \theta < 1$ sufficiently close to 1 and using second order Korn inequality we conclude. $\square$

Now applying Theorem B.20 for each $p \in \mathcal{Q}$ there exists $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta} \rangle = \langle l, \mathcal{J}(\tilde{\zeta}) \rangle \quad \text{for all } \tilde{\zeta} \in \mathcal{X}.$$

Moreover ξ is unique. Indeed by strong monotonicity if ξ_1 and ξ_2 are two solutions we have

$$M \|\xi_2 - \xi_1\|_{\mathcal{X}}^2 \leq \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle = 0.$$

We call $\Xi: \mathcal{Q} \rightarrow \mathcal{X}$ the map that assign to $p \in \mathcal{Q}$ the unique ξ . Now to conclude the existence of a solution to Problem 7.6 we need to solve in $p \in \mathcal{Q}$ the equation

$$\langle \mathcal{A}(p, \mathcal{J}(\Xi(p))), (0, 0, \tilde{\mu}) \rangle + \langle Bp, (0, 0, \tilde{\mu}) \rangle = \langle l, (0, 0, \tilde{\mu}) \rangle \quad \text{for all } \tilde{\mu} \in L^2(\Omega)$$

that is

$$\int_{\Omega} \text{tr} [\tau \mathfrak{C} (E\dot{u} - \dot{U})] \frac{\tilde{\mu}}{d} dx - \int_{\Omega} p \tilde{\mu} dx = \int_{\Omega} \text{tr} [\mathfrak{C} (Eu - U)] \frac{\tilde{\mu}}{d} dx$$

for all $\tilde{\mu} \in L^2(\Omega)$, where $\dot{y} = (\dot{u}, \dot{U}, \dot{\lambda}) = \mathcal{J}(\Xi(p))$. This is equivalent to

$$(7.3) \quad p = \frac{1}{d} \text{tr} [\tau \mathfrak{C} (E\dot{u} - \dot{U})] - \frac{1}{d} \text{tr} [\mathfrak{C} (Eu - U)] = \mathcal{S}(p)$$

where we have introduced the operator $\mathcal{S}: \mathcal{Q} \rightarrow \mathcal{Q}$. We want to prove the existence of a fixed point of $\mathcal{S}$ using the Schauder fixed point theorem (Theorem B.15). To prove that $\mathcal{S}$ satisfy the required hypotheses we need to establish some regularity results regarding Ξ . We first need the following lemma.

LEMMA 7.10. The operator $\mathcal{A}$ is Lipschitz continuous in p uniformly in $\dot{y}$.

PROOF. By (Y, a) and (G, c) we have

$$\begin{aligned} |\langle \mathcal{A}(p, \dot{y}) - \mathcal{A}(q, \dot{y}), \tilde{z} \rangle| &\leq \int_{\Omega} |Y(p) - Y(q)| |\partial G(\dot{U})| |\tilde{V}| dx \leq \\ &\leq \text{Lip}(Y) \|\partial G\|_{\infty} \int_{\Omega} |p - q| |\tilde{V}| dx \leq L \|p - q\|_2 \|\tilde{V}\|_2 \end{aligned}$$

for some constant $L > 0$. Then we have

$$\|\mathcal{A}(p, \dot{y}) - \mathcal{A}(q, \dot{y})\|_{\mathcal{X}'} \leq L \|p - q\|_{\mathcal{Q}}$$

This concludes the proof. $\square$

Now we can prove the continuity of Ξ .

LEMMA 7.11. The operator Ξ is continuous and has bounded range.

PROOF. First we prove that Ξ has bounded range. We assume by contradiction that there exists $p_n \in \mathcal{Q}$ such that $\|\Xi(p_n)\|_{\mathcal{X}} \geq n$. Let $\xi_n = \Xi(p_n)$. Now

$$\frac{\langle \mathcal{T}_{p_n}(\xi_n), \xi_n \rangle}{\|\xi_n\|_{\mathcal{X}}} \leq \|\langle l, \mathcal{J}(\cdot) \rangle\|_{\mathcal{X}'}. \quad (7.4)$$

But by Lemma 7.8

$$\frac{\langle \mathcal{T}_{p_n}(\xi_n), \xi_n \rangle}{\|\xi_n\|_{\mathcal{X}}} \rightarrow \infty$$

and this is a contradiction. Now we prove that Ξ is Lipschitz continuous. Indeed let $p_i \in \mathcal{Q}$ and $\xi_i = \Xi(p_i)$ with i equals to 1 or 2. Since $\mathcal{T}_{p_1}(\xi_1) = \mathcal{T}_{p_2}(\xi_2)$, $\mathcal{T}_p$ is strictly monotone uniformly in p and $\mathcal{A}$ is uniformly Lipschitz in p we have

$$\begin{aligned} M \|\xi_2 - \xi_1\|^2 &\leq \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_1}(\xi_1), \xi_2 - \xi_1 \rangle = \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_2}(\xi_2), \xi_2 - \xi_1 \rangle \leq \\ &\leq L \|p_2 - p_1\|_{\mathcal{Q}} \|\xi_2 - \xi_1\|_{\mathcal{X}}. \end{aligned}$$

This concludes the proof. $\square$

The following lemma establish the regularity of $\mathcal{J}$.

LEMMA 7.12. The map $\mathcal{J}: \mathcal{X} \rightarrow \mathcal{X}$ is continuous and bounded.

PROOF. It is trivial since $\mathcal{J}(\xi) = (\dot{u}, \dot{U}, 0)$. $\square$

Now we are ready to prove the desired properties of $\mathcal{S}$.

LEMMA 7.13. The map $\mathcal{S}: \mathcal{Q} \rightarrow \mathcal{Q}$ is continuous and has a bounded and compact range.

PROOF. We notice that since Ξ is continuous and has compact range then also $\mathcal{S}$ is continuous and has compact range. We notice that by Rellich–Kondarchov theorem then $\dot{u} \mapsto E\dot{u}$ from H^2 to L^2 and $\dot{U} \mapsto \dot{U}$ from H^1 to L^2 are compact. Then $\mathcal{S}$ is compact. Since $\mathcal{S}$ has bounded range and is compact then it has compact range. $\square$

Now we are in position to apply Schauder fixed point theorem and we get the existence of a fixed point of $\mathcal{S}$. We have obtained the desired result.

PROPOSITION 7.14. Problem 7.6 admits at least a solution.

Now we redefine the operator $\mathcal{A}: \mathcal{X} \rightarrow \mathcal{X}'$ by

$$\langle \mathcal{A}y, \tilde{z} \rangle = \int_{\Omega} \mathfrak{C} \left(Eu - U - \frac{\lambda}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \int_{\Omega} \mathfrak{A} \operatorname{grad} u : \operatorname{grad} \tilde{v} dx + \int_{\Omega} \mathfrak{B} U \circ \tilde{V} dx$$

and we define operator $\mathcal{E}: \mathcal{X} \rightarrow \mathcal{X}'$ by

$$\langle \mathcal{E}y, \tilde{z} \rangle = \int_{\Omega} \mathfrak{D} E\dot{u} : E\tilde{v} dx + \int_{\Omega} \mathfrak{C} \operatorname{grad} \dot{u} : \operatorname{grad} \tilde{v} dx + \int_{\Omega} \mathfrak{F} \dot{U} \circ \tilde{V} dx.$$

We define also $\mathcal{D}: \mathcal{Q} \times \mathcal{X} \rightarrow \tilde{\mathcal{Z}}'$ by

$$\langle \mathcal{D}(p, \dot{y}), \tilde{z} \rangle = \int_{\Omega} Y(p) \partial G(\dot{U}) : \tilde{V} dx.$$

The operator $\mathcal{B}$ is defined exactly as before and and $\mathcal{C}: \mathcal{X} \rightarrow \mathcal{Q}'$ is redefined by

$$\langle \mathcal{C}y, \tilde{q} \rangle = \int_{\Omega} \lambda \tilde{q} dx = 0.$$

Let also $l: [0, T] \rightarrow \mathcal{X}'$ redefined by

$$\langle l(t), \tilde{z} \rangle = \int_{\Omega} f(t) \cdot \tilde{v} dx + \int_{\Gamma_N} h(t) \cdot \tilde{v} dx.$$

Proposition 7.14 ensures that that for each N there exists $\{(\dot{y}_n^N, p_n^N, y_n^N)\}_{n=0}^N$ such that $(\dot{y}_0^N, p_0^N, y_0^N) = 0$, $y_n^N = y_{n-1}^N + \tau \dot{y}_n^N$ and

$$(7.4) \quad \begin{cases} \mathcal{A}y_n^N + \mathcal{E}\dot{y}_n^N + \mathcal{D}(p_n^N, \dot{y}_n^N) + \mathcal{B}p_n^N = l_n^N \\ \mathcal{C}y_n^N = 0 \end{cases}$$

for all $n \in \{1, \dots, N\}$. Here $l_n^N = l(t_n^N)$.

Construction of the continuous in time solution. Now we want to construct the continuous in time solution passing to the limit $N \rightarrow \infty$. We define

$$\begin{aligned} l^N &\in L^2((0, T), \mathcal{X}') && \text{piecewise constant interpolation of } \{l_n^N\}_{n=1}^N \\ \bar{y}^N &\in L^2((0, T), \mathcal{X}) && \text{piecewise constant interpolation of } \{y_n^N\}_{n=1}^N \\ y^N &\in H^1((0, T), \mathcal{X}) && \text{piecewise linear interpolation of } \{y_n^N\}_{n=1}^N \\ \dot{y}^N &\in L^2((0, T), \mathcal{X}) && \text{piecewise constant interpolation of } \{\dot{y}_n^N\}_{n=1}^N \\ p^N &\in L^2((0, T), \mathcal{Q}) && \text{piecewise constant interpolation of } \{p_n^N\}_{n=1}^N. \end{aligned}$$

We will need the following lemma.

LEMMA 7.15. We have that $l^N \rightarrow l$ in $L^2((0, T), \mathcal{X}')$ and in particular

$$\|l^N\|_{L^2((0, T), \mathcal{X}')}^2 = \sum_{n=1}^N \tau_N \|l_n^N\|_{\mathcal{X}'}^2$$

is bounded.

PROOF. For all $t \in [t_{n-1}^N, t_n^N]$ we have

$$l^N(t) = l_n^N = l(t_n^N) = l(t) + \int_t^{t_n^N} \dot{l}(s) ds.$$

By Hölder inequality

$$\|l^N(t) - l(t)\|_{\mathcal{X}'} \leq \int_t^{t_n^N} \|\dot{l}(s)\|_{\mathcal{X}'} ds \leq \int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{X}'} ds \leq \sqrt{\tau_N} \left[\int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{X}'}^2 ds \right]^{1/2}.$$

Taking the square and integrating for $t \in [t_{n-1}^N, t_n^N]$ we have

$$\int_{t_{n-1}^N}^{t_n^N} \|l^N(t) - l(t)\|_{\mathcal{X}'}^2 dt \leq \tau_N^2 \int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{X}'}^2 ds.$$

Adding this inequalities for $n \in \{1, \dots, N\}$ we have

$$\|l^N - l\|_{L^2((0,T), \mathcal{X}')}^2 \leq \tau_N^2 \|\dot{l}\|_{L^2((0,T), \mathcal{X}')}^2 \rightarrow 0.$$

This concludes the proof. $\square$

We now obtain a priori estimates on $\bar{y}^N$, y^N and $\dot{y}^N$.

LEMMA 7.16. The functions $\bar{y}^N$ and y^N are bounded in $L^\infty((0, T), \mathcal{X})$ and the function $\dot{y}^N$ is bounded in $L^2((0, T), \mathcal{X})$.

PROOF. We first observe that y_n^N and $\dot{y}_n^N$ belongs to $\mathcal{N}$ and that $\mathcal{A}$ and $\mathcal{E}$ are coercive there. Next we test the first equation in (7.4) with $\tilde{z} = \dot{y}_n^N$. We notice that $\langle \mathcal{D}(p_n^N, \dot{y}_n^N), \dot{y}_n^N \rangle \geq 0$ and $\langle \mathcal{B}p_n^N, \dot{y}_n^N \rangle = 0$. Then

$$\frac{1}{\tau_N} \langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N \rangle + \langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N \rangle \leq \langle l_n^N, \dot{y}_n^N \rangle.$$

Multiplying by $2\tau_N$ we have

$$2\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N \rangle + 2\tau_N \langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N \rangle \leq 2\tau_N \langle l_n^N, \dot{y}_n^N \rangle.$$

Now

$$\begin{aligned} 2\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N \rangle &= \langle \mathcal{A}y_n^N, y_n^N \rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N \rangle + \langle \mathcal{A}(y_n^N - y_{n-1}^N), (y_n^N - y_{n-1}^N) \rangle \geq \\ &\geq \langle \mathcal{A}y_n^N, y_n^N \rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N \rangle. \end{aligned}$$

Let $E > 0$ be its coercivity constant of $\mathcal{E}$ in $\mathcal{N}$. Then

$$2\tau_N \langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N \rangle \geq 2\tau_N E \|\dot{y}_n^N\|_{\mathcal{X}}^2$$

and using Young inequality

$$2\tau_N \langle l_n^N, \dot{y}_n^N \rangle \leq \frac{\tau_N}{E} \|l_n^N\|_{\mathcal{X}'}^2 + \tau_N E \|\dot{y}_n^N\|_{\mathcal{X}}^2.$$

Then

$$(7.5) \quad \langle \mathcal{A}y_n^N, y_n^N \rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N \rangle + \tau_N E \|\dot{y}_n^N\|_{\mathcal{X}}^2 \leq \frac{\tau_N}{E} \|l_n^N\|_{\mathcal{X}'}^2.$$

Let $A > 0$ the coercivity constant of $\mathcal{A}$ on $\mathcal{N}$. Then adding the inequalities (7.5) we have

$$A\|y_n^N\|_{\mathcal{X}}^2 + E \sum_{k=1}^n \tau_N \|\dot{y}_k^N\|_{\mathcal{X}}^2 \leq \langle \mathcal{A}y_n^N, y_n^N \rangle + E \sum_{k=1}^n \tau_N \|\dot{y}_k^N\|_{\mathcal{X}}^2 \leq \frac{1}{E} \sum_{k=1}^n \tau_N \|l_k^N\|_{\mathcal{X}'}^2.$$

Using Lemma 7.15 we have that

$$\|y_n^N\|_{\mathcal{X}} \quad \text{and} \quad \sum_{n=1}^N \tau_N \|\dot{y}_n^N\|_{\mathcal{X}}^2$$

are bounded uniformly in $N \in \mathbb{N}$ and $n \in \{0, \dots, N\}$. This implies immediately the thesis. $\square$

Using that lemma, up to extracting a subsequence we have that

$$\begin{aligned} \bar{y}^N &\overset{*}{\rightharpoonup} \bar{y} \quad \text{in } L^\infty((0, T), \mathcal{X}) \\ y^N &\overset{*}{\rightharpoonup} y \quad \text{in } L^\infty((0, T), \mathcal{X}) \\ \dot{y}^N &\rightharpoonup z \quad \text{in } L^2((0, T), \mathcal{X}). \end{aligned}$$

We notice that from $\lambda_n^N = 0$ it follows immediately that $\lambda = 0$.

LEMMA 7.17. We have that $\bar{y} = y \in H^1((0, T), \mathcal{X})$ and $\dot{y} = z$. Moreover $y(0) = 0$.

PROOF. We prove that $\bar{y}^N - y^N \rightarrow 0$ in $L^\infty((0, T), \mathcal{X})$. Indeed let $t \in [t_{n-1}^N, t_n^N]$. Then

$$\|\bar{y}^N(t_n^N) - y(t_n^N)\|_{\mathcal{X}} = \|y_n^N - y_n^N - (t_n^N - t)\dot{y}_n^N\|_{\mathcal{X}} \leq \tau_N \|\dot{y}_n^N\|_{\mathcal{X}}.$$

It follows that

$$\|\bar{y}^N(t_n^N) - y(t_n^N)\|_{\mathcal{X}}^2 \leq \tau_N^2 \|\dot{y}_n^N\|_{\mathcal{X}}^2 \leq \tau_N \|\dot{y}^N\|_{L^2((0, T), \mathcal{X})}^2 \rightarrow 0.$$

This establish the equality $\bar{y} = y$. Now let $\phi \in C_c^\infty([0, T])$ and $\zeta \in \mathcal{X}'$ arbitrary. Then

$$\int_0^T \phi \langle \zeta, \dot{y}^N \rangle dt = -\phi(0) \langle \zeta, y^N(0) \rangle - \int_0^T \dot{\phi} \langle \zeta, y^N \rangle dt.$$

Choosing first $\phi \in C_c^\infty((0, T))$ and exploiting (7.6) we have

$$\int_0^T \phi \langle \zeta, \dot{y} \rangle dt = - \int_0^T \dot{\phi} \langle \zeta, y \rangle dt.$$

This means exactly that $y \in H^1((0, T), \mathcal{X})$ and $\dot{y} = z$. In particular for $\phi \in C_c^\infty([0, T])$ generic we have

$$\int_0^T \phi \langle \zeta, \dot{y} \rangle dt = -\phi(0) \langle \zeta, y(0) \rangle - \int_0^T \dot{\phi} \langle \zeta, y \rangle dt.$$

Then we get also

$$\langle \zeta, y^N(0) \rangle \rightarrow \langle \zeta, y(0) \rangle.$$

Since $y^N(0) = 0$ also $y(0) = 0$. □

We now discuss the convergence of p^N .

LEMMA 7.18. Up to extracting a further subsequence $u^N \rightarrow u$ in $C([0, T], H^1(\Omega)^d)$ and $U^N \rightarrow U$ in $C([0, T], L^2(\Omega, \text{DevSym}))$. Moreover $p^N \rightarrow p$ in $C([0, T], \mathcal{Q})$ for some p .

PROOF. Since $y^N \rightarrow y$ in $L^\infty((0, T), \mathcal{X})$ and $\dot{y}^N \rightarrow \dot{y}$ in $L^2((0, T), \mathcal{X})$ then also

$$\begin{aligned} u^N &\rightharpoonup u \text{ in } L^\infty(\Omega, H^2) \quad \text{and} \quad \dot{u}^N \rightharpoonup \dot{u} \text{ in } L^2(\Omega, H^2), \\ U^N &\rightharpoonup U \text{ in } L^\infty(\Omega, H^1) \quad \text{and} \quad \dot{U}^N \rightharpoonup \dot{U} \text{ in } L^2(\Omega, H^1). \end{aligned}$$

By Aubin–Lions–Simon lemma (Theorem B.5)

$$u^N \rightarrow u \text{ in } C([0, T], H^1) \quad \text{and} \quad U^N \rightarrow U \text{ in } C([0, T], L^2).$$

We recall that by (7.3) we have

$$p_n^N = \frac{1}{d} \text{tr} [\tau_N \mathfrak{C} (E \dot{u}_n^N - \dot{U}_n^N)] - \frac{1}{d} \text{tr} [\mathfrak{C} (E u_n^N - U_n^N)].$$

Since $\tau_N \rightarrow 0$ and $\dot{y}_n^N$ is bounded in $L^2((0, T), \mathcal{X})$ we have

$$p^N \rightarrow p = -\frac{1}{d} \text{tr} [\mathfrak{C} (Eu - U)] \quad \text{in } L^2((0, T), \mathcal{X}).$$

This concludes the proof. □

REMARK 7.19. The fact that $p \in C^0([0, T], \mathcal{Q})$ suggest that there are not bifurcation modes of pressure.

Now we notice that (7.4) implies that for all $\tilde{z} \in L^2((0, T), \mathcal{X})$ we have

$$\int_0^T \langle A \bar{y}^N, \tilde{z} \rangle dt + \int_0^T \langle \mathcal{E} \dot{y}^N, \tilde{z} \rangle dt + \int_0^T \langle \mathcal{D}(P^N, \dot{y}^N), \tilde{z} \rangle dt + \int_0^T \langle \mathcal{B} p^N, \tilde{z} \rangle dt = \int_0^T \langle l^N, \tilde{z} \rangle dt.$$

By Lemma 7.15 and (7.6) we have

$$\begin{aligned} \int_0^T \langle l^N, \tilde{z} \rangle dt &\rightarrow \int_0^T \langle l, \tilde{z} \rangle dt \\ \int_0^T \langle A \bar{y}^N, \tilde{z} \rangle dt &\rightarrow \int_0^T \langle Ay, \tilde{z} \rangle dt \\ \int_0^T \langle \mathcal{E} \dot{y}^N, \tilde{z} \rangle dt &\rightarrow \int_0^T \langle \mathcal{E} \dot{y}, \tilde{z} \rangle dt \\ \int_0^T \langle \mathcal{B} p^N, \tilde{z} \rangle dt &\rightarrow \int_0^T \langle \mathcal{B} p, \tilde{z} \rangle dt. \end{aligned}$$

Moreover we have

$$\begin{aligned} \int_0^T \|\mathcal{D}(p^N, \dot{y}^N)\|_{\mathcal{X}}^2 dt &\leq \int_0^T \int_{\Omega} Y(p^N)^2 |\partial G(\dot{U}^N)|^2 dx dt \leq \\ &\leq \int_0^T \int_{\Omega} (Y(0) + \text{Lip}(Y)|p^N|)^2 \|\partial G\|_{\infty}^2 dx dt \lesssim \\ &\lesssim (1 + \|p^N\|_{L^2((0,T),\mathcal{Q})})^2. \end{aligned}$$

But since p^N converges in $C([0, T], \mathcal{Q})$ then $\mathcal{D}(p^N, \dot{y}^N)$ is bounded in $L^2((0, T), \mathcal{X}')$. Then up to extracting a further subsequence

$$\mathcal{D}(p^N, \dot{y}^N) \rightharpoonup \delta \text{ in } L^2((0, T), \mathcal{X}')$$

for some $\delta \in L^2((0, T), \mathcal{X}')$. Then we have

$$(7.7) \quad \int_0^T \langle \mathcal{A}y, \tilde{z} \rangle dt + \int_0^T \langle \mathcal{E}\dot{y}, \tilde{z} \rangle dt + \int_0^T \langle \delta, \tilde{z} \rangle dt + \int_0^T \langle \mathcal{B}p, \tilde{z} \rangle dt = \int_0^T \langle l, \tilde{z} \rangle dt$$

for all $\tilde{z} \in L^2((0, T), \mathcal{X})$. This implies also

$$\langle \mathcal{A}y, \tilde{z} \rangle + \langle \mathcal{E}\dot{y}, \tilde{z} \rangle + \langle \delta, \tilde{z} \rangle + \langle \mathcal{B}p, \tilde{z} \rangle = \langle l, \tilde{z} \rangle$$

for almost every $t \in (0, T)$. If we prove that $\delta = \mathcal{D}(p, \dot{y})$ we have that (y, p) is a global in time solution to Problem 7.4 and we have concluded. To this aim let us define $\mathcal{X}_T = L^2((0, T), \mathcal{X})$ and $\mathcal{Q}_T = L^2((0, T), \mathcal{Q})$. Let $\mathcal{J}: \mathcal{Q} \times \mathcal{X} \rightarrow \mathbb{R}$ be defined by

$$\mathcal{J}(p, \dot{y}) = \int_{\Omega} Y(p)G(\dot{U})dx.$$

Let also $\mathcal{J}_T: \mathcal{Q}_T \times \mathcal{X}_T \rightarrow \mathbb{R}$ and $\mathcal{D}_T: \mathcal{Q}_T \times \mathcal{X}_T \rightarrow \mathcal{X}'_T$ defined by

$$\mathcal{J}_T(p, \dot{y}) = \int_0^T \mathcal{J}(p, \dot{y})dt \quad \text{and} \quad \langle \mathcal{D}_T(p, \dot{y}), \tilde{z} \rangle = \int_0^T \langle \mathcal{D}(p, \dot{y}), \tilde{z} \rangle dt$$

for all $\tilde{z} \in L^2((0, T), \mathcal{X})$. We notice that for all $y \in \mathcal{X}_T$ and $p \in \mathcal{Q}_T$ we can identify $\mathcal{D}_T(p, \dot{y}) \in \mathcal{X}'_T$ with $\mathcal{D}(p, \dot{y}) \in L^2((0, T), \mathcal{X}')$.

LEMMA 7.20. Let $\dot{y} \in L^2((0, T), \mathcal{X})$, $p \in L^2((0, T), \mathcal{Q})$ and $\delta \in L^2((0, T), \mathcal{X}')$. Then $\delta = \mathcal{D}(p, \dot{y})$ if and only if

$$(7.8) \quad \mathcal{J}_T(p, \tilde{z}) \geq \mathcal{J}_T(p, \dot{y}) + \int_0^T \langle \delta, \tilde{z} - \dot{y} \rangle dt$$

for all $\tilde{z} \in L^2((0, T), \mathcal{X})$.

PROOF. It is easy to see that $\mathcal{J}$ is convex in $\dot{y}$ and is Gateaux differentiable in $\dot{y}$ with Gateaux differential

$$\langle D_{\dot{y}}\mathcal{J}_T(p, \dot{y}), \tilde{z} \rangle = \int_0^T \langle \mathcal{D}(p, \dot{y}), \tilde{z} \rangle dt = \langle \mathcal{D}_T(p, \dot{y}), \tilde{z} \rangle$$

for all $\tilde{z} \in L^2((0, T), \mathcal{X})$. By [14, Proposition 5.3] then $\mathcal{J}_T$ is sub differentiable and $D_{\dot{y}}^{\text{sub}}\mathcal{J}_T(p, \dot{y}) = \{D_{\dot{y}}\mathcal{J}_T(p, \dot{y})\}$. Then $\delta = \mathcal{D}(p, \dot{y})$ if and only if $\delta \in D_{\dot{y}}^{\text{sub}}\mathcal{J}_T(p, \dot{y})$ but this last inclusion is equivalent to (7.8). $\square$

By that lemma we have

$$(7.9) \quad \mathcal{J}_T(p^N, \tilde{z}) \geq \mathcal{J}_T(p^N, \dot{y}) + \langle \mathcal{D}_T(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N \rangle.$$

Now we want to pass to the limit in this inequality. We will need the following lemma.

LEMMA 7.21. For each $\dot{y} \in \mathcal{X}_T$ the functional $\mathcal{J}_T(\cdot, \dot{y})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{y}\|_{\mathcal{X}_T}$. Moreover for each $p \in \mathcal{Q}_T$ we have that $\mathcal{J}_T(p, \cdot)$ is convex and continuous. In particular $\mathcal{J}_T(p, \cdot)$ is weakly lower semicontinuous.

PROOF. We have that

$$\begin{aligned} |\mathcal{J}_T(p_2, \dot{y}) - \mathcal{J}_T(p_1, \dot{y})| &\leq \int_0^T \int_{\Omega} |Y(p_2) - Y(p_1)| |G(\dot{U})| dx \leq \\ &\leq \|\partial G\|_{\infty} \text{Lip}(Y) \int_0^T \int_{\Omega} |p_2 - p_1| |\dot{U}| dt dx \lesssim \\ &\lesssim \|p_2 - p_1\|_{\mathcal{Q}_T} \|\dot{y}\|_{\mathcal{X}_T}. \end{aligned}$$

This proves that for each $\dot{y} \in \mathcal{X}_T$ the functional $\mathcal{J}_T(\cdot, \dot{y})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{y}\|_{\mathcal{X}_T}$. The convexity of $\mathcal{J}_T(p, \cdot)$ follows immediately from the convexity of G . Moreover

$$\begin{aligned} |\mathcal{J}_T(p, \dot{y}_2) - \mathcal{J}_T(p, \dot{y}_1)| &\leq \int_0^T \int_{\Omega} |Y(p)| |G(\dot{U}_2) - G(\dot{U}_1)| dx \leq \\ &\leq \|\partial G\|_{\infty} \int_0^T \int_{\Omega} (Y(0) + \text{Lip}(Y)|p|) |\dot{U}_2 - \dot{U}_1| dt dx \lesssim \\ &\lesssim (1 + \|p\|_{\mathcal{Q}_T}) \|\dot{y}_2 - \dot{y}_1\|_{\mathcal{X}_T}. \end{aligned}$$

This proves the continuity of $\mathcal{J}_T(p, \cdot)$ for each $p \in \mathcal{Q}_T$ fixed. $\square$

Now we have

$$|\mathcal{J}_T(p^N, \tilde{z}) - \mathcal{J}_T(p, \tilde{z})| \leq C \|p^N - p\|_{\mathcal{Q}_T} \rightarrow 0$$

for some constant $C > 0$. Moreover

$$\mathcal{J}_T(p^N, \dot{y}^N) = [\mathcal{J}_T(p^N, \dot{y}^N) - \mathcal{J}_T(p, \dot{y}^N)] + \mathcal{J}_T(p, \dot{y}^N).$$

But since $p^N \rightharpoonup$ in $\mathcal{Q}_T$ and $\dot{y}^N$ are bounded in $\mathcal{X}_T$

$$|\mathcal{J}_T(p^N, \dot{y}^N) - \mathcal{J}_T(p, \dot{y}^N)| \leq C \|\dot{y}^N\|_{\mathcal{X}_T} \|p^N - p\|_{\mathcal{Q}_T} \rightarrow 0$$

and

$$\liminf_{N \rightarrow \infty} \mathcal{J}_T(p, \dot{y}^N) \geq \mathcal{J}_T(p, \dot{y}).$$

Then

$$\liminf_{N \rightarrow \infty} \mathcal{J}_T(p^N, \dot{y}^N) \geq \mathcal{J}_T(p, \dot{y}).$$

Now by construction

$$\begin{aligned} \int_0^T \langle Ay^N, \tilde{z} - \dot{y}^N \rangle dt + \int_0^T \langle \mathcal{E} \dot{y}^N, \tilde{z} - \dot{y}^N \rangle dt + \int_0^T \langle \mathcal{D}(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N \rangle dt + \\ + \int_0^T \langle \mathcal{B} p^N, \tilde{z} - \dot{y}^N \rangle dt = \int_0^T \langle l^N, \tilde{z} - \dot{y}^N \rangle dt. \end{aligned}$$

But by Lemma 7.15, Lemma 7.18 and (7.6) we have

$$\begin{aligned} \int_0^T \langle Ay^N, \tilde{z} - \dot{y}^N \rangle dt &\rightarrow \int_0^T \langle Ay, \tilde{z} - \dot{y} \rangle dt \\ \int_0^T \langle \mathcal{B} p^N, \tilde{z} - \dot{y}^N \rangle dt &\rightarrow \int_0^T \langle \mathcal{B} p, \tilde{z} - \dot{y} \rangle dt \\ \int_0^T \langle l^N, \tilde{z} - \dot{y}^N \rangle dt &\rightarrow \int_0^T \langle l, \tilde{z} - \dot{y} \rangle dt \end{aligned}$$

since the convergence of y^N , p^N and l^N is strong. Moreover since

$$\dot{y} \rightarrow \int_0^T \langle \mathcal{E} \dot{y}^N, \dot{y}^N - \tilde{z} \rangle dt$$

is continuous and convex in $L^2((0, T), \mathcal{X})$ it is weakly lower semicontinuous and

$$\liminf_{N \rightarrow \infty} \int_0^T \langle \mathcal{E} \dot{y}^N, \dot{y}^N - \tilde{z} \rangle dt \geq \int_0^T \langle \mathcal{E} \dot{y}, \dot{y} - \tilde{z} \rangle dt.$$

This implies that

$$\begin{aligned} \liminf_{N \rightarrow \infty} \langle \mathcal{D}_T(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N \rangle &\geq \\ &\geq \int_0^T \langle l, \tilde{z} - \dot{y} \rangle dt - \int_0^T \langle Ay, \tilde{z} - \dot{y} \rangle dt - \int_0^T \langle \mathcal{E} \dot{y}, \tilde{z} - \dot{y} \rangle dt - \int_0^T \langle \mathcal{B} p, \tilde{z} - \dot{y} \rangle dt = \\ &= \int_0^T \langle \delta, \tilde{z} - \dot{y} \rangle dt \end{aligned}$$

where the last equality holds for (7.7). Collecting these results when we take the inferior limit for $N \rightarrow \infty$ of (7.9) and we get

$$\mathcal{J}_T(p, \tilde{z}) \geq \mathcal{J}_T(p, \dot{y}) + \int_0^T \langle \delta, \tilde{z} - \dot{y} \rangle dt.$$

Since now δ satisfy (7.8) we have by Lemma 7.8 that $\delta = \mathcal{D}(p, \dot{y})$ as desired.

2.5. Example of constitutive relations. In this section we give examples of constitutive relations satisfying the hypotheses of Section 2.1. Let $g: [0, \infty) \rightarrow \mathbb{R}$ be defined by

$$g(r) = \begin{cases} \frac{r^2}{4c} & \text{if } r \leq c \\ r - \frac{3}{4}c - \frac{c}{2} \log \frac{r}{c} & \text{if } r > c \end{cases}$$

where $c > 0$ so that

$$g'(r) = \begin{cases} \frac{r}{2c} & \text{if } r \leq c \\ 1 - \frac{c}{2r} & \text{if } r > c. \end{cases}$$

We set $G(\dot{U}) = g(|\dot{U}|)$. We let the yield strength be given by

$$Y(p) = \begin{cases} Y_0 & \text{if } p < -p_0 \\ Y_0 + \alpha(p + p_0) & \text{if } p \geq -p_0 \end{cases}$$

where $Y_0 \geq 0$, $p_0 \geq 0$ and $\alpha > 0$. We set $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{D}$, $\mathfrak{E}$ and $\mathfrak{F}$ positive constants and $\mathfrak{C}$ and $\mathfrak{D}$ positive definite isotropic fourth order tensors.

2.6. Further regularity hypotheses. The local existence and uniqueness theorem is obtained under the following additional hypotheses.

$$(Y') \quad Y \in C^{1,1}(\mathbb{R})$$

$$(G') \quad G \in C^{2,1}(\text{DevSym})$$

2.7. Local existence and uniqueness and global uniqueness.

PROPOSITION 7.22. The map $\mathcal{F}$ is of class C^1 and the Fréchet derivative is given by

$$\begin{aligned} \langle D_{\dot{y}} \mathcal{F}(\dot{y}, p, y, t)z, (\tilde{z}, \tilde{q}) \rangle &= \int_{\Omega} \mathfrak{D}Ev : E\tilde{v}dx + \int_{\Omega} \mathfrak{E} \text{grad}^2 v : \text{grad}^2 \tilde{v}dx + \\ &\quad + \int_{\Omega} \mathfrak{F} \mathbb{V} \circ \tilde{\mathbb{V}}dx + \int_{\Omega} Y(p) \partial^2 G(\dot{U})V : \tilde{V}dx \\ \langle D_p \mathcal{F}(\dot{y}, p, y, q)q, (\tilde{z}, \tilde{q}) \rangle &= \int_{\Omega} Y'(p)q \partial G(\dot{U}) : \tilde{V}dx - \int_{\Omega} q \tilde{\mu}dx \\ \langle D_y \mathcal{F}(\dot{y}, p, y, t)z, (\tilde{z}, \tilde{q}) \rangle &= \int_{\Omega} \mathfrak{C} \left(Ev - V - \frac{\mu}{d}I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d}I \right) dx + \\ &\quad + \int_{\Omega} \mathfrak{A} \text{grad}^2 v : \text{grad}^2 \tilde{v}dx + \int_{\Omega} \mathfrak{B} \mathbb{V} \circ \tilde{\mathbb{V}}dx \\ \langle D_t \mathcal{F}(\dot{y}, p, y, t), (\tilde{z}, \tilde{q}) \rangle &= - \int_{\Omega} \dot{f} \cdot \tilde{v}dx - \int_{\Omega} \dot{h} \cdot \tilde{v}dx. \end{aligned}$$

PROOF. I still need to report in $\text{\LaTeX}$ the boring check but it should be true. First we prove the Gateaux differentiability everywhere that ensure also the continuity. Then we prove that the Gateaux differential is continuous. To this aim we use Lemma B.12, the following lemma and the embedding of H^1 in L^6 for $d = 2$ or $d = 3$.

LEMMA 7.23. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $\phi: \mathbb{R}^m \rightarrow \mathbb{R}^l$ be Lipschitz continuous and bounded. Let $f_n \rightarrow f$ in $L^p(\Omega, \mathbb{R}^m)$ for some $p \in [1, \infty)$. Then $\phi(f_n) \rightarrow \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$ for all $q \in [p, \infty)$.

PROOF. Let $A_n = \{x \in \Omega \mid |f_n(x) - f(x)| < 1\}$. Since convergence in L^p implies convergence in measure then $|\Omega \setminus A_n| \rightarrow 0$. Now let $q \in [p, \infty)$. Then if we call $L > 0$ the Lipschitz constant of ϕ and $C = \sup_{\mathbb{R}^m} |\phi|$ we have

$$\begin{aligned} \|\phi(f_n) - \phi(f)\|_q^q &= \int_{A_n} |\phi(f_n) - \phi(f)|^q dx + \int_{\Omega \setminus A_n} |\phi(f_n) - \phi(f)|^q dx \leq \\ &\leq \int_{A_n} L|f_n - f|^q dx + 2C|\Omega \setminus A_n|. \end{aligned}$$

But on A_n we have $|f_n - f|^q \leq |f_n - f|^p$ so that

$$\|\phi(f_n) - \phi(f)\|_q^q \leq \int_{A_n} L|f_n - f|^p q dx + 2C|\Omega \setminus A_n| \leq L\|f_n - f\|_p^p + 2C|\Omega \setminus A_n|.$$

This proves that $\phi(f_n) \rightarrow \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$. $\square$

$\square$

We notice that $\mathcal{F}$ does not depend on λ . We call $\xi = (\dot{u}, \dot{U}, \lambda, p)$, $\zeta = (v, V, \mu, q) = (z, q)$ and $\tilde{\zeta} = (\tilde{v}, \tilde{V}, \tilde{\mu}, \tilde{q}) = (\tilde{z}, \tilde{q})$ all belonging to $\mathcal{Z} \times \mathcal{Q}$.

PROPOSITION 7.24. For any $\dot{y} \in \mathcal{X}$, $p \in \mathcal{Q}$, $y \in \mathcal{X}$ and $t \in [0, T]$ we have $D_\xi \mathcal{F}(\dot{y}, p, y, t) \in \text{Iso}(\mathcal{X} \times \mathcal{Q}, \mathcal{X}' \times \mathcal{Q}')$.

PROOF. We define the bilinear form $a: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ by

$$\begin{aligned} a(z, \tilde{z}) = & \int_{\Omega} \mathfrak{D}Ev : E\tilde{v} dx - \int_{\Omega} \mathfrak{E} \frac{\mu}{d} I : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \int_{\Omega} \mathfrak{E} \text{grad}^2 v : \text{grad}^2 \tilde{v} dx + \\ & + \int_{\Omega} \mathfrak{F} \mathbb{V} \circ \tilde{\mathbb{V}} dx + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) V : \tilde{V} dx. \end{aligned}$$

We define also the bilinear forms $b: \mathcal{Q} \times \mathcal{X} \rightarrow \mathbb{R}$ and $c: \mathcal{Q} \times \mathcal{X} \rightarrow \mathbb{R}$ by

$$b(p, \tilde{z}) = \int_{\Omega} (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q dx$$

and

$$c(\tilde{q}, z) = - \int_{\Omega} \mu \tilde{q} dx = 0.$$

To prove that $D_\xi \mathcal{F}(\dot{y}, p, y, t) \in \text{Iso}(\mathcal{X} \times \mathcal{Q}, \mathcal{X}' \times \mathcal{Q}')$ we have to prove that the mixed elliptic problem

$$\begin{aligned} a(z, \tilde{z}) + b(p, \tilde{z}) &= \langle l, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \mathcal{X} \\ c(\tilde{q}, z) &= \langle m, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \mathcal{Q} \end{aligned}$$

is well posed for all $l \in \mathcal{X}'$ and $m \in \mathcal{Q}'$. To this aim it is sufficient to check the hypotheses of Theorem B.10. We first check the hypotheses on b . We have

$$\mathcal{N} = \{ \tilde{z} \in \mathcal{Z} \mid b(p, \tilde{z}) = 0 \text{ for all } p \in \mathcal{Q} \} = \{ \tilde{z} \in \mathcal{Z} \mid \tilde{\mu} = Y'(p) \partial G(\dot{U}) : \tilde{V} \}.$$

Notice that $|Y'(p) \partial G(\dot{U}) : \tilde{V}| \leq \text{Lip}(Y) \|\partial G\|_{\infty} |\tilde{V}| \in L^2$. Now

$$\sup_{q \in \mathcal{Q} \setminus \{0\}} \frac{\int_{\Omega} (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q dx}{\|q\|_{\mathcal{Q}}} = \|Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}\|_2.$$

On the other hand since $\tilde{z}_0 = (\tilde{v}_0, \tilde{V}_0, \tilde{\mu}_0) \in \mathcal{N}$ if and only if $\tilde{\mu}_0 = Y'(p) \partial G(\dot{U}) : \tilde{V}_0$ we have

$$\begin{aligned} \|\tilde{z}\|_{\mathcal{X}/\mathcal{N}} &= \inf_{\tilde{z}_0 \in \mathcal{N}} [\|\tilde{v} + \tilde{v}_0\|_{1,2} + \|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p) \partial G(\dot{U}) : \tilde{V}_0\|_2] = \\ &= \inf_{\tilde{V}_0 \in H^1} [\|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p) \partial G(\dot{U}) : \tilde{V}_0\|_2] \leq \\ &\leq \|\tilde{\mu} - Y'(p) \partial G(\dot{U}) : \tilde{V}\|_2 \end{aligned}$$

where the last inequality holds since we can choose $\tilde{V}_0 = -\tilde{V}$. Then we have

$$\sup_{q \in \mathcal{Q} \setminus \{0\}} \frac{\int_{\Omega} (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q dx}{\|q\|_{\mathcal{Q}}} \geq \|\tilde{z}\|_{\mathcal{X}/\mathcal{N}}.$$

Moreover if $q \in \mathcal{Q}$ is such that

$$b(p, \tilde{z}) = \int_{\Omega} (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q dx = 0 \quad \text{for all } \tilde{z} \in \mathcal{X}$$

then choosing $\tilde{V} = 0$ and $\tilde{\mu} \in L^2(\Omega)$ arbitrary we have $q = 0$. This completes the check of the hypotheses on b . The check of the hypotheses on c is similar. In particular we have

$$\mathcal{N} = \{ z \in \mathcal{Z} \mid c(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \mathcal{Q} \} = \{ z \in \mathcal{Z} \mid \mu = 0 \}.$$

Now we check the hypotheses on a . For any $z = (v, V, 0) \in \mathcal{N}$ we consider $\tilde{z} = (v, V, Y'(p)\partial G(\dot{U}) : V) \in \tilde{\mathcal{N}}$ and conversely. We notice that with that choice of z and $\tilde{z}$ there exist $K > 0$ and $C > 0$ such that $\|\tilde{z}\|_{\mathcal{Z}} \leq K\|z\|_{\mathcal{X}}$ and

$$a(z, \tilde{z}) \geq \delta\|Ev\|_2^2 + \epsilon\|\text{grad}^2 v\|_2^2 + \phi\|\mathbb{V}\|_2^2 \geq C\|z\|_{\mathcal{X}}^2 \geq \frac{C}{K^2}\|\tilde{z}\|_{\mathcal{Z}}^2.$$

This immediately implies that for all $z \in \mathcal{N}$ we have

$$\sup_{\tilde{z} \in \tilde{\mathcal{Z}}} \frac{a(z, \tilde{z})}{\|\tilde{z}\|_{\mathcal{Z}}} \geq \frac{C}{K}\|z\|_{\mathcal{X}}$$

and if $\tilde{z} \in \tilde{\mathcal{N}}$ satisfy $a(\cdot, \tilde{z}) = 0$ then $\tilde{z} = 0$. Then also the hypotheses on a are verified. This completes the proof. $\square$

By Proposition 7.24 in a neighborhood of every $(\dot{y}_0, p_0, y_0, t_0)$ with $\lambda_0 = 0$ we can explicit $\xi = (\dot{u}, \dot{U}, \lambda, p)$ from the equation

$$\mathcal{F}(\dot{y}, p, y, t) = 0.$$

We get

$$(\dot{u}, \dot{U}) = \mathcal{G}(u, U, t) \quad \text{and} \quad (\lambda, p) = \mathcal{H}(u, U, t)$$

where $\mathcal{G}$ and $\mathcal{H}$ are maps of class C^1 . Then we can apply Theorem B.16 to get locally in time a unique solution (u, U) of class C^2 to the Cauchy problem

$$\begin{cases} (\dot{u}, \dot{U}) = \mathcal{G}(u, U, t) \\ u(0) = u_0 \\ U(0) = U_0. \end{cases}$$

Then we set $(\lambda, p) = \mathcal{H}(u, U, t)$ which is of class C^1 . It is clear that $\mathcal{F}(\dot{y}, p, y, t) = 0$. In conclusion we have obtained the following theorem.

THEOREM 7.25. Let $(\dot{y}_0, p_0, y_0, t_0) \in \mathcal{X} \times \mathcal{Q} \times \mathcal{X} \times [0, T]$ with $\lambda_0 = 0$. Then there exists a unique C^1 solution (y, p) to Problem 7.4 defined locally around t_0 .

Considering also Theorem 7.5 we obtain global uniqueness.

COROLLARY 7.26. Assume the hypotheses of Section 2.1 and of Section 2.6. Then there exists a unique $y \in C^1([0, T], \mathcal{X})$ and $p \in C^1([0, T], \mathcal{Q})$ solution to Problem 7.4.

2.8. Bifurcation analysis.

REMARK 7.27. This section is not definitive but contains only the ideas that we have collected in the hope of being able to reach a more complete result regarding the phenomenon of localization of deformation.

Some considerations. Since we have proved in Corollary 7.26 that the solution is unique no bifurcation can occur with nonzero viscosity. As the viscosity approaches to zero we expect that the evolution of the system reveals it extreme sensibility on the problem data when strain localization occur. To explain better the ideas we present the following example that describe what is the behavior that we expect. Consider a perfectly symmetrical specimen under perfectly symmetrical compressive loads. By uniqueness and symmetry in presence of viscosity the specimen will experience a perfectly symmetrical deformation pattern during the entire evolution (a distributed deformation or at most a symmetrical pattern of localized shear bands). However if the loads, the geometry or the material properties of the specimen are not perfectly symmetric then the system experience an unsymmetrical and completely different deformation pattern (for example an unsymmetrical pattern of shear bands). When viscosity decreases, the threshold beyond which perturbations from the condition of perfect symmetry trigger a localization with a significantly altered pattern is reduced. In some sense in the case of null viscosity the system is sensible also to vanishing perturbations. This suggest a lost of uniqueness at a stage of the loading program also in case of perfect symmetry.

We also expect that the bifurcation is actually associated with a jump in the time evolution. The reasons are the following.

- (i) When passing to the limit $N \rightarrow \infty$ in the Rothe method in Section 2.4 we used the viscosity to get the a priori estimates on $\dot{y}^N$. We where not able to get them without viscosity, using only the $H^1((0, T))$ regularity of the load, as can be done for Von Mises plasticity [25]. This suggest that without viscosity the solution of our model cannot not lies in $H^1((0, T))$ but must lie in $BV((0, T))$.

- (ii) Similar models of materials with pressure dependent yield strength exhibit jumps. It is the case of Cam–Clay model studied in [13] or the model studied analytically in [12] coupling the associative Drucker–Prager model with damage.
- (iii) We expect that when the viscosity is small the evolution became very fast where the deformation occur. In the limit of null viscosity this became a jump.

REMARK 7.28. My desire is to understand better the phenomenon of localization when the viscosity if small but nonzero trying to understand how the sensitivity of the system to imperfection increases with decreasing viscosity. Anyway, if it is useful in some way, I will try to study also the case of zero viscosity. But I would like to avoid the use of the notion of energetic solution. Indeed the choice made by the system at the jump point in the energetic solution approach is determined by a global minimizing approach that seems nonphysical to us. This view is confirmed in [30]. The approach that I could accept is that of balanced viscosity solutions where is the viscosity that drive the evolution at the jumps.

Study on the invertibility of the stiffness operator. In what follows are reported our efforts to understand better the phenomenon of localization. In particular we have confined ourself to the time discrete problem (based on the implicit Euler scheme). We computed the “stiffness operator” (the one that would be assembled in the numerical approximation of the model with Newton method). It seems to became singular during plastic flow in absence of viscosity suggesting that the system reaches indeed a bifurcation point (where the jump occur). This analysis should be intended only as a first trail and not as a final result. We hope that the analysis can be extended also to the the time continuous case.

We consider $\mathfrak{D}_\eta$, $\mathfrak{E}_\eta$ and $\mathfrak{F}_\eta$ as dependent continuously a viscosity parameter $\eta > 0$ and vanishing for $\eta = 0$. We call $\mathcal{F}_\eta$ the corresponding functional. We consider the discrete in time problem (Problem 7.6). We focus on a single time step and to simplify the notation we set $y = y_{n-1}^N$, $t = t_{n-1}^N$, $p = p_n^N$ and $\dot{y} = \tau_N^{-1}(y_n^N - y_{n-1}^N)$ and $\tau = \tau_N$. Then $y_n^N = y + \tau \dot{y}$. The incremental problem reduces to find $\dot{y} \in \mathcal{Z}$ and $p \in \mathcal{Q}$ such that

$$\mathcal{F}_{(\tau,\eta)}^{(y,t)}(\dot{y}, p) = \mathcal{F}_\eta(\dot{y}, p, y + \tau \dot{y}, t + \tau) = 0.$$

We compute the Fréchet differential $\mathcal{K}_{(\tau,\eta)}^{(y,t)}(\dot{y}, p)$ of $\mathcal{F}_{(\tau,\eta)}^{(y,t)}$ with respect to $(\dot{y}, p)$. It can be written as

$$\langle \mathcal{K}_{(\tau,\eta)}^{(y,t)}(\dot{y}, p)(z, q), (\tilde{z}, \tilde{q}) \rangle = a_{(\tau,\eta)}^{(\dot{y},p,y,t)}(z, \tilde{z}) + b_{(\tau,\eta)}^{(\dot{y},p,y,t)}(q, \tilde{z}) + c_{(\tau,\eta)}^{(\dot{y},p,y,t)}(\tilde{q}, z).$$

We have introduced the bilinear form

$$\begin{aligned} a_{(\tau,\eta)}^{(\dot{y},p,y,t)}(z, \tilde{z}) &= \int_{\Omega} \mathfrak{D}_\eta E v : E \tilde{v} dx + \int_{\Omega} \tau \mathfrak{C} \left(E v - V - \frac{\mu}{d} I \right) : \left(E \tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ &\quad + \int_{\Omega} (\mathfrak{E}_\eta + \tau \mathfrak{A}) \operatorname{grad}^2 v : \operatorname{grad}^2 \tilde{v} dx + \int_{\Omega} (\mathfrak{F}_\eta + \tau \mathfrak{B}) \nabla \circ \tilde{\nabla} dx + \\ &\quad + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) V : \tilde{V} dx \\ b_{(\tau,\eta)}^{(\dot{y},p,y,t)}(q, \tilde{z}) &= \int_{\Omega} q(Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) dx \\ c_{(\tau,\eta)}^{(\dot{y},p,y,t)}(q, \tilde{z}) &= - \int_{\Omega} \tau \mu \tilde{q} dx. \end{aligned}$$

We want to study if $\mathcal{K}_{(\tau,\eta)}^{(y,t)}$ is in $\operatorname{Iso}(\mathcal{Z} \times \mathcal{Q}, \mathcal{Z}' \times \mathcal{Q}')$. To this aim we can apply Theorem B.10 to the bilinear forms $a_{(\tau,\eta)}^{(\dot{y},p,y,t)}$, $b_{(\tau,\eta)}^{(\dot{y},p,y,t)}$ and $c_{(\tau,\eta)}^{(\dot{y},p,y,t)}$. Reasoning as in the proof of Proposition 7.24 we have that $b_{(\tau,\eta)}^{(\dot{y},p,y,t)}$ and $c_{(\tau,\eta)}^{(\dot{y},p,y,t)}$ satisfies the hypotheses. Moreover the sets $\mathcal{N}$ and $\tilde{\mathcal{N}}$ are defined exactly in the same way. We need to check the hypotheses on $a_{(\tau,\eta)}^{(\dot{y},p,y,t)}$. We reason again as in the proof of Proposition 7.24. For any $z = (v, V, 0) \in \mathcal{N}$ we consider $\tilde{z} = (v, V, Y'(p) \partial G(\dot{U}) : V) \in \tilde{\mathcal{N}}$ and conversely. We have that with that

choice of z and $\tilde{z}$ there exist $K > 0$ such that $\|\tilde{z}\|_{\mathcal{X}} \leq K\|z\|_{\mathcal{X}}$. Then

$$\begin{aligned} a_{(\tau,\eta)}^{(\dot{y},p,y,t)}(z,\tilde{z}) &= \int_{\Omega} \mathfrak{D}_{\eta} E v : E v dx + \int_{\Omega} \tau \mathfrak{E} (E v - V) : \left(E v - V - \frac{Y'(p) \partial G(\dot{U}) : V}{d} I \right) dx + \\ &\quad + \int_{\Omega} (\mathfrak{E}_{\eta} + \tau \mathfrak{A}) \text{grad}^2 v : \text{grad}^2 v dx + \int_{\Omega} (\mathfrak{F}_{\eta} + \tau \mathfrak{B}) \mathbb{V} \circ \mathbb{V} dx + \\ &\quad + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) V : V dx \\ &\leq \delta_{\eta} \|E v\|_2^2 + \tau \gamma \|E v - V\|_2^2 - 2\tau \kappa \|E v - V\|_2 \|V\|_2 + \\ &\quad + (\epsilon_{\eta} + \tau \alpha) \|\text{grad}^2 v\|_2^2 + (\phi_{\eta} + \tau \beta) \|\mathbb{V}\|_2^2 + \sigma(p, \dot{U}) \|V\|_2^2. \end{aligned}$$

Here

$$\kappa = \frac{|\mathfrak{E}| \text{Lip}(Y) \|\partial G\|_{\infty}}{2\sqrt{d}} \quad \text{and} \quad \sigma(p, \dot{U}) = \text{essinf}_{\Omega} [Y(p) \text{se}(\partial^2 G(\dot{U}))] \geq 0$$

where se denotes the smallest eigenvalue. Then by Young inequality for all $\theta_1 \in (0, \kappa^{-1}\gamma)$ and $\theta_2 \in (0, \infty)$ we have

$$\begin{aligned} a_{(\tau,\eta)}^{(\dot{y},p,y,t)}(z,\tilde{z}) &\leq \delta_{\eta} \|E v\|_2^2 + \tau(\gamma - \kappa\theta_1) \|E v - V\|_2^2 - \tau \frac{\kappa}{\theta_1} \|V\|_2^2 + \\ &\quad + (\epsilon_{\eta} + \tau \alpha) \|\text{grad}^2 v\|_2^2 + (\phi_{\eta} + \tau \beta) \|\mathbb{V}\|_2^2 + \sigma(p, \dot{U}) \|V\|_2^2. \\ &\leq [\delta_{\eta} + \tau(\gamma - \kappa\theta_1)(1 - \theta_2)] \|E v\|_2^2 + (\epsilon_{\eta} + \tau \alpha) \|\text{grad}^2 v\|_2^2 + \\ &\quad + \left\{ \phi_{\eta} + \sigma(p, \dot{U}) + \tau \left[(\gamma - \kappa\theta_1) \left(1 - \frac{1}{\theta_2} \right) - \frac{\kappa}{\theta_1} + \beta \right] \right\} \|V\|_2^2 + \\ &\quad + (\phi_{\eta} + \tau \beta) \|\text{grad} V\|_2^2. \end{aligned}$$

The hypotheses on $a_{(\tau,\eta)}^{(\dot{y},p,y,t)}$ are satisfied if we can find $\theta = (\theta_1, \theta_2) \in (0, \kappa^{-1}\gamma) \times (0, \infty)$ such that

$$\begin{cases} \delta_{\eta} + \tau(\gamma - \kappa\theta_1)(1 - \theta_2) > 0 \\ \phi_{\eta} + \sigma(p, \dot{U}) + \tau \left[(\gamma - \kappa\theta_1) \left(1 - \frac{1}{\theta_2} \right) - \frac{\kappa}{\theta_1} + \beta \right] > 0. \end{cases}$$

If $\eta > 0$ then for each θ fixed there exists $\tau > 0$ sufficiently small such that these conditions are satisfied. This connected to the fact that the solution is unique as already established in Section 2.7. When $\eta = 0$ then $\delta_{\eta} = 0$ and $\phi_{\eta} = 0$. If $\sigma(p, \dot{U}) > 0$ then we can choose $\theta_2 < 1$ so that the first inequality is satisfied and for $\tau > 0$ sufficiently small also the second inequality is satisfied. If instead $\sigma(p, \dot{U}) = 0$ then the inequalities does not depend on τ and can be solved only if $\beta\gamma > \kappa^2$. Indeed from the first we have $\theta_2 < 1$ and from the second we get

$$\frac{1}{\theta_2} < 1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1}.$$

Then we can find $\theta_2 \in (0, 1)$ if and only if

$$1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1} > 1 \quad \text{or equivalently} \quad \beta - \frac{\kappa}{\theta_1} > 0.$$

Then we can find $\theta_1 \in (0, 1)$ if and only if $\beta\gamma > \kappa^2$. This means that hardening must be strong enough compared to the sensitivity of the yield strength on pressure (notice that κ is proportional to $\text{Lip}(Y)$).

REMARK 7.29. In the case of Von Mises plasticity the stiffness operator is an isomorphism for any positive hardening. This due to the fact that the additional term that comes from the dependence of the yield strength on the plastic pressure is absent.

Intuition about the localization process. We now assume that $\eta = 0$ and that G is a smooth and convex dissipation potential that satisfy $G(\dot{U}) = |\dot{U}| + c$ for all $\dot{U} \in \text{DevSym}$ with $|\dot{U}| > R$ where $c \in \mathbb{R}$ and $R > 0$ are two constants. This gives a plastic force that is independent on the rate and equal to the yield strength when the rate is sufficiently high. So it is a regularization of the rate independent dissipation potential that preserve the existence of a yield limit. We notice that

$$\partial^2 G(\dot{U}) = \frac{|\dot{U}|^2 I - \dot{U} \otimes \dot{U}}{|\dot{U}|^2} \quad \text{for } |\dot{U}| > R$$

so that $\text{se}(\partial^2 G(\dot{U})) = 0$ there. On the contrary $\text{se}(\partial^2 G(\dot{U}))$ must be very high for $\dot{U}$ near the origin if G is a very close approximation of the rate independent dissipation potential $|\cdot|$. We now consider a

loading process. When the load is still small the plastic force is under the yield limit and $\sigma(p, \dot{U})$ is high. By the analysis we have just performed this ensure that $\mathcal{K}_{(\tau, \eta)}^{(y, t)}(\dot{y}, p)$ is invertible and this suggest that the incremental problem is uniquely solved. Moreover we expect that the deformation rate it is not localized in some region but distributed all over the body. As the load increases $\text{se}(\partial^2 G(\dot{U}))$ decreases (uniformly in all the body). Then $\mathcal{K}_{(\tau, \eta)}^{(y, t)}(\dot{y}, p)$ is invertible only for small time steps τ . Assuming that the hardening is sufficiently small, we can conclude that $\mathcal{K}_{(\tau, \eta)}^{(y, t)}(\dot{y}, p)$ became singular (or is nonsingular only for very small time increments so that the evolution stagnates). This is the signal that we have reached the bifurcation point. Now the system to restore the invertibility of $\mathcal{K}_{(\tau, \eta)}^{(y, t)}(\dot{y}, p)$ and overcome the singularity point makes $\dot{U}$ small everywhere except that in a small shear band. This cause an elastic unloading elsewhere. Then out of the shear band $\text{se}(\partial^2 G(\dot{U}))$ becomes big again. This makes $a_{(\tau, \eta)}^{(\dot{y}, p, y, t)}$ coercive because out of the shear band the plastic strain is held back by the coercivity of $\partial^2 G(\dot{U})$. In the shear band the plastic strain is held back by the surrounding regions through the plastic strain gradient.

3. Numerical approximation

3.1. Formulation.

3.2. Validation. To validate the numerical scheme we solve the following boundary value problem that admits an analytical solution. Consider the domain $\Omega = (0, L) \times (-H, H)$ and for all $t \in [0, T]$ impose the boundary conditions $u_2(x, t) = \epsilon(t) \text{sign } x_2$ for $x_2 \in \{-H, H\}$ and $u_1(x, t) = 0$ for $x_1 = 0$. Here $\epsilon(t) = \frac{t}{T} \epsilon_0$ where $\epsilon_0 > 0$. Assume that no external volume forces or external tractions are imposed. We look for uniform solutions of the form

$$u(x, t) = \begin{pmatrix} \zeta(t)x_1 \\ \epsilon(t)x_2 \end{pmatrix}, \quad U_p(x, t) = \begin{pmatrix} -\gamma(t) & 0 \\ 0 & \gamma(t) \end{pmatrix} \quad \text{and} \quad p(x, t) = p(t).$$

The unknowns are ζ, γ, p from $[0, T]$ to $\mathbb{R}$. We easily compute

$$Eu = \begin{pmatrix} \zeta & 0 \\ 0 & \epsilon \end{pmatrix}, \quad \text{and} \quad E_e = Eu - U_p = \begin{pmatrix} \zeta + \gamma & 0 \\ 0 & \epsilon - \gamma \end{pmatrix}.$$

Assuming an isotropic elastic response, then

$$\begin{aligned} T_e &= 2\mu E_e + \lambda \text{tr } E_e I = \begin{pmatrix} 2\mu(\zeta + \gamma) + \lambda(\zeta + \epsilon) & 0 \\ 0 & 2\mu(\epsilon - \gamma) + \lambda(\zeta + \epsilon) \end{pmatrix} = \\ &= \begin{pmatrix} (2\mu + \lambda)\zeta + \lambda\epsilon + 2\mu\gamma & 0 \\ 0 & \lambda\zeta + (2\mu + \lambda)\epsilon - 2\mu\gamma \end{pmatrix}. \end{aligned}$$

The condition that the lateral traction vanish, impose that $(2\mu + \lambda)\zeta + \lambda\epsilon + 2\mu\gamma = 0$ so that

$$\zeta = -\frac{\lambda\epsilon + 2\mu\gamma}{2\mu + \lambda}.$$

The micropressure can be immediately computed as

$$p = -\frac{1}{2} \text{tr } T_e = -(\lambda + \mu)(\zeta + \epsilon) = -E(\epsilon - \gamma),$$

where $E = \frac{2\mu(\lambda + \mu)}{2\mu + \lambda}$. Now we consider the microforce balance. Since the solution is uniform the microstress vanish and we get $-\text{dev } T_e + T_p = 0$. We assume that

$$Y(p) = \begin{cases} 0 & \text{if } p \leq -\frac{Y_0}{\alpha} \\ Y_0 + \alpha p & \text{if } p > -\frac{Y_0}{\alpha} \end{cases}$$

Now

$$\text{dev } T_e = 2\mu \text{dev } E_e = \begin{pmatrix} \mu(\zeta - \epsilon + 2\gamma) & 0 \\ 0 & \mu(\epsilon - \zeta - 2\gamma) \end{pmatrix} = E(\epsilon - \gamma) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Before the yield point is reached, $\gamma = 0$ and the yield point is reached when $|\text{dev } T_e| = Y(p)$. This can be written explicitly as $\sqrt{2}E\epsilon = Y_0 - \alpha E\epsilon$, so that at yield point

$$\epsilon = \frac{Y_0}{(\sqrt{2} + \alpha)E}.$$

After that the yield point is reached $\dot{\gamma} > 0$ and the microforce balance reads

$$-\operatorname{dev} T_e + \mathfrak{B}U_p + \frac{1}{\sqrt{2}}Y(p) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = 0,$$

or equivalently

$$-\sqrt{2}E\epsilon + \sqrt{2}E\gamma + \sqrt{2}\mathfrak{B}\gamma + Y_0 - \alpha E\epsilon + \alpha E\gamma = 0.$$

This allows to compute γ as

$$\gamma = \frac{(\sqrt{2} + \alpha)E\epsilon - Y_0}{(\sqrt{2} + \alpha)E + \sqrt{2}\mathfrak{B}}.$$

This completes the computation of the solution.

Large deformation model

This chapter is not updated and incomplete. If I have time I will try to get an existence theorem also for the large strain model but I give the priority to understand better the localization of deformation in the small strain model.

1. Formulation

1.1. Kinematics. Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic deformation is a map $\chi: \Omega \rightarrow \mathbb{R}^d$ satisfying $\det \text{grad } \chi > 0$. We set $F = \text{grad } \chi$, $E = (F^T F - I)/2$ and $M = \text{grad}^2 \chi$. We subdivide $\partial\Omega$ in two disjoint parts, Γ_D and Γ_N . On Γ_D we impose the kinematical constrain $\chi = \bar{\chi}$. The distortion of the microscopic structure is described by the plastic distortion $F_p: \Omega \rightarrow \text{Lin}^+$. We regard F_p as the elastically relaxed distortion so we define the elastic distortion as $F_e = F F_p^{-1}$. The volumetric distortion of the microscopic structure is described by $J_p = \det F_p$ while the shear distortion by $U_p = J_p^{-1} F_p$. We impose the kinematical constrain $J_p = 1$ and to this aim we introduce the Lagrange multiplier p with the meaning of a plastic pressure. The kinematical constrain can be written also as

$$\int_{\Omega} (J_p - 1) \tilde{q} dx = 0 \quad \text{for all } \tilde{q}: \Omega \rightarrow \mathbb{R}.$$

1.2. Statics. We denote with $\tilde{v}$ the virtual rate of u .

Let $\Omega \subset \mathbb{R}^3$ be the reference configuration of the body. The deformation of the body is a map $\chi: \Omega \rightarrow \mathbb{R}^d$. We set $v = \dot{\chi}$ and $F = \text{Grad } \chi$ and $L = \dot{F} F^{-1}$. We subdivide $\partial\Omega$ into disjoint parts Γ_D and Γ_N . The plastic distortion is a map $F_p: \Omega \rightarrow \text{Lin}^+$. We set $L_p = \dot{F}_p F_p^{-1}$ and we assume that $L_p = D_p \in \text{Sym}$. We set $U_p = \text{dev } D_p$ and $M_p = \text{Grad } U_p \cdot F_p^{-1}$. The elastic distortion is $F_e = F F_p^{-1}$ and we set $L_e = \dot{F}_e F_e^{-1}$.

LEMMA 8.1. It holds $L = L_e + F_e D_p F_e^{-1}$.

PROOF. We have $L = \dot{F} F^{-1} = (\dot{F}_e F_p + F_e \dot{F}_p) F_p^{-1} F_e^{-1} = \dot{F}_e F_e^{-1} + F_e \dot{F}_p F_p^{-1} F_e^{-1} = L_e + F_e D_p F_e^{-1}$. $\square$

We add the kinematical constrain $\text{tr } D_p = 0$. To impose this constrain we consider a Lagrange multiplier $p: \Omega \rightarrow \mathbb{R}$. The constrain can be written in weak form as

$$- \int_{\Omega} \text{tr } L_p \tilde{q} dX = 0 \quad \text{for all } \tilde{q}: \Omega \rightarrow \mathbb{R}.$$

1.3. Statics. We choose an internal virtual power of the form

$$\tilde{\mathcal{W}} = \int_{\Omega} T : \tilde{L}_e dX + \int_{\Omega} \left(T_p : \tilde{U}_p - p \text{tr } \tilde{L}_p + K_p : \tilde{M}_p \right) dX.$$

Here T is the macrostress, T_p is the microstress which can be assumed deviatoric and K_p is the microhyperstress. We choose an external virtual power of the form

$$\tilde{\mathcal{P}} = \int_{\Omega} f \cdot \tilde{v} dX + \int_{\Gamma_N} h \cdot \tilde{v} dA + \int_{\Gamma_D} r \cdot \tilde{v} dA.$$

DEFINITION 8.2 (Principle of Virtual Powers). Equilibrium holds when $\tilde{\mathcal{W}} = \tilde{\mathcal{P}}$ for all virtual rates.

To get the local version of the equilibrium condition we apply Lemma 8.1 and we write the internal virtual power as

$$\tilde{\mathcal{W}} = \int_{\Omega} T F^{-T} : \text{Grad } \tilde{v} dX + \int_{\Omega} \left(-T_e : \tilde{L}_p + T_p : \tilde{U}_p - p \text{tr } \tilde{L}_p + K_p : \tilde{M}_p \right) dX$$

where $T_e = F_e^T T F_e^{-T}$ is the Mandel stress. Then we apply the Green theorem

$$\begin{aligned} \tilde{W} = & - \int_{\Omega} \text{Div}(TF^{-T}) : \text{Grad } \tilde{v} dX + \int_{\partial\Omega} (TF^{-T})N \cdot \tilde{v} dA + \\ & + \int_{\Omega} \left\{ \left[T_p - \text{dev } T_e - \text{Div} \left(K_p \cdot F_p^{-T} \right) \right] : \tilde{U}_p - \left[\frac{1}{d} \text{tr } T_e + p \right] \text{tr } \tilde{L}_p \right\} dX + \\ & + \int_{\partial\Omega} \left(K_p \cdot F_p^{-T} N \right) : \tilde{U}_p dA. \end{aligned}$$

Then the local form of the equilibrium condition is

$$\begin{aligned} \text{Div}(TF^{-T}) + f &= 0 \quad \text{in } \Omega \\ (TF^{-T})N &= r \quad \text{in } \Gamma_D \\ (TF^{-T})N &= h \quad \text{in } \Gamma_N \\ T_p - \text{dev } T_e - \text{Div} \left(K_p \cdot F_p^{-T} \right) &= 0 \quad \text{in } \Omega \\ \frac{1}{d} \text{tr } T_e + p &= 0 \quad \text{in } \Omega \\ K_p \cdot F_p^{-T} N &= 0 \quad \text{in } \partial\Omega. \end{aligned}$$

Collection of useful results

APPENDIX A

Notation

1. Tensors

$\text{Lin} = \mathbb{R}^{d \times d}$	$d \times d$ matrices
Lin^+	$d \times d$ matrices with positive determinant
Sym	$d \times d$ symmetric matrices
Dev	$d \times d$ deviatoric matrices
DevSym	$d \times d$ symmetric and deviatoric matrices

APPENDIX B

Tools from mathematical analysis

In this chapter we collect some tools of linear and nonlinear analysis used in the main body of this thesis.

1. Some classical results

1.1. A general form of the Gronwall inequality.

PROPOSITION B.1. (Gronwall inequality) Let $z \in L^\infty((0, T))$. Let $C \in \mathbb{R}$ and α, β in $L^1((0, T), [0, \infty))$ such that

$$y(t) \leq C + \int_0^t (\alpha(s)y(s) + \beta(s)) ds \quad \text{for a.e. } t \in (0, T).$$

Then

$$z(t) \leq e^{\int_0^t \alpha(s) ds} \left(C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r) dr} ds \right) \quad \text{for a.e. } t \in (0, T).$$

PROOF. Let

$$Z(t) = C + \int_0^t (\alpha(s)z(s) + \beta(s)) ds$$

so that $z(t) \leq Z(t)$. Clearly $Z \in AC([0, T])$. It holds that

$$\dot{Z}(t) = \alpha(t)z(t) + \beta(t) \leq \alpha(t)Z(t) + \beta(t) \quad \text{for a.e. } t \in (0, T).$$

Multiplying by $e^{-\int_0^t \alpha(s) ds}$ one gets

$$\frac{d}{dt} \left(Z(t) e^{-\int_0^t \alpha(s) ds} \right) \leq \beta(t) e^{-\int_0^t \alpha(s) ds}$$

and integrating on $(0, t)$

$$Z(t) e^{-\int_0^t \alpha(s) ds} \leq C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r) dr} ds$$

Then

$$z(t) \leq Z(t) \leq e^{\int_0^t \alpha(s) ds} \left(C + \int_0^t \beta(s) e^{-\int_0^s \alpha(r) dr} ds \right) \quad \text{for a.e. } t \in (0, T).$$

□

2. Linear functional analysis

2.1. Other.

PROPOSITION B.2. Let $\Omega \subset \mathbb{R}^d$ be open, bounded and Lipschitz. Let $p \in [1, \infty]$ and let $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$. Then $u_n \rightarrow u$ in $L^p(\Omega)$. Moreover if $p > d$ then $u_k \rightarrow u$ in $C^{0,\alpha}(\overline{\Omega})$ where $\alpha = 1 - \frac{d}{p}$.

PROOF. Since $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$ then u_n are bounded in $W^{1,p}(\Omega)$. By Rellich–Kondrakov theorem each subsequence of u_n admits a further subsequence that converges strongly in $L^p(\Omega)$. Since we already know the weak convergence in $W^{1,p}(\Omega)$ then the limit is always u . Then necessarily the whole sequence u_n converges strongly to u in $L^2(\Omega)$.

The result when $p > d$ follows from Morrey’s theorem and a similar argument. □

2.2. Bochner spaces.

DEFINITION B.3. Let V be a Banach space. We say that $u: [0, T] \rightarrow V$ is absolutely continuous and we write $u \in AC([0, T], V)$ if there exists $m \in L^1((0, T), (0, \infty))$ such that

$$\|u(t) - u(s)\| \leq \int_0^T m(r) dr \quad \text{for all } 0 \leq s < t \leq T.$$

Given $u: [0, T] \rightarrow V$ and $0 \leq a < b \leq T$ we define the total variation on $[a, b]$

$$\text{Var}(u, [a, b]) = \sup \left\{ \sum_{k=1}^N \|u(t_k) - u(t_{k-1})\| \mid a = t_0 < \dots < t_N = b \right\}.$$

We say that u has bounded variation and we write $u \in BV([0, T], V)$ if $\text{Var}(u, [0, T]) < \infty$.

PROPOSITION B.4. Let V be a Banach space. Then $W^{1,1}((0, T), V) \subset AC([0, T], V)$. If V is separable and reflexive the equality holds.

PROOF. Let $u \in W^{1,1}((0, T), V)$. Then

$$\|u(t) - u(s)\| = \left\| \int_s^t \dot{u}(r) dr \right\| \leq \int_s^t \|\dot{u}(r)\| dr.$$

This proves that $u \in AC([0, T], V)$ with $m = \|\dot{u}\|$.

Now suppose that V is reflexive. We have that

$$\left\| \frac{u(t+h) - u(t)}{h} \right\| \leq \frac{1}{h} \int_t^{t+h} m(r) dr.$$

By the Lebesgue differentiation theorem the left hand side is bounded as $h \rightarrow 0$ for a.e. $t \in (0, T)$. Then for each $h_n \rightarrow 0$, up to extracting a subsequence,

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{for some } v \in V.$$

We prove that v is common to each sequence. Since V^* is separable and reflexive then there exists ξ_i dense in V^* . Now for each i the map $t \mapsto \langle \xi_i, u(t) \rangle$ is $AC([0, T])$. Then for a.e. $t \in (0, T)$

$$(B.1) \quad \lim_{h \rightarrow 0} \left\langle \xi_i, \frac{u(t+h) - u(t)}{h} \right\rangle \quad \text{exists for all } i.$$

Then if $h_n \rightarrow 0$ and $k_n \rightarrow 0$ are such that

$$\frac{u(t+h_n) - u(t)}{h_n} \rightharpoonup v \quad \text{and} \quad \frac{u(t+k_n) - u(t)}{k_n} \rightharpoonup w$$

from (B.1) it follows that $\langle \xi_i, v - w \rangle = 0$ for all i . This implies $v = w$. Up to now we have proved that there exists $v: (0, T) \rightarrow V$ such that

$$\frac{u(t+h) - u(t)}{h} \rightharpoonup v(t) \quad \text{for a.e. } t \in (0, T).$$

Now one should prove that this convergence is actually strong. This will show automatically that $u \in W^{1,1}((0, T), V)$. □

THEOREM B.5 (Aubin–Lions–Simon lemma). Let $T > 0$ and $V \subset H \subset E$ be Banach spaces. Assume that the embedding of V in H is compact and the embedding of H in E is compact. Let p and q in $[1, \infty]$. Then the space

$$\{u \in L^p((0, T), V) \mid \dot{u} \in L^q((0, T), E)\}$$

is compactly embedded in

- (i) $L^p((0, T), H)$ if $p < \infty$
- (ii) $C([0, T], H)$ if $p = \infty$ and $q > 1$.

PROOF. See [8, Theorem II.5.16]. □

2.3. Generalized mixed elliptic problems. We start with the following theorem.

THEOREM B.6 (Babuška). Let V and Q be reflexive Banach spaces. Let $b: Q \times V \rightarrow \mathbb{R}$ be bilinear form. Let $Z = \{z \in V \mid b(q, z) = 0 \text{ for all } q \in Q\}$ and define $W = V/Z$. Assume that b is continuous so that there exists $M > 0$ such that

$$b(q, v) \leq M \|q\|_Q \|v\|_V \quad \text{for all } (q, v) \in Q \times V,$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, w)}{\|q\|_Q} \geq \beta \|w\|_W$$

for all $w \in W$ and that if $q \in Q$ satisfies

$$b(q, v) = 0 \quad \text{for all } v \in V$$

then $q = 0$. Then for each $f \in Q'$ there exists a unique $w \in W$ such that

$$b(q, w) = \langle f, q \rangle \quad \text{for all } q \in Q$$

and

$$\|w\|_W \leq \frac{1}{\beta} \|f\|_{Q'}.$$

PROOF. *Step 1.* Consider the linear operator $B: W \rightarrow Q'$ defined by

$$\langle Bw, q \rangle = b(w, q).$$

We notice that

$$|\langle Bw, q \rangle| = \inf_{z \in Z} |b(w + z, q)| \leq M \|q\|_Q \inf_{z \in Z} \|w + z\|_V = M \|q\|_Q \|w\|_W$$

so that $B \in \mathcal{L}(W, Q')$.

Step 2. We prove that B is injective. Indeed if $w \in W$ satisfies $Bw = 0$ we have

$$\beta \|w\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q \rangle}{\|q\|_Q} = 0$$

so that $w = 0$.

Step 3. We prove that $\text{range}(B)$ is closed. Indeed let $f_n = Bw_n$ be such that $f_n \rightarrow f$ in Q' . Then f_n is a Cauchy sequence. Now

$$\beta \|w_n - w_m\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle B(w_n - w_m), q \rangle}{\|q\|_Q} \leq \|f_n - f_m\|_{Q'}.$$

Then also w_n is a Cauchy sequence in W and $w_n \rightarrow w$ in W for some $w \in W$. Now by continuity of B we have $Bw = f$ so that $f \in \text{range}(B)$.

Step 4. We prove that $\text{range}(B) = Q'$. Indeed let $(B^T) = \{q \in Q \mid b(q, v) = 0 \text{ for all } v \in V\} = \{0\}$ by hypotheses. On the other hand $(B^T)^\perp = \text{range}(B)$. Since Q is reflexive $\text{range}(B) = (\text{range}(B)^\perp)^\perp = Q'$.

Step 5. We have proved that B is a bijection from W to Q' . Then for each $f \in Q'$ there exists a unique $w \in W$ such that $Bw = f$. Moreover

$$\beta \|w\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q \rangle}{\|q\|_Q} = \sup_{q \in Q \setminus \{0\}} \frac{\langle f, q \rangle}{\|q\|_Q} \leq \|f\|_{Q'}.$$

This completes the proof. □

LEMMA B.7. Assume that the hypotheses of Theorem B.6 holds. Then

$$\sup_{w \in W \setminus \{0\}} \frac{b(q, w)}{\|w\|_W} \geq \beta \|q\|_Q$$

for all $q \in Q$.

PROOF. For each $f \in Q'$ let $w_f \in W$ the solution to

$$b(q, w) = \langle f, q \rangle \quad \text{for all } q \in Q.$$

Notice that $\|f\|_{Q'} \geq \beta \|w\|_W$. Now

$$\|q\|_Q = \sup_{f \in Q' \setminus \{0\}} \frac{\langle f, q \rangle}{\|f\|_{Q'}} = \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\|f\|_{Q'}} \leq \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\beta \|w_f\|_W} \leq \sup_{w \in W \setminus \{0\}} \frac{b(q, w)}{\beta \|w\|_W}.$$

This completes the proof. □

Then we get immediately the following proposition.

COROLLARY B.8. Assume the hypotheses of Theorem B.6. Then for each $g \in W'$ there exists a unique $p \in Q$ such that

$$b(p, w) = \langle g, w \rangle \quad \text{for all } w \in W$$

and

$$\|p\|_Q \leq \frac{1}{\beta} \|g\|_{W'}.$$

PROOF. Apply first Lemma B.7 and then Theorem B.6 with the role of W and Q swapped. $\square$

REMARK B.9. Assume that $g \in V'$ is such that $\langle g, z \rangle = 0$ for all $z \in Z$. Then we can define g as an element of W' . Indeed for each $w \in W$ we take a representative $v \in V$ still denoted with w and we define $\langle g, w \rangle = \langle g, v \rangle$. The definition does not depend on the choice of the representative. Moreover

$$\|g\|_W = \sup_{w \in W \setminus \{0\}} \frac{\langle g, w \rangle}{\|w\|_W} = \sup_{v \in V \setminus \{0\}} \frac{\langle g, v \rangle}{\|v\|_V} = \|g\|_{V'}.$$

where we use that since V is reflexive for each $w \in W$ there exists a representative $v \in V$ such that $\|w\|_W = \|v\|_V$ and that for any other representative $u \in V$ we have $\|w\|_W \leq \|u\|_V$. In this case the Corollary B.8 can be formulated as follows. There exists a unique $p \in Q$ such that

$$b(p, v) = \langle g, v \rangle \quad \text{for all } v \in V$$

and

$$\|p\|_Q \leq \frac{1}{\beta} \|g\|_{V'}.$$

The following theorem is the main result of this Section. The result is proved in [33] in the Hilbert spaces setting and we have generalized it for Banach spaces.

THEOREM B.10. Let $V, \tilde{V}, Q$ and $\tilde{Q}$ be reflexive Banach spaces. Let $b: Q \times \tilde{V} \rightarrow \mathbb{R}$ be a bilinear form. Let $\tilde{Z} = \{\tilde{z} \in \tilde{V} \mid b(q, \tilde{z}) = 0 \text{ for all } q \in Q\}$ and define $\tilde{W} = \tilde{V} / \tilde{Z}$. Assume that there exists $M > 0$ such that

$$b(q, \tilde{v}) \leq M \|q\|_Q \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } q \in Q \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, \tilde{w})}{\|q\|_Q} \geq \beta \|\tilde{w}\|_{\tilde{W}} \quad \text{for all } \tilde{w} \in \tilde{W}$$

and that if $q \in Q$ satisfies

$$b(q, \tilde{v}) = 0 \quad \text{for all } \tilde{v} \in \tilde{V}$$

then $q = 0$. Let $c: \tilde{Q} \times V \rightarrow \mathbb{R}$ be a bilinear form. Let $Z = \{z \in V \mid b(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \tilde{Q}\}$ and define $W = V / Z$. Assume that there exists $N > 0$ such that

$$c(\tilde{q}, v) \leq N \|\tilde{q}\|_{\tilde{Q}} \|v\|_V \quad \text{for all } \tilde{q} \in \tilde{Q} \text{ and } v \in V.$$

Assume also that there exists $\gamma > 0$ such that

$$\sup_{\tilde{q} \in \tilde{Q} \setminus \{0\}} \frac{c(\tilde{q}, w)}{\|\tilde{q}\|_{\tilde{Q}}} \geq \gamma \|w\|_W \quad \text{for all } w \in W$$

and that if $\tilde{q} \in \tilde{Q}$ satisfies

$$c(\tilde{q}, v) = 0 \quad \text{for all } v \in V$$

then $\tilde{q} = 0$. Let $a: V \times \tilde{V} \rightarrow \mathbb{R}$ be a bilinear form. Assume that there exists a constant L such that

$$a(v, \tilde{v}) \leq L \|v\|_V \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } v \in V \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\alpha > 0$ such that

$$\sup_{\tilde{v} \in \tilde{V} \setminus \{0\}} \frac{a(z, \tilde{v})}{\|\tilde{v}\|_{\tilde{V}}} \geq \alpha \|z\|_V \quad \text{for all } z \in Z$$

and that if $\tilde{z} \in \tilde{Z}$ satisfies

$$a(v, \tilde{z}) = 0 \quad \text{for all } v \in V$$

then $\tilde{z} = 0$. Then for all $f \in \tilde{V}'$ and $g \in \tilde{Q}'$ there exists a unique $(u, p) \in V \times Q$ such that

$$\begin{aligned} a(u, \tilde{v}) + b(p, \tilde{v}) &= \langle f, \tilde{v} \rangle \quad \text{for all } \tilde{v} \in \tilde{V} \\ c(\tilde{q}, u) &= \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}. \end{aligned}$$

Moreover

$$\begin{aligned} \|u\|_V &\leq \frac{1}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'} + \frac{1}{\alpha} \|f\|_{\tilde{V}'}, \\ \|p\|_Q &\leq \frac{1}{\beta} \left(1 + \frac{L}{\alpha}\right) \|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'}. \end{aligned}$$

PROOF. By Theorem B.6 there exists a unique $w \in W$ such that

$$c(\tilde{q}, w) = \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}$$

and

$$\|w\|_W \leq \frac{1}{\gamma} \|g\|_{\tilde{Q}'}.$$

We identify w with one of its representative in V and we can choose it such that $\|w\|_V = \|w\|_W$. We notice that

$$a(w, \tilde{v}) \leq L \|w\|_V \|\tilde{v}\|_{\tilde{V}}.$$

Then $a(w, \cdot) \in \tilde{V}' \subset \tilde{Z}'$. Then $\langle f, \cdot \rangle - a(w, \cdot) \in \tilde{Z}'$. Then by Theorem B.6 the equation

$$a(z, \tilde{z}) = \langle f, \tilde{z} \rangle - a(w, \tilde{z}) \quad \text{for all } \tilde{z} \in \tilde{Z}$$

has a unique solution $z \in Z$ that satisfies

$$\|z\|_V \leq \frac{1}{\alpha} \left(\|f\|_{\tilde{V}'} + \frac{L}{\gamma} \|g\|_{\tilde{Q}'} \right).$$

Now define $u = w + z \in V$ and we notice that

$$(B.2) \quad a(u, \tilde{z}) = \langle f, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \tilde{Z}$$

and

$$\|u\|_V \leq \frac{1}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'} + \frac{1}{\alpha} \|f\|_{\tilde{V}'}.$$

Now we notice by (B.2) we have

$$\langle f, \tilde{z} \rangle - a(u, \tilde{z}) = \langle f, \tilde{z} \rangle - a(w, \tilde{z}) - a(z, \tilde{z}) = 0 \quad \text{for all } \tilde{z} \in \tilde{Z}.$$

Then by Remark B.9 we have $\langle f, \cdot \rangle - a(u, \cdot) \in \tilde{W}'$ and the equation

$$b(p, \tilde{v}) = \langle f, \tilde{v} \rangle - a(u, \tilde{v}) \quad \text{for all } \tilde{v} \in \tilde{V}$$

has a unique solution $p \in Q$ that satisfies

$$\|p\|_Q \leq \frac{1}{\beta} (\|f\|_{\tilde{V}'} + L \|u\|_V) \leq \frac{1}{\beta} \left(1 + \frac{L}{\alpha}\right) \|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'}.$$

This concludes the proof. $\square$

3. Nonlinear functional analysis

3.1. Other.

LEMMA B.11. Let X and Y be Banach spaces. Let U an open subset of X and $F: U \rightarrow Y'$. Let $x \in U$. Assume that there exists $A \in \mathcal{L}(X, Y')$ and for all $v \in X$ there exists $\omega: \mathbb{R} \rightarrow [0, \infty)$ with $\omega(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ such that

$$\left\| \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av, y \right\| \leq \omega(\epsilon) \|y\|_Y$$

for all $y \in Y$ and $\epsilon \in \mathbb{R}$ sufficiently small. Then F is Gateaux differentiable in x with Gateaux differential A .

PROOF. Fix $v \in X$. Then by hypotheses there exists $\omega: \mathbb{R} \rightarrow [0, \infty)$ with $\omega(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ such that

$$\left\| \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av \right\|_{Y'} = \sup_{y \in Y \setminus \{0\}} \frac{\left\langle \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av, y \right\rangle}{\|y\|_Y} \leq \omega(\epsilon).$$

Then

$$\left\| \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av \right\|_{Y'} \rightarrow 0$$

as $\epsilon \rightarrow 0$. This is exactly the thesis. $\square$

LEMMA B.12. Let X and Y be two Banach spaces. Let A_n and A in $\mathcal{L}(X, Y')$ such that for all $x \in X$ and $y \in Y$ we have

$$|\langle A_n x - Ax, y \rangle| \leq \omega_n \|x\|_X \|y\|_Y$$

where $\omega_n \rightarrow 0$. Then $A_n \rightarrow A$ in $\mathcal{L}(X, Y')$.

PROOF. We have

$$\|A_n - A\|_{\mathcal{L}(X, Y')} = \sup_{x \in X \setminus \{0\}} \frac{\|A_n x - Ax\|_{Y'}}{\|x\|_X} = \sup_{x \in X \setminus \{0\}} \sup_{y \in Y \setminus \{0\}} \frac{|\langle A_n x - Ax, y \rangle|}{\|x\|_X \|y\|_Y} \leq \omega_n \rightarrow 0.$$

This concludes the proof. $\square$

PROPOSITION B.13. Let V a Banach space and let $f \in C^1(V)$ and $K \subset V$ compact. Then f is Lipschitz continuous on K .

PROOF. By contradiction let u_n and v_n two sequences in K such that

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \geq n.$$

By compactness of K up to extracting subsequences we can assume that $u_n \rightarrow u$ and $v_n \rightarrow v$. If $u \neq v$ then

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} \rightarrow \frac{|F(u) - F(v)|}{\|u - v\|} < \infty$$

which is a contradiction. If $u = v$ then

$$\frac{|F(u_n) - F(v_n)|}{\|u_n - v_n\|} = \frac{|\langle DF(u), u_n - v_n \rangle + o(\|u_n - v_n\|)|}{\|u_n - v_n\|} \leq \|DF(u)\| + o(1) \rightarrow \|DF(u)\| < \infty$$

which is a contradiction. $\square$

3.2. Implicit function theorem.

THEOREM B.14. Let X, Y and E be Banach spaces. Let U and open set of $X \times Y$ and $(x_0, y_0) \in U$. Let $F \in C^1(U, E)$ such that $F(x_0, y_0) = 0$ and $D_x F(x_0, y_0) \in \text{Iso}(X, E)$. Then we can find $r > 0$ and $s > 0$ with $B(x_0, r) \times B(y_0, s) \subset U$ such that there exist a unique $\chi \in C^1(B(y_0, s), B(x_0, r))$ satisfying

$$F(\chi(y), y) = 0 \quad \text{for all } y \in B(y_0, s).$$

Moreover if F is of class C^r with $r \geq 2$ also χ is of class C^r .

PROOF. See [34]. $\square$

3.3. Fixed point theorems.

THEOREM B.15 (Schauder fixed point theorem). Let X be a Banach space and $C \subset X$ closed and convex. Let $F: C \rightarrow C$ continuous and with compact range. Then F admits a fixed point.

PROOF. See [34]. $\square$

3.4. Local existence theorem for Cauchy problems in Banach spaces.

THEOREM B.16. Let X be a Banach space and let U be an open subset of $X \times \mathbb{R}$ with $(x_0, 0) \in U$. Let $G \in C^1(U, X)$. Then there exists $T > 0$ such that the Cauchy problem

$$\begin{cases} \dot{x}(t) = G(x(t), t) & \text{for } t \in [0, T] \\ x(0) = x_0 \end{cases}$$

admits a unique solution $x \in C^2([0, T], X)$.

PROOF. Since G is of class C^1 there exists a neighborhood $V \subset U$ of $(x_0, 0)$ such that G and $D_x G$ are bounded in V . Let $M = \sup_V \|G\|_X < \infty$ and $L = \sup_V \|D_x G\|_{\mathcal{L}(X, X)}$. We notice that G is Lipschitz continuous with respect to x and uniformly in t on V with Lipschitz constant L . Let $B = B(x_0, r)$ for some $r > 0$ and $T > 0$ such that $B(x_0, r) \times (-T, T) \subset V$. We introduce the normed space $\mathcal{X} = C^0([0, T], B)$ endowed with the norm $\|x\|_{\mathcal{X}} = \sup_{t \in [0, T]} \|x(t)\|_X$. Since $\mathcal{X}$ is a closed subset of $C^0([0, T], B)$ it is a complete metric space. Now introduce the map

$$T: B \rightarrow C^0([0, T], B)$$

defined by

$$T(x)(t) = x_0 + \int_0^t G(x(s), s) ds.$$

We notice that $\|T(x) - x_0\|_X \leq TM$. Up to choose a smaller T we can assume that $TM < r$ so that actually $T: B \rightarrow B$. Moreover for any x_1 and x_2 in X we have

$$\|T(x_2) - T(x_1)\|_{\mathcal{X}} \leq \int_0^T \|G(x_2(s), s) - G(x_1(s), s)\|_X ds \leq TL \|x_2 - x_1\|_{\mathcal{X}}.$$

Up to choose a smaller T we can assume that $TL < 1$. In this case T is a strict contraction. By the Banach–Caccioppoli fixed point theorem there exists a unique $x \in B$ such that $Tx = x$. Since Tx is of class C^2 then also x is of class C^2 . $\square$

3.5. Browder–Minty theorem.

The results of this section are taken from [38] and [39]. Let X be a Banach space and $T: X \rightarrow X'$ be a nonlinear operator and $f \in X'$. The Browder–Minty theorem allows to study the existence of solutions to the equation $T(x) = f$ when there is not any variational formulation. The key assumption is that T is coercive and monotone, a property that makes T similar in some sense to an elliptic operator.

DEFINITION B.17. Let X be a Banach space and let $T: X \rightarrow X'$. We say that

- (i) T is bounded if maps bounded sets of X into bounded sets of X' ,
- (ii) T is coercive if

$$\frac{\langle T(x), x \rangle}{\|x\|} \rightarrow \infty \quad \text{as } \|x\| \rightarrow \infty$$

and that

- (iii) T is monotone if

$$\langle T(x) - T(y), x - y \rangle \geq 0 \quad \text{for all } x \text{ and } y \text{ in } X.$$

We now prove a lemma that establish the equivalence between the equation $T(x) = f$ and a variational inequality under the assumption that T is monotone.

LEMMA B.18. Let $T: X \rightarrow X'$ be continuous and $f \in X'$. Then if $x \in X$ is a solution to the variational inequality

$$\langle T(y) - f, y - x \rangle \geq 0 \quad \text{for all } y \in X$$

it is also a solution to the equation $T(x) = f$. Moreover if T is monotone then any solution to the equation $T(x) = f$ is also a solution to the variational inequality.

PROOF. Let $y = x + tv$ with $t > 0$ and $v \in X$. Then if $x \in X$ is a solution to the variational inequality we have

$$\langle T(x + tv) - f, v \rangle \geq 0.$$

Letting $t \rightarrow 0$ we have $\langle T(x) - f, v \rangle \geq 0$. By arbitrariness of v we have $T(x) = f$. Conversely let $x \in X$ be such that $T(x) = f$. By monotonicity

$$\langle T(y) - f, y - x \rangle = \langle T(y) - T(x), y - x \rangle \geq 0.$$

Then x is a solution to the variational inequality. $\square$

Now we study the equation $T(x) = f$ when the Banach space is finite dimensional. This results will be used to get the general result using the Galerkin method.

PROPOSITION B.19. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous and coercive. Then for every $f \in \mathbb{R}^n$ the equation $T(x) = f$ has at least one solution.

PROOF. Without loss of generality we can assume that $f = 0$. Indeed otherwise define $S(x) = T(x) - f$ and notice that S is still continuous and coercive. Indeed

$$\frac{S(x) \cdot x}{|x|} = \frac{T(x) \cdot x}{|x|} - \frac{f \cdot x}{|x|} \geq \frac{T(x) \cdot x}{|x|} - |f|.$$

Now define $G: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $G(x) = x - T(x)$. We notice that

$$G(x) \cdot x = |x|^2 - T(x) \cdot x.$$

Since T is coercive there exists an $r > 0$ such that $|x|^2 - G(x) \cdot x = T(x) \cdot x > 0$ for all $x \in \mathbb{R} \setminus B(0, r)$. We define $\Pi: \mathbb{R}^n \rightarrow C$ with $C = \overline{B}(0, r)$ by

$$\Pi(x) = \begin{cases} x & \text{if } |x| \leq r \\ r \frac{x}{|x|} & \text{if } |x| > r. \end{cases}$$

We set $H(x) = \Pi(G(x))$ and we notice that $H: C \rightarrow C$ and it is continuous. By Brouwer fixed point theorem there exists $x \in C$ such that $H(x) = x$. If $x \in \overset{\circ}{C}$ then $x = H(x) = \Pi(G(x)) = G(x)$ so that $T(x) = 0$. If we prove that $x \notin \partial C$ we have concluded. By contradiction assume that $|x| = r$. Then $|G(x)| \geq r$ and

$$|x|^2 = x \cdot x = H(x) \cdot x = \frac{r}{|G(x)|} G(x) \cdot x < \frac{r}{|G(x)|} |x|^2 \leq |x|^2$$

where we used that $|x|^2 > G(x) \cdot x$ for $|x| \geq r$. This is a contradiction. $\square$

We are now in position to prove the Browder–Minty theorem.

THEOREM B.20. Let X be a reflexive and separable Banach space. Let $T: X \rightarrow X'$ be bounded, continuous, coercive and monotone. Then for every $f \in X'$ there exists a solution $x \in X$ to the equation $T(x) = f$.

PROOF. As anticipated the proof relies on the Galerkin method.

Step 1. Let $\{y_n\} \subset X$ such that the linear combinations of y_n are dense in X . Let

$$X_n = \text{span} \{y_m\}_{m=1}^n.$$

We define $f_n \in X'_n$ by $\langle f_n, x \rangle = \langle f, x \rangle$ for all $x \in X_n$. We also define $T_n: X_n \rightarrow X'_n$ by $\langle T_n(x), y \rangle = \langle T(x), y \rangle$ for all x and y in X_n .

Step 2. We prove that T_n are still bounded, continuous and coercive. For all $x \in X_n$ we have

$$\|T_n(x)\|_{X'_n} = \sup_{y \in X_n \setminus \{0\}} \frac{\langle T_n(x), y \rangle}{\|y\|_{X_n}} \leq \sup_{y \in X \setminus \{0\}} \frac{\langle T(x), y \rangle}{\|y\|_X} = \|T(x)\|_X$$

so the boundedness of T_n follows from the boundedness of T . Moreover let $x_k \rightarrow x$ in X_n . Then

$$\begin{aligned} \|T_n(x_k) - T_n(x)\|_{X'_n} &= \sup_{y \in X_n \setminus \{0\}} \frac{\langle T_n(x_k) - T_n(x), y \rangle}{\|y\|_{X_n}} \leq \sup_{y \in X \setminus \{0\}} \frac{\langle T(x_k) - T(x), y \rangle}{\|y\|_X} = \\ &= \|T(x_k) - T(x)\|_{X'} \rightarrow 0 \end{aligned}$$

and this proves that T_n is continuous. Moreover for $x \in X_n$

$$\frac{\langle T_n(x), x \rangle}{\|x\|_{X_n}} = \frac{\langle T(x), x \rangle}{\|x\|_X}$$

so the coercivity of T_n follows immediately from the coercivity of T . By B.19 we have that for each n there exists $x_n \in X_n$ that solves $T_n(x_n) = f_n$.

Step 3. We have the estimate

$$\frac{\langle T(x_n), x_n \rangle}{\|x_n\|_X} \leq \frac{\langle T_n(x_n), x_n \rangle}{\|x_n\|_X} = \frac{\langle f_n, x_n \rangle}{\|x_n\|_X} = \frac{\langle f, x_n \rangle}{\|x_n\|_X} \leq \|f\|.$$

Since T is coercive then x_n are bounded in X and since T is bounded then $T(x_n)$ are bounded in X' . Then up to a subsequence

$$\begin{aligned} x_n &\rightharpoonup x \quad \text{in } X \\ T(x_n) &\overset{*}{\rightharpoonup} g \quad \text{in } X' \end{aligned}$$

for some $x \in X$ and $g \in X'$.

Step 4. We first prove that $g = f$. To this aim we fix any y_m and we notice that

$$\langle g - f, y_m \rangle = \lim_{n \rightarrow \infty} \langle T(x_n) - f, y_m \rangle = 0.$$

Then by density we have indeed $g = f$. Next we have

$$\langle T(x_n), x_n \rangle = \langle T_n(x_n), x_n \rangle = \langle g_n, x_n \rangle = \langle g, x_n \rangle \rightarrow \langle g, x \rangle.$$

Step 5. Eventually using the monotonicity of T we have

$$0 \geq \langle T(y) - T(x_n), y - x_n \rangle \rightarrow \langle T(y) - g, y - x \rangle = \langle T(y) - f, y - x \rangle.$$

Applying Lemma B.18 we get the thesis. $\square$

Actually to obtain the same existence result the monotonicity assumption can be weakened.

DEFINITION B.21. Let X be a Banach space and let $T: X \rightarrow X'$. We say that T is pseudomonotone if it is bounded and every time

$$x_k \rightharpoonup x \text{ in } X \quad \text{and} \quad \limsup_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \leq 0$$

then

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$.

THEOREM B.22. Let X be a separable and reflexive Banach space. Let $T: X \rightarrow X'$ be continuous, coercive and pseudomonotone. Then for every $f \in X'$ there exists $x \in X$ such that $T(x) = f$.

PROOF. The proof is exactly the same as the one of Theorem B.20 except to Step 5. We know that $x_k \rightharpoonup x$ in X , that $T(x_k) \xrightarrow{*} f$ in X' and $\langle T(x_k), x_k \rangle \rightarrow \langle f, x \rangle$. Then in particular

$$\langle T(x_k), x_k - x \rangle = \langle T(x_k), x_k \rangle - \langle T(x_k), x \rangle \rightarrow \langle f, x \rangle - \langle f, x \rangle = 0.$$

Since T is pseudomonotone then

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$. But

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle = \langle f, x - y \rangle.$$

Then

$$\langle f, x - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$. By arbitrariness of y we have $T(x) = f$. $\square$

The following proposition establish the relationship between monotone and pseudomonotone operators.

PROPOSITION B.23. Let X be a Banach space. Let $T: X \rightarrow X'$ be continuous, bounded and monotone. Then T is pseudomonotone.

PROOF. Let $x_k \rightharpoonup x$ with

$$\limsup_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \leq 0.$$

Since T is monotone then

$$\langle T(x_k), x_k - x \rangle \geq \langle T(x), x_k - x \rangle \rightarrow 0.$$

This implies also

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \geq 0.$$

Then

$$\lim_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle = 0.$$

Now let $y \in X$ and set $y_\epsilon = (1 - \epsilon)x + \epsilon y$ for $\epsilon > 0$. By monotonicity

$$0 \leq \langle T(x_k) - T(y_\epsilon), x_k - y_\epsilon \rangle = \langle T(x_k) - T(y_\epsilon), \epsilon(x - y) + x_k - x \rangle.$$

Rearranging the terms we have

$$\epsilon \langle T(x_k), x - y \rangle \geq \langle T(y_\epsilon), x_k - x \rangle - \langle T(x_k), x_k - x \rangle + \epsilon \langle T(y_\epsilon), x - y \rangle.$$

Passing to the limit $k \rightarrow \infty$ with $\epsilon > 0$ fixed

$$\epsilon \liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq \epsilon \langle T(y_\epsilon), x - y \rangle.$$

Dividing by ϵ and passing to the limit $\epsilon \rightarrow 0^+$ we obtain using the continuity of T that

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq \langle T(x), x - y \rangle.$$

Eventually

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle = \lim_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle + \liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq 0.$$

This proves that T is pseudomonotone. $\square$

The following result is useful.

PROPOSITION B.24. Let X be a Banach space. Let T_1 and T_2 from X into X' be pseudomonotone operators. Then also $T_1 + T_2$ is pseudomonotone.

PROOF. Let $x_k \rightharpoonup x$ with

$$\limsup_{k \rightarrow \infty} \langle T_1(x_k) + T_2(x_k), x_k - x \rangle \leq 0.$$

We claim that

$$(B.3) \quad \limsup_{k \rightarrow \infty} \langle T_i(x_k), x_k - x \rangle \leq 0$$

for both $i = 1$ and $i = 2$. By contradiction

$$\limsup_{k \rightarrow \infty} \langle T_2(x_k), x_k - x \rangle = \epsilon > 0.$$

Up to taking a subsequence we can assume that

$$\lim_{k \rightarrow \infty} \langle T_2(x_k), x_k - x \rangle = \epsilon > 0.$$

Then

$$(B.4) \quad \limsup_{k \rightarrow \infty} \langle T_1(x_k), x_k - x \rangle \leq -\epsilon < 0.$$

Then since T_1 is pseudomonotone

$$\liminf_{k \rightarrow \infty} \langle T_1(x_k), x_k - x \rangle \geq 0$$

but this contradicts (B.4). Then (B.3) holds. Since T_i is pseudomonotone then for all $y \in X$ we have

$$\liminf_{k \rightarrow \infty} \langle T_i(x_k), x_k - y \rangle \geq \langle T_i(x), x - y \rangle.$$

It follows that

$$\liminf_{k \rightarrow \infty} \langle T_1(x_k) + T_2(x_k), x_k - y \rangle \geq \langle T_1(x) + T_2(x), x - y \rangle.$$

This means that $T_1 + T_2$ is pseudomonotone. $\square$

3.6. Lower semicontinuity of integral functionals.

THEOREM B.25 (Lower semicontinuity of integral functionals). Let $\Omega \subset \mathbb{R}^d$ open bounded. Let $f: \Omega \times \mathbb{R}^n \times \mathbb{R}^m \rightarrow (-\infty, \infty]$ such that

- (i) $f(\cdot, z, p)$ is measurable for all $(z, p) \in \mathbb{R}^n \times \mathbb{R}^m$
- (ii) $f(x, \cdot, \cdot)$ is continuous for a.e. $x \in \Omega$
- (iii) $f(x, z, \cdot)$ is convex for all $x \in \Omega$ and $z \in \mathbb{R}^n$
- (iv) there exists $\phi \in L^1(\Omega)$ such that $f(x, z, p) \geq \phi(x)$ for a.e. $x \in \Omega$ and all $(z, p) \in \mathbb{R}^n \times \mathbb{R}^m$.

Then for all measurable $y: \Omega \rightarrow \mathbb{R}^n$ and $u: \Omega \rightarrow \mathbb{R}^m$ is well defined

$$I(y, u) = \int_{\Omega} f(x, y, u) dx.$$

Moreover if $y_k, y: \Omega \rightarrow \mathbb{R}^n$ measurable satisfy

$$y_k \rightarrow y \quad \text{in measure}$$

and $u_k, u \in L^1(\Omega, \mathbb{R}^m)$ satisfy

$$u_k \rightarrow u \quad \text{in } L^1(\Omega, \mathbb{R}^m)$$

then

$$I(y, u) \leq \liminf_{k \rightarrow \infty} I(y_k, u_k).$$

4. Convex analysis

EXAMPLE B.26. Let $f(v) = \frac{1}{2}\|v\|^2$. Then $f^*(\xi) = \frac{1}{2}\|\xi\|^2$. Indeed

$$\langle \xi, v \rangle - \frac{1}{2}\|v\|^2 \leq \frac{2\|\xi\|\|v\| - \|v\|^2}{2} \leq \frac{\|\xi\|^2}{2}.$$

This implies $f^*(\xi) \leq \frac{\|\xi\|^2}{2}$. To get the opposite inequality, fix $\epsilon > 0$ and let $v \in V$ such that $\|v\| = \|\xi\|$ and $\langle \xi, v \rangle \geq \|\xi\|^2 - \epsilon$. Then

$$f^*(\xi) \geq \|\xi\|^2 - \epsilon - \frac{1}{2}\|\xi\|^2 \geq \frac{1}{2}\|\xi\|^2 - \epsilon.$$

By arbitrariness of ϵ we conclude.

Let f and g from V to $(-\infty, \infty]$. Their inf-convolution $f \square g: V \rightarrow (-\infty, \infty]$ is

$$f \square g(v) = \inf_{w \in V} (f(w) + g(v - w)).$$

PROPOSITION B.27. (inf-convolution duality formula) Let f and g from V to $(-\infty, \infty]$. Then

- (i) $(f \square g)^* = f^* + g^*$
- (ii) $(f + g)^* \leq f^* \square g^*$
- (iii) if f and g are convex and proper and there exists $v \in \text{dom } f \cap \text{dom } g$ and f is continuous at v then $(f + g)^* = f^* \square g^*$.

PROOF. *Step 1.* It holds that

$$\begin{aligned} (f \square g)^* &= \sup_{v \in V} \left(\langle \xi, v \rangle - \inf_{w \in V} (f(w) + g(v - w)) \right) = \\ &= \sup_{v \in V} \sup_{w \in V} (\langle \xi, w \rangle - f(w) + \langle \xi, v - w \rangle - g(v - w)) = \\ &= \sup_{w \in V} \sup_{v \in V} (\langle \xi, w \rangle - f(w) + \langle \xi, v - w \rangle - g(v - w)) = \\ &= \sup_{w \in V} (\langle \xi, w \rangle - f(w) + g^*(\xi)) = f^*(\xi) + g^*(\xi). \end{aligned}$$

This establish (i).

Step 2. By the Young–Fenchel inequality

$$f^*(\zeta) + g^*(\xi - \zeta) \geq \langle \xi, v \rangle - f(v) - g(v).$$

Taking the supremum over $v \in V$

$$f^*(\zeta) + g^*(\xi - \zeta) \geq (f + g)^*$$

and taking the infimum over $\zeta \in V$

$$f^* \square g^* \geq (f + g)^*.$$

This proves (ii).

Step 3. Now we prove (iii). If $(f + g)^*(\xi) = \infty$ then (iii) is satisfied at ξ . Then suppose that $(f + g)^*(\xi) \in [-\infty, \infty)$. Since $\text{dom } f \cap \text{dom } g \neq \emptyset$ then actually $(f + g)^*(\xi) = c \in (-\infty, \infty)$. Let

$$A = \{(a, v) \in \mathbb{R} \times V \mid a \leq \langle \xi, v \rangle - g(v) - c\}.$$

Clearly A is nonempty and convex. We show that A and $\text{int}(\text{epi } f)$ are disjoint. Indeed if by contradiction $(a, v) \in A \cap \text{int}(\text{epi } f)$ then

$$f(v) < a \leq \langle \xi, v \rangle - g(v) - c$$

and

$$c < \langle \xi, v \rangle - f(v) - g(v) \leq (f + g)^*(\xi) = c.$$

Then by Hahn–Banach theorem there exists $(\gamma, \zeta) \in \mathbb{R} \times V^*$ such that

$$(B.5) \quad \gamma a + \langle \zeta, v \rangle \leq \gamma b + \langle \zeta, w \rangle$$

for all $(a, v) \in \text{int}(\text{epi } f)$ and $(b, w) \in A$. If we fix $v = w \in \text{dom } f \cap \text{dom } g$, $b \in \mathbb{R}$ sufficiently small such that $(b, v) \in A$ and we let $a \rightarrow \infty$, we discover that $\gamma \leq 0$ necessarily. If $\gamma = 0$ then inequality (B.5) and Hahn–Banach theorem imply that $\text{dom } f$ and $\text{dom } g$ are separated by a closed hyperplane. This in turn

implies that $\text{int}(\text{dom } f)$ and $\text{dom } g$ are disjoint, but this contradicts the hypotheses so that $\beta < 0$. In this case we set $\theta = |\beta|^{-1}f\zeta$ and from inequality (B.5) we get

$$\begin{aligned} f^*(\theta) &= \sup \{ \langle \theta, v \rangle - f(v) \mid v \in V \} = \\ &= \sup \{ \langle \theta, v \rangle - a \mid (a, v) \in \text{epi } f \} \leq \\ &\leq \inf \{ \langle \theta, v \rangle - a \mid (a, v) \in A \} = \\ &= c + \inf \{ \langle \theta - \xi, v \rangle + g(v) \mid v \in V \} = c - g^*(\xi - \theta). \end{aligned}$$

Then

$$f^* \square g^*(\xi) \leq f^*(\theta) + g^*(\xi - \theta) \leq c = (f + g)^*(\xi).$$

This concludes the proof. $\square$

PROPOSITION B.28 (properties of positively 1-homogeneous convex functions). Let $f: V \rightarrow (-\infty, \infty]$ be a proper, l.s.c., positively 1-homogeneous and convex function. Then

- (i) $f(0) = 0$ and f is subadditive
- (ii) for each $x \in \text{dom } f$ it holds $\partial f(x) = \{ \xi \in \partial f(0) \mid f(x) = \langle \xi, x \rangle \}$
- (iii) $f^* = \iota_K$ with $K = \partial f(0)$ and $f = \sigma_K$.

PROOF. *Step 1.* Let $v \in \text{dom } f$. Since f is l.s.c. and is positive 1-homogeneous

$$f(0) \leq \liminf_{n \rightarrow \infty} f\left(\frac{1}{n}v\right) = \liminf_{n \rightarrow \infty} \frac{f(v)}{n} = 0.$$

By positively 1-homogeneity and convexity we have

$$\frac{1}{2}f(v_0 + v_1) = f\left(\frac{v_0 + v_1}{2}\right) \leq \frac{f(v_0) + f(v_1)}{2}.$$

This proves (i).

Step 2. We prove that for all $\xi \in V^*$ it holds that $f^*(\xi) \in \{0, \infty\}$. Indeed let $\xi \in V^*$ such that $f^*(\xi) < \infty$. Then since f is positively 1-homogeneous, it holds $f^*(\xi) = 2^{-1}f^*(\xi)$ and this implies $f^*(\xi) = 0$.

Step 3. Let $v \in \text{dom } f$ and $\xi \in \partial f(v)$. This is equivalent to the equality $f(v) + f^*(\xi) = \langle \xi, v \rangle$ which in turn implies that $f^*(\xi) < \infty$ and by previous step $f^*(\xi) = 0$. Then $f(v) = \langle \xi, v \rangle$ and for all $w \in V$

$$f(w) \geq f(v) + \langle \xi, w - v \rangle = \langle \xi, w \rangle.$$

This is equivalent to $\xi \in \partial f(0)$. We have proved that if $\xi \in \partial f(v)$ then $f(v) = \langle \xi, v \rangle$ and $v \in \partial f(0)$. It is immediate to check that also the converse is true. This proves (ii).

Step 4. It holds that $\xi \in K$ if and only if $f^*(\xi) = f(0) + f^*(\xi) = \langle \xi, 0 \rangle = 0$. This proves that $f^* = \iota_K$. Since f is l.s.c. then $f = f^{**} = \iota_K^* = \sigma_K$. This establishes (iii). $\square$

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